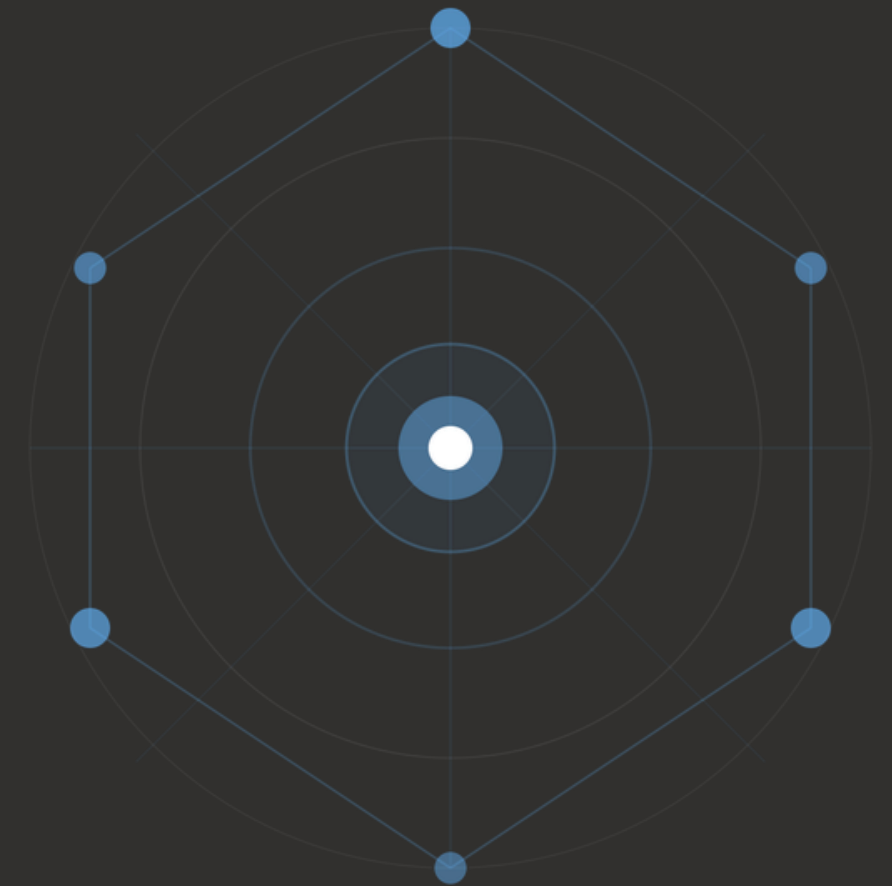




Interactive Workshop on **Data & AI** Fundamentals

Building a shared understanding of data, its
value, and how AI transforms it into intelligence.



DATA SPARKS!



Assoc. Prof. Ir. Dr. Ahmad Jazlan Haja Mohideen

Department of Mechatronics
Kulliyah of Engineering
International Islamic University Malaysia

Education

BEng (Mechatronics), IIUM
MSc (Mechatronics), IIUM
PhD, Control Systems Engineering, University of Western Australia

Area of Expertise

Machine learning
Data analytics



Training Experience

Deep Learning for Machine Vision AI and Data Management
AI for Data Analysis
AI for Business Writing
Power Apps and Power Automate



Passionate about merging **technology** & **design** for smarter, interactive systems!

MEET THE TRAINER



DR DWI PEBRIANTI

**Center for Unmanned
Technology (CUTe)**



Education

PhD. (Artificial System Science)

Chiba University, Japan

M.Eng (Precision Engineering)

The University of Tokyo, Japan

BEng. (Control), Universitas Indonesia

Area of Expertise

**Robotics, Unmanned Aerial Vehicle,
Machine Learning and Artificial Intelligence**

Training Experience

Autonomous Robot

Intelligent Control

Optimization for Engineering

Data science

AI for Business

Quick Poll · Mentimeter



menti.com

1510 1750

Introduction

Why Data?

- Data is the **new currency** — it underpins every modern business decision, product, and service.
- The volume of data being generated is **accelerating exponentially** — and most organisations are barely using what they have.
- Understanding data — and AI's relationship with it — is now a **core professional skill**, not just a technical one.

328.77M

terabytes of new data created every single day

Source: Exploding Topics, 2024

\$274B

Global Big Data & Analytics market value

RM1.9B

Lost to fraud in Malaysia by Sept 2025

The Stakes

Data Breaches Have Real Costs

Online fraud is accelerating — and **data is both the vulnerability and the solution** . Understanding data literacy isn't optional.

47,854

Online fraud & scam cases

Recorded in Malaysia as of September 2025 — largely facilitated by data breaches.

RM 1.919 B

Total losses in the same period — demonstrating that **data security is a financial imperative** .

What is Dark Data?

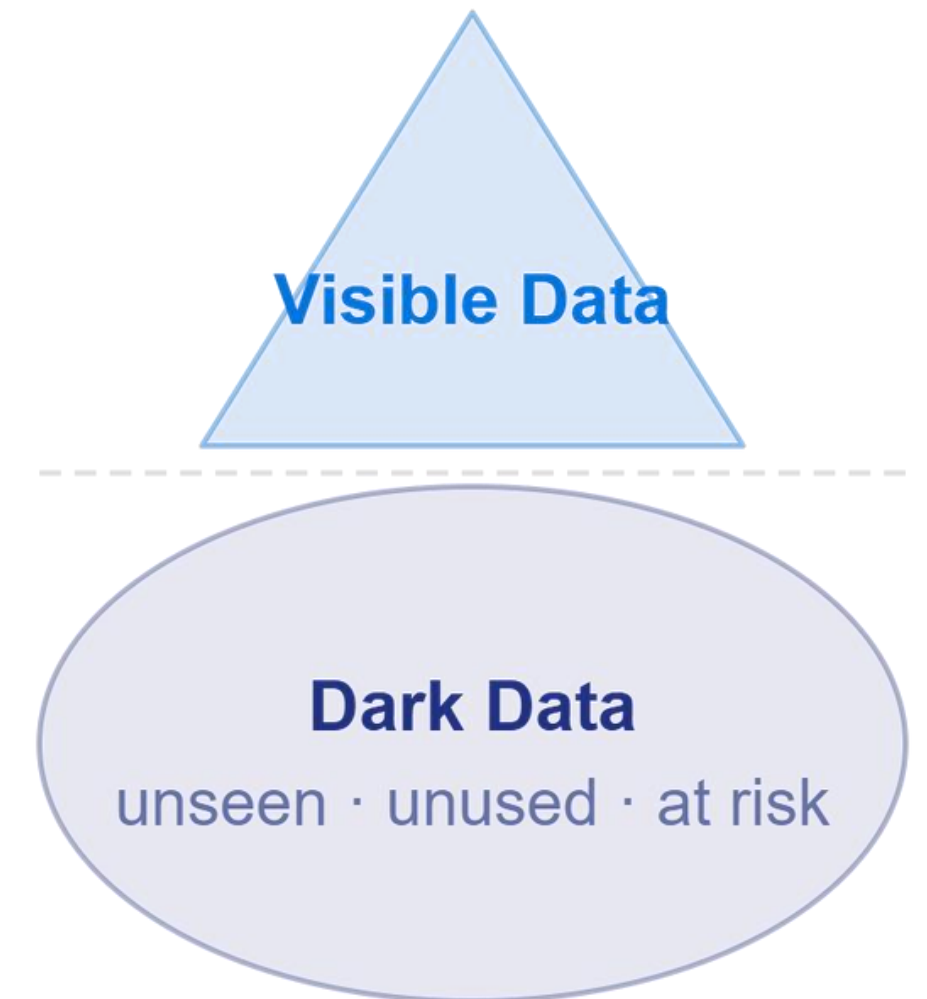
Dark Data are information assets that organisations **collect, process and store** during regular business activities, but generally **fail to use** for analytics, business insights, or direct value creation.

What

Emails, logs, forms, documents, sensor feeds — captured but never analysed.

Why

Lack of tools, awareness, or strategy to extract and activate the value within.



The Dark Data Problem Is Bigger Than You Think

60%

of organisations report

that half or more of their data is considered dark — invisible, unanalysed, unused.

n = 1,300+ business & IT leaders

33%

say it's even higher —

a full third of respondents reported that **75% or more** of their data is dark.

Massive

Illuminating dark data with AI-driven analytics is one of the highest-ROI investments an organisation can make today.

Table Discussion:

Where does OUR dark data live?

- **claims adjuster notes**
- **call logs**
- **underwriting notes (scanned)**
- **servicing emails**
- **call transcripts**
- **customer feedback survey notes**
- **agent chat & WhatsApp messages**
- **branch visit records**

Data Detective

Work in groups. Choose one source of underused insurance data and identify a practical use for it.

CHOOSE

Claims notes, call transcripts, servicing emails, customer feedback or agent messages

DECIDE

What useful pattern, risk or opportunity could AI help reveal?

CHECK

What privacy, consent or accuracy concern must the team address?

YOUR OUTPUT

One data source, one AI use case and one safeguard

02

Part 2

AI in Action

From raw data to intelligence — how organisations are using AI to create real-world value right now.

USE CASES COVERED

Content Recommendation

Search Engine Algorithms

Inventory Management

Auto-Claim Processing

Client Onboarding

Personalised Coverage

AI Turning Data into Value

Content Recommendation

Machine learning analyses demographic and behavioural data to surface the most relevant content for each individual user.

Search Engine Algorithms

ML understands user intent and delivers ranked, relevant results from vast search data — improving with every query.

Inventory Management

Predictive analytics and ML forecast product demand and optimise stock levels using historical purchase data.

Auto-Claim Processing

MyInfo & PayNow integration (e.g. Singapore) enables automatic flight-delay claims — zero manual intervention.

Client Onboarding

AI bots accelerate onboarding by identifying gaps in census files — improving speed and quality of new benefit setups.

Personalised Coverage

AI leverages customer data to recommend tailored products and coverage levels — at the right moment.



What We Covered

- ① **Data is the new currency** — and its volume is growing faster than most organisations can process.
- ② **Dark data is a hidden liability** — 60% of organisations have more unused data than they realise.
- ③ **AI activates that data** — turning raw information into recommendations, automation, and insight.



Workshop Series

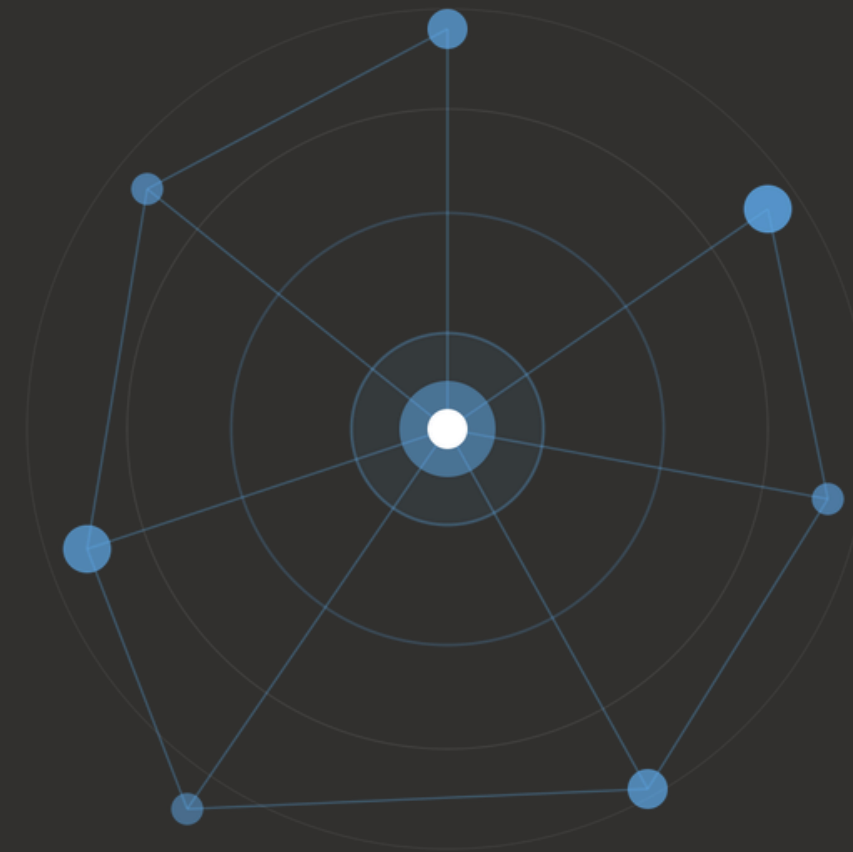




Fundamentals of AI

Building a Shared Understanding of
the Modern Intelligence Era

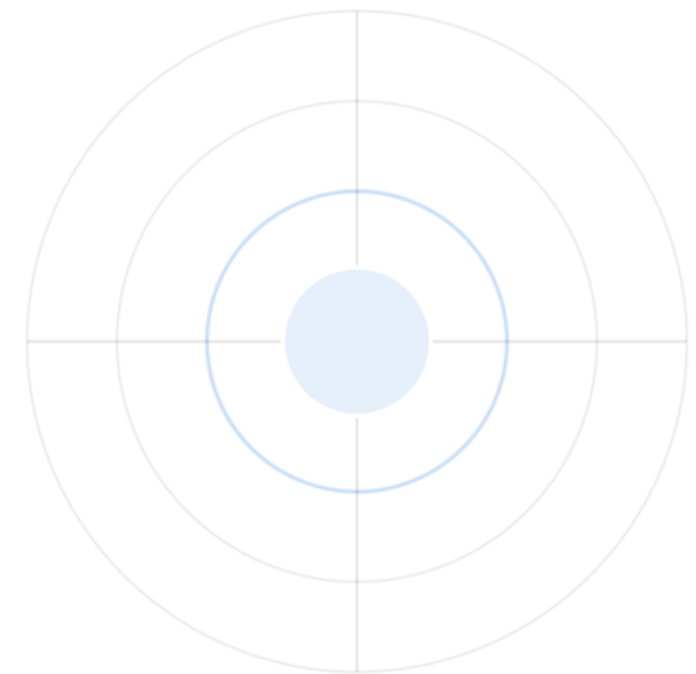
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Session Goal

Why Are We Here?

- Build a **shared vocabulary** around AI so every team member can participate in the conversation.
- Understand the **mechanics** behind modern AI — how systems learn and why they behave the way they do.
- Explore AI's **critical role** in today's workplace and the tools already available to your team.



Defining the "Big Three"

Artificial Intelligence

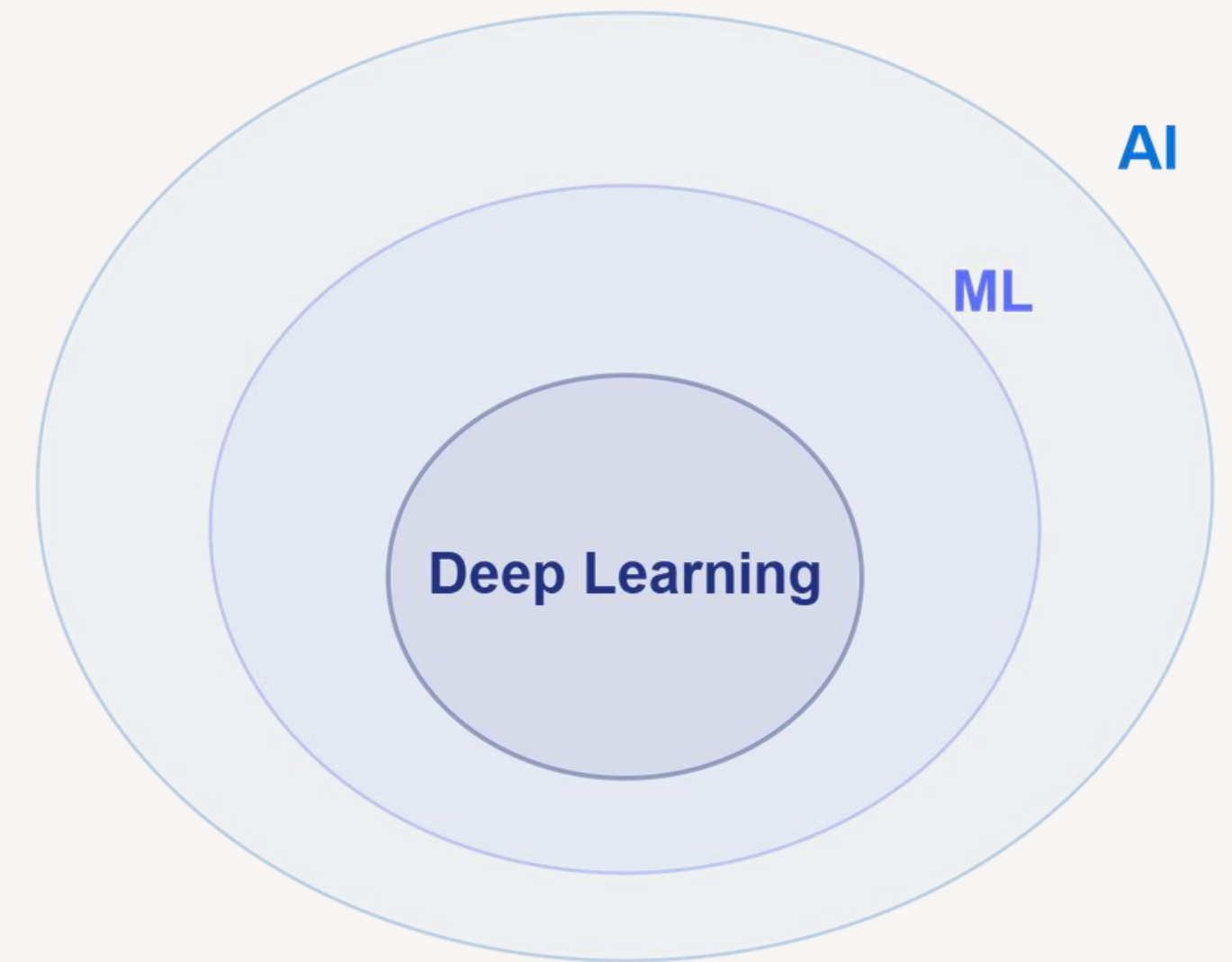
The broad umbrella — any technique enabling machines to simulate human-like intelligence.

Machine Learning

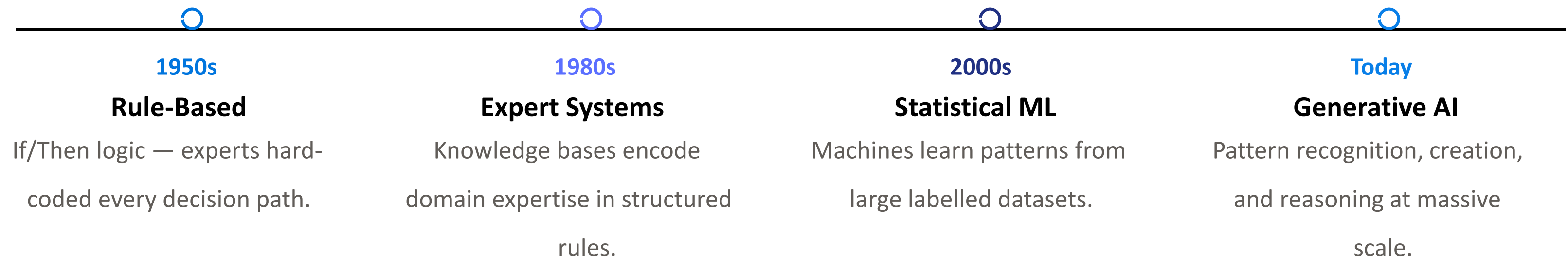
A subset of AI — systems that learn patterns directly from data without being explicitly programmed.

Deep Learning

A subset of ML — layered neural networks modelled on the structure of the human brain.



From Rules to Reason



Speak the Language

Model

A trained mathematical function that maps inputs to outputs based on learned patterns.

Training Data

The dataset used to teach a model — quality and volume directly shape performance.

Inference

Applying a trained model to new inputs to generate a result — what happens at runtime.

Parameters

Internal numerical weights adjusted during training to minimise prediction error.

Tokens

The basic units of text (words or sub-words) that language models read and generate.

Three Pillars of Learning

Supervised

Classification & Prediction

Learns from labelled examples — every input paired with a correct output. Used for spam detection, fraud classification, sales forecasting.

Unsupervised

Clustering & Anomalies

Finds hidden structure in unlabelled data. Groups, patterns and outliers emerge naturally. Used for customer segmentation, anomaly detection.

Reinforcement

Trial & Error

An agent learns by taking actions and receiving rewards or penalties. Used in game AI, robotics, and optimisation problems.

GenAI: The Creative Leap

It doesn't just classify — it **creates** new content: text, code, images, and structured data, all from a simple prompt.

The Creative Engine

Generative models learn **statistical relationships** across billions of tokens.

Given a prompt, they predict the most likely continuation — producing fluent, coherent, novel output at speed.

Understanding Hallucinations

AI outputs are **probability predictions, not fact lookups**. When the model is uncertain, it can generate plausible-sounding but incorrect information. Critical outputs must always be verified.

Section 1.3

AI in the Enterprise

Corporate's strategy: embed intelligence directly into the tools
your team already uses every day — no new app to learn.

● Word

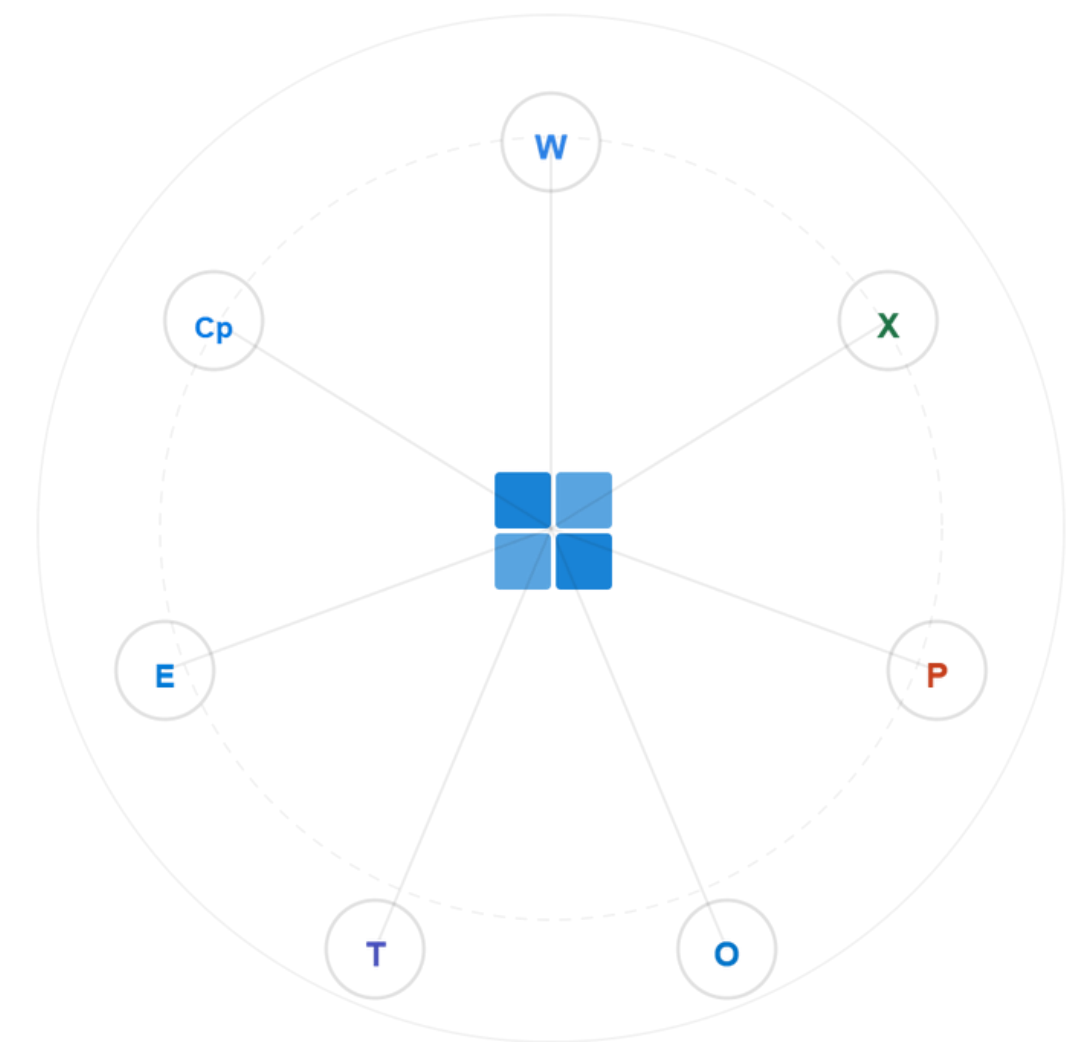
● Excel

● PowerPoint

● Outlook

● Teams

● Edge + Copilot



How Copilot Works For You



LLM + Microsoft Graph + Your Data = Grounded Intelligence

AI or Not AI?

Classify each scenario before the facilitator reveals the answers.

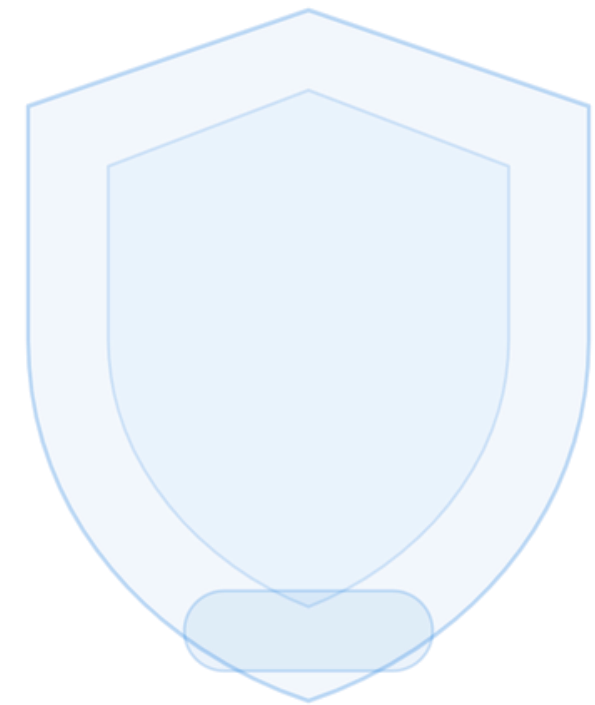
- A Predict whether a policyholder may lapse
- B Group customers by behaviour patterns
- C Generate a personalised email draft
- D Apply a fixed eligibility rule from a policy table

YOUR OUTPUT

Label each item: rule-based, machine learning or generative AI

Your Data, Your Boundary

- **Stays within your tenant.** Your data never leaves the Microsoft 365 boundary — it is not used to train public AI models.
- **Enterprise-grade encryption** at rest and in transit, consistent with your existing Microsoft 365 compliance policies.
- **Existing permissions honoured.** Copilot only surfaces content you already have access to — no escalation of privileges.



02

Continuing

Going Deeper

Expanding the fundamentals — from how models work internally to responsible deployment in the enterprise.

PART 2 COVERS

Training Data & Bias

GenAI Output Types

Copilot Workflow Detail

Responsible AI Principles

Training Data: The Foundation

A model is only as good as the data it learned from. The pipeline has three critical stages.

- **Pre-training**
Vast corpora of internet text, books and code — the model learns general language and reasoning.
- **Fine-tuning**
Curated, domain-specific examples sharpen the model for a particular task or style.
- **RLHF**
Human feedback (Reinforcement Learning from Human Feedback) aligns outputs with helpfulness and safety goals.

Garbage In, Garbage Out

Biased, low-quality or
unrepresentative training data
produces biased, unreliable
models — regardless of
architecture quality.

Learning Types in the Real World

Supervised

Email spam filter

Classifies messages as spam or not based on labelled examples

Sales forecasting

Predicts next quarter revenue from historical data

Fraud detection

Flags anomalous transactions vs known-fraud patterns

Unsupervised

Customer segmentation

Groups customers by purchase behaviour without predefined labels

Anomaly detection

Identifies unusual patterns in network traffic or sensor data

Topic modelling

Discovers recurring themes across thousands of support tickets

Reinforcement

Recommendation engine

Learns which content maximises engagement through feedback

Supply chain optimiser

Learns to minimise cost through trial-and-error simulations

RLHF alignment

Trains LLMs to produce helpful, safe outputs via human ratings

What Generative AI Can Produce

Text

Reports, emails,
summaries, translations,
code documentation,
legal drafts

Copilot in Word · Outlook

Code

Functions, scripts, SQL
queries, test cases,
debugging explanations,
API wrappers

GitHub Copilot · Azure AI

Structured Data

Tables, JSON,
categorised outputs,
data extraction from
unstructured sources
Copilot in Excel · Power BI

Insights

Meeting recaps, action
items, trend analysis,
sentiment summaries
from conversations
Copilot in Teams · Loop

The Copilot Request Lifecycle

1. User Prompt Natural language request entered in any M365 app	2. Safety Check Responsible AI filters applied before processing	3. Graph Query Microsoft Graph fetches relevant files, emails and data	4. LLM Reasoning Model synthesises context + prompt to generate a response	5. Grounding Response anchored to your real data with citations	6. Delivery Output surfaced in- app — draft, summary, or insight
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Throughout the entire lifecycle: **your data never leaves your Microsoft 365 tenant** and is processed under your organisation's existing compliance boundaries.

Spot the Hallucination

Review this synthetic AI answer. It contains claims that may be fabricated.

“Prudential Malaysia’s XYZ Medical Plan provides RM5 million annual coverage, covers all pre-existing conditions after 30 days and guarantees approval for applicants below age 60.”

REVIEW

Underline every statement that requires verification. Name the authoritative source you would check.

YOUR OUTPUT

Three risks and the first correction you would make

Microsoft's 6 Responsible AI Principles

Fairness

AI systems should treat all people equitably — actively auditing for bias in outcomes.

Reliability & Safety

Systems should perform as intended across all conditions and fail safely.

Privacy & Security

Personal data must be respected, protected, and used only for its stated purpose.

Inclusiveness

AI should empower everyone — including people with disabilities and underserved communities.

Transparency

People should understand how AI systems work and what influences their outputs.

Accountability

Humans remain responsible for the design, deployment and outcomes of AI systems.

Important Context

What AI Cannot Do

Understanding limitations
is as important as
understanding capabilities.
The best AI users know
when *not* to use AI.

No True Reasoning

AI produces statistically likely outputs
— it doesn't reason, plan, or
understand cause-and-effect the way
humans do.

No Real-time Knowledge

Models have a training cutoff. Without
retrieval tools, they cannot access
current events or live data.

Hallucinations

Confident-sounding errors are a
structural characteristic. Critical
outputs — especially facts and figures
— must be verified.

Ethical Judgement

AI reflects patterns in its training data.
It cannot exercise moral or contextual
judgement — humans must remain in
the loop.



Key Takeaways

- 01 AI is a broad field — ML and Deep Learning are subsets, each with distinct strengths.
- 02 Generative AI creates — it doesn't look up. Always verify outputs, especially facts and figures.
- 03 Prompt quality directly determines output quality. Role + Context + Task + Format.
- 04 Copilot keeps your data within your tenant boundary — your existing permissions apply.
- 05 Responsible AI requires human accountability. AI augments judgement — it doesn't replace it.

UP NEXT

Module 2

10:15 – 10:30 Break 10:30
Module 2 begins



IIUM Academy

AI for Productivity Using Microsoft Copilot

Delivered By Dr. Dwi Pebrianti
Kulliyyah of Engineering
International Islamic University Malaysia





AI & data analysis

Be heard, collaborate, and share ideas—make meetings and classes more engaging with real conversations.



The 'Will Smith Eating Spaghetti' Metric



Generative AI Observation Challenge

Study the AI examples shown previously. Identify what improved and what still looks unreliable.

NOTICE

Realism, consistency, text quality, lighting, hands and physical behaviour

DISCUSS

Which improvement matters most for business use?

CAUTION

Which remaining weakness could cause reputational or compliance risk?

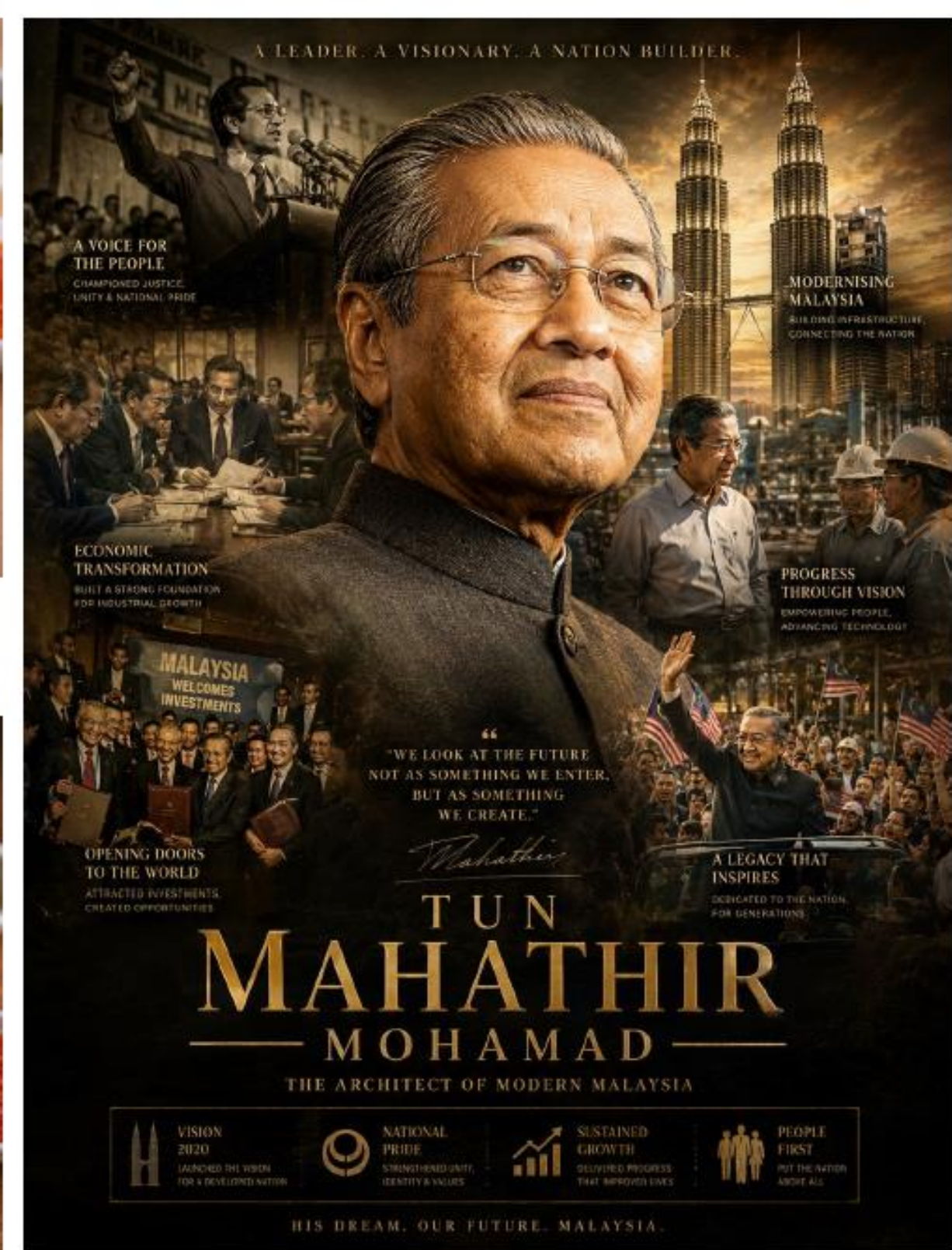
YOUR OUTPUT

Two improvements and one remaining risk

LATEST TREND



Using the images I'm going to upload to create a soft, emotional black and white art. Let the formation be as follows: On the left: a child version of me (from a childhood photo) looking with an innocent smile to the right, sits with her hand under her chin, with 1984 written above it. On the right: A present version of me (from a recent photo) sits with her hands under her chin and looks at the child with a calm smile with 2026" written above her. Studio background: plain and soft (studio background The lighting: Soft, cinematic, warm (even a black-and-white photo) Style: Professional, minimal emotional photography, focusing on feelings and visual communication between the two versions. Make the picture look so real like a between the two versions. Make the picture look so real like a real photoshoot, same background and don't split it. Keep my original features without changing. photo size 4:5



This poster retrospectively showcases the classic achievements of (Name) in (Language), featuring a realistic movie poster style. The vertical composition centers on a large close-up portrait of (Name), surrounded by multiple scenes representing key moments from (Name)'s classic achievements, arranged in layers. The figures are interspersed, with clear foreground and background obscuring each other, creating a strong sense of 3D collage. The overall effect is grand and impactful, yet not chaotic, with rich layers, cinematic lighting, and a gold, brown, black, and gold color scheme. It combines retro and action elements, boasting a realistic texture, high detail, and a theatrical poster quality. It avoids cartoonish, anime-style, exaggerated illustrations, fabricated characters, and childish content.

GENERATIVE AI EVOLUTION 2022–2025 & BEYOND

FROM CHATBOTS TO AGENTS • FROM TOOLS TO PARTNERS • FROM DIGITAL TO REAL WORLD IMPACT



2026 & BEYOND: 3 MAJOR BREAKTHROUGHS AHEAD



CROSS-CUTTING TRENDS (2022–2025)



FROM MODEL → SYSTEM
Shift from isolated models to integrated AI systems



FROM KNOWLEDGE → REASONING
Shift from memorizing information to reasoning and problem solving



FROM TOOL → AUTONOMOUS AGENT
Shift from passive tools to agents that plan and act independently



FROM SINGLE-MODAL → MULTIMODAL
Shift from text-only to understanding all forms of human communication



FROM DEMO → REAL DEPLOYMENT
Shift from experimental demos to production-grade, real-world impact



THE JOURNEY CONTINUES
The next wave of AI will be more autonomous, more personal, and more impactful than ever before.

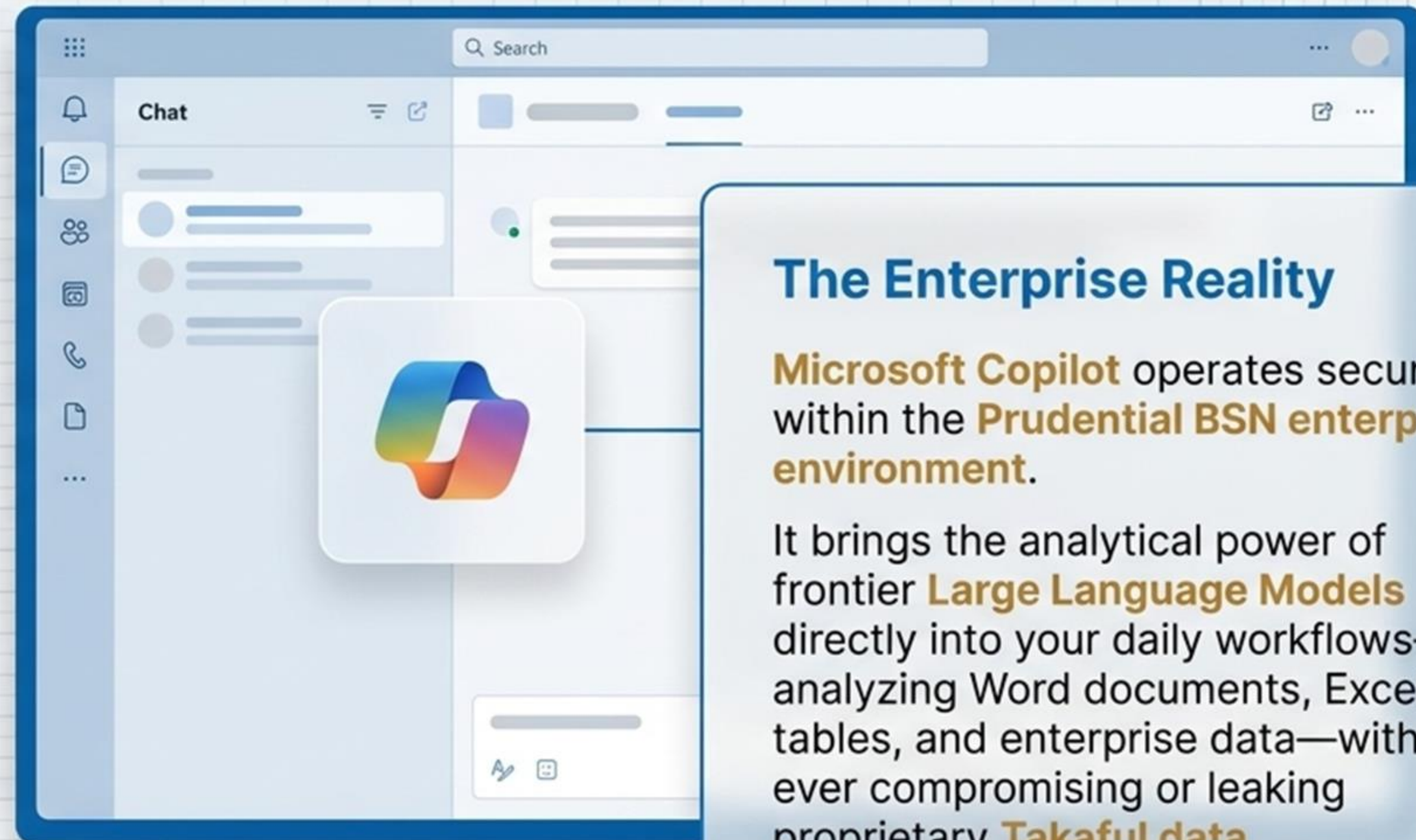
Your Untiring Digital Teammate: Microsoft Copilot

The Shift

"Robots don't take coffee breaks."

"Robots aren't distracted by the latest drama."

"Robots can process in a fraction of the time."

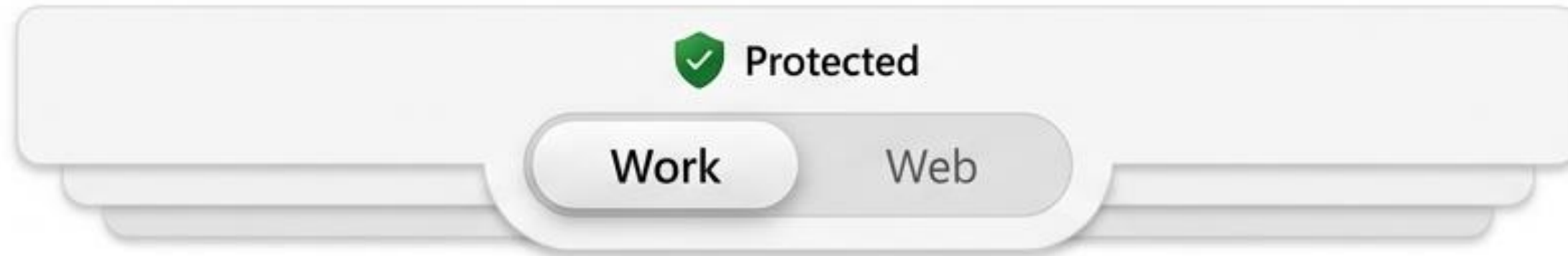


The Enterprise Reality

Microsoft Copilot operates securely within the **Prudential BSN enterprise environment**.

It brings the analytical power of frontier **Large Language Models** directly into your daily workflows—analyzing Word documents, Excel tables, and enterprise data—without ever compromising or leaking proprietary **Takaful data**.

The Microsoft Copilot Ecosystem: Web vs. Work



Copilot Web (Baseline)

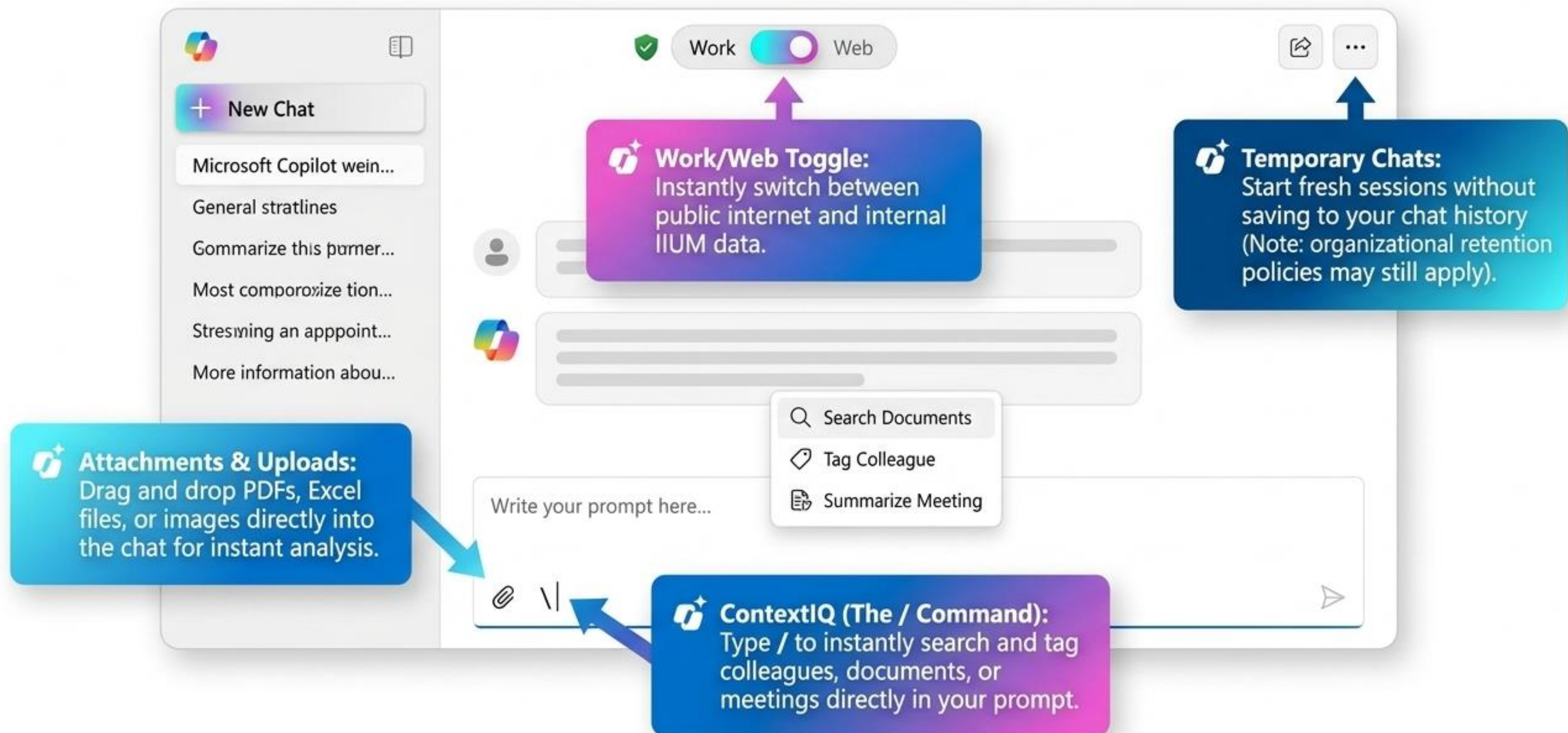
- ✓ Grounded in public internet data (like ChatGPT).
- ✓ Secured by Enterprise Data Protection (EDP)—your prompts are not used to train—your prompts are not used to train global AI models.
- ✓ Best for: General research, broad questions, market trends.



Copilot Work (M365 Copilot)

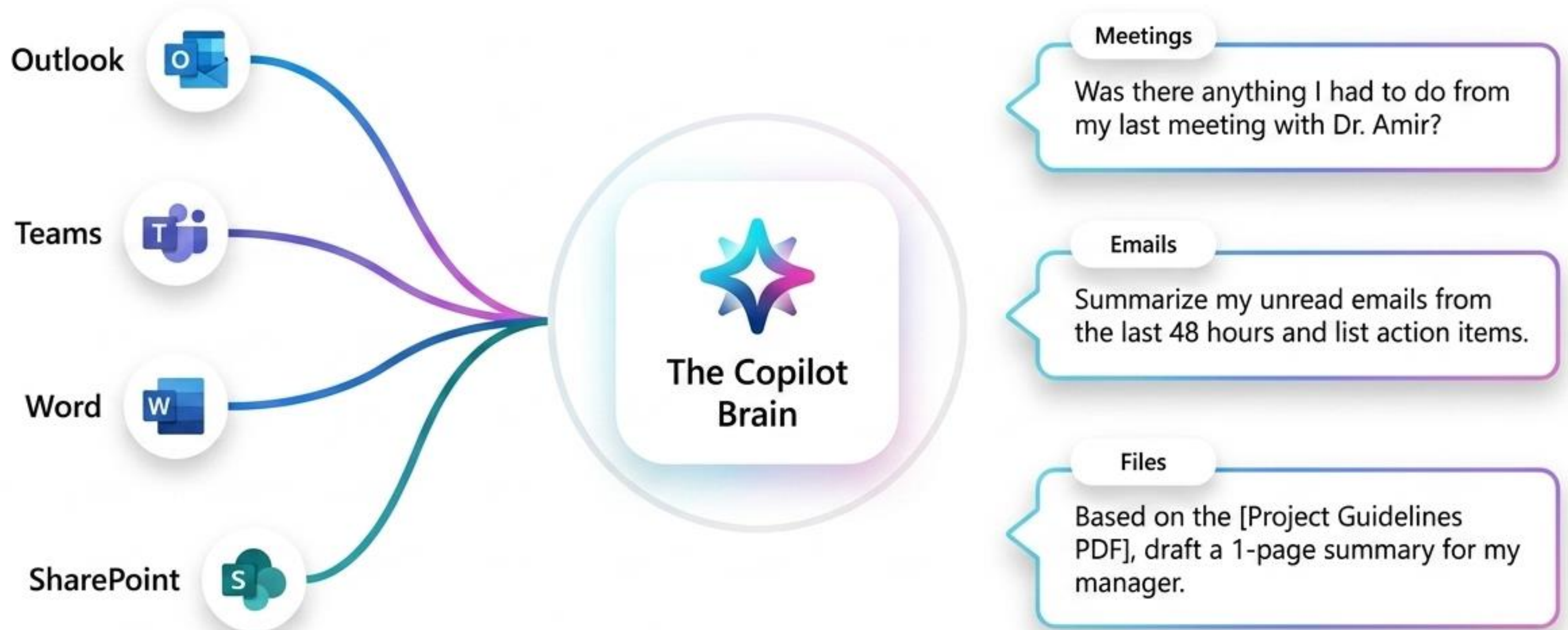
- ✓ Grounded in your private organizational data via Microsoft Graph.
- ✓ Reads your secure emails, Teams chats, and SharePoint files.
- ✓ Best for: Summarize my missed emails, Draft a proposal based on this internal PDF.

Navigating the Copilot Chat Interface



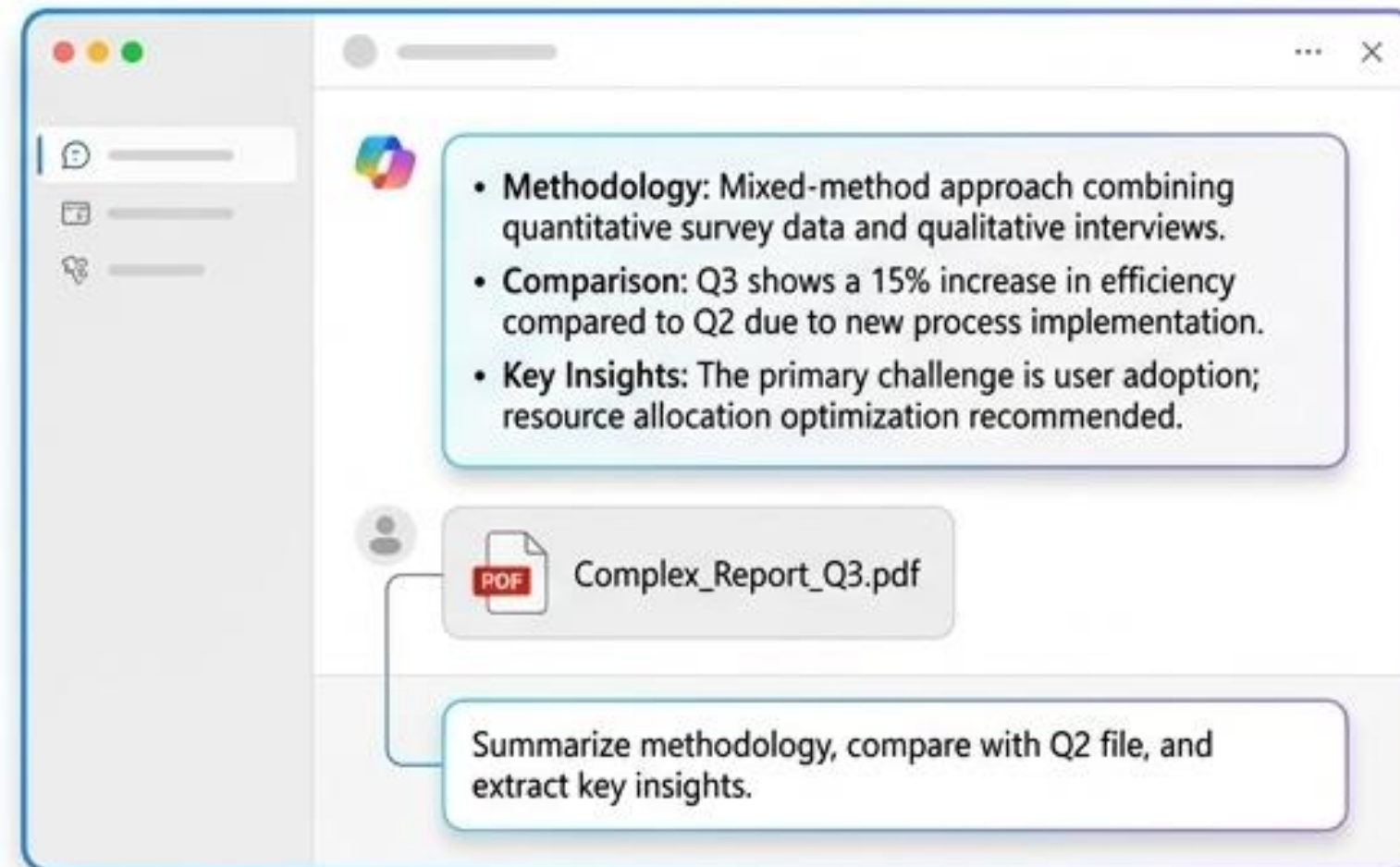
Grounding AI in Your Organizational Data

- Access Emails, Teams Chats, and SharePoint seamlessly without switching apps.



Document Analysis & Media Generation

Analyze Complex Files



Upload large PDFs to summarize methodology, compare files, or extract key insights.

Generate Documents: Draft a PowerPoint outline based on these 3 Word documents.

Generate Images & Video

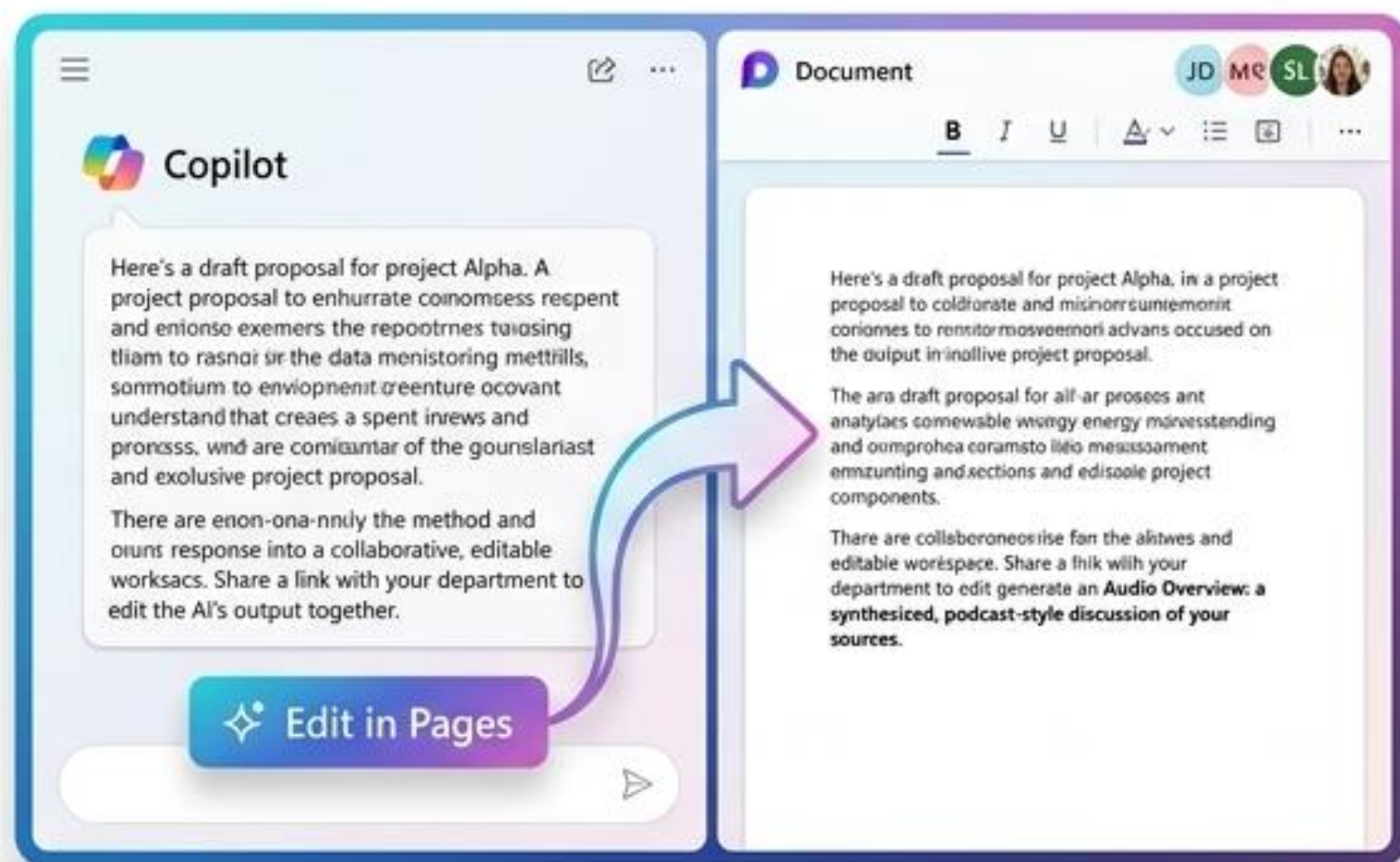


Designer: Create custom diagrams, infographics, or stock images.

Clipchamp: Sequence stock clips and audio narrations into video presentations.

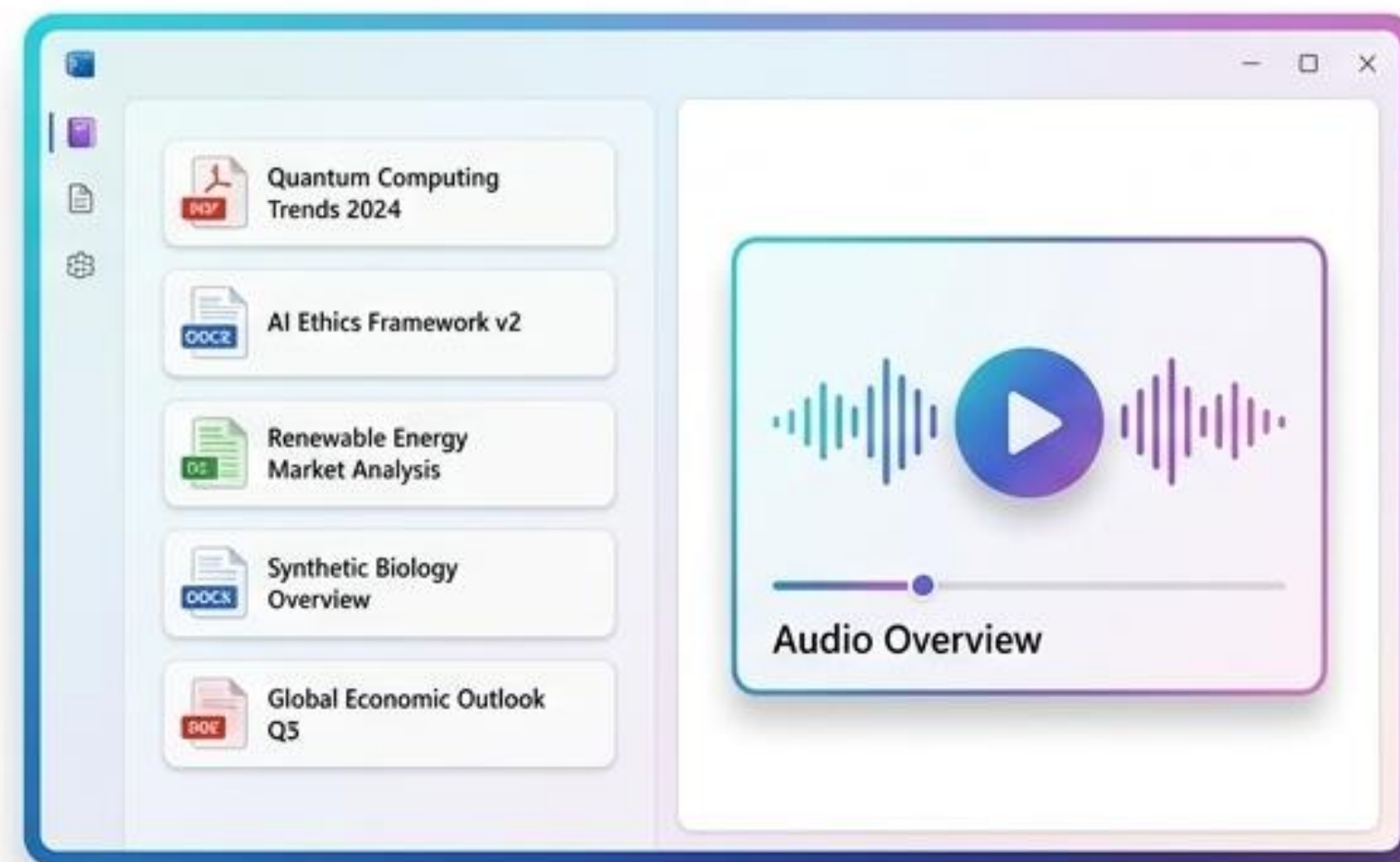
Collaborative AI: Copilot Pages & Notebooks

Copilot Pages



Turn a great AI chat response into a collaborative, editable workspace. Share a link with your department to edit the AI's output together.

Copilot Notebooks



A dedicated workspace for complex, multi-source research. Upload multiple documents simultaneously and generate an Audio Overview: a synthesized, podcast-style discussion of your sources.

Drafting & Designing in M365 Apps

Microsoft Word

- ✧ **Drafting:** Use the floating Copilot box to generate first drafts instantly.
- ✧ **Rewriting:** Highlight text, click the “Sparkle” icon, and choose “Rewrite” to adjust tone or length.



Microsoft PowerPoint

- ✧ **Creation:** Type “Create a presentation from [Document Name]” to build full slide decks.
- ✧ **Design:** Generate custom infographics or summarize entire presentations.



Data Analysis & Agent Mode in Excel

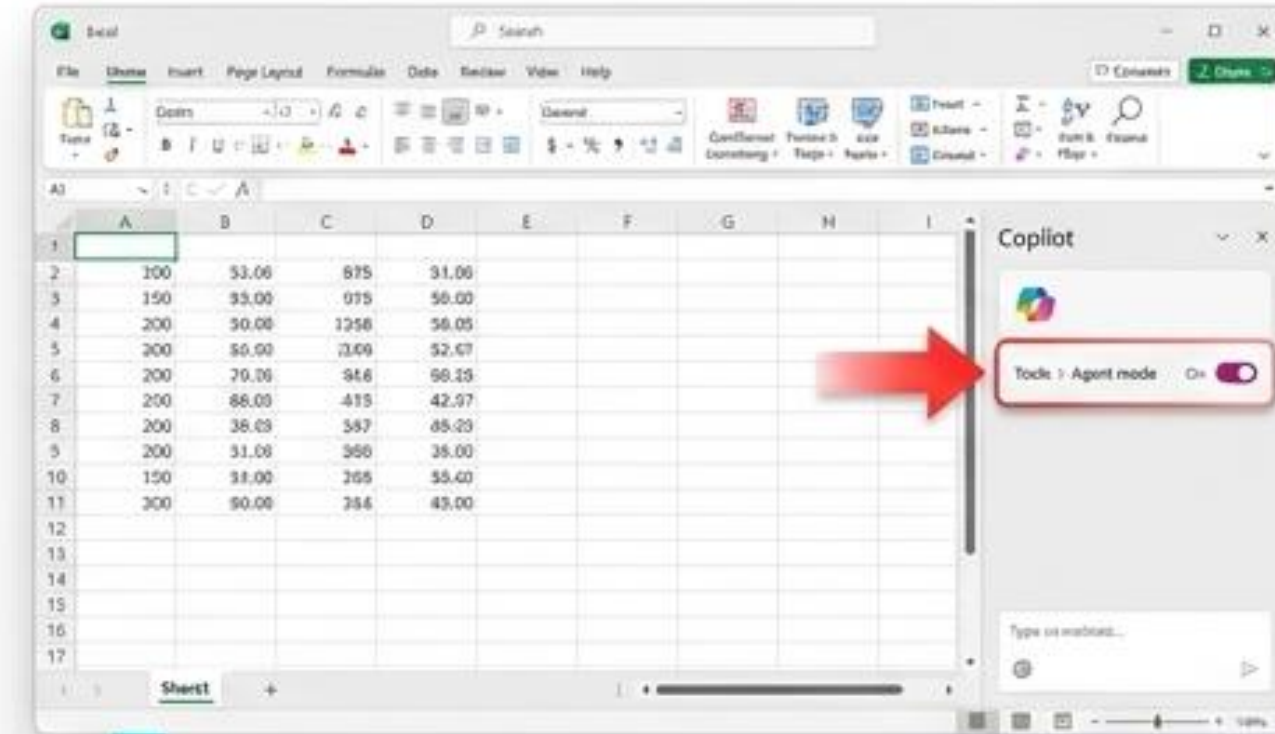
Chat Mode

Ask conversational questions to extract insights (e.g., "What are the top 5 highest costs?"). Copilot reads, but does not alter, the sheet.

Agent Mode (Advanced)

Toggle "Agent Mode" to allow Copilot to actively edit the workbook.

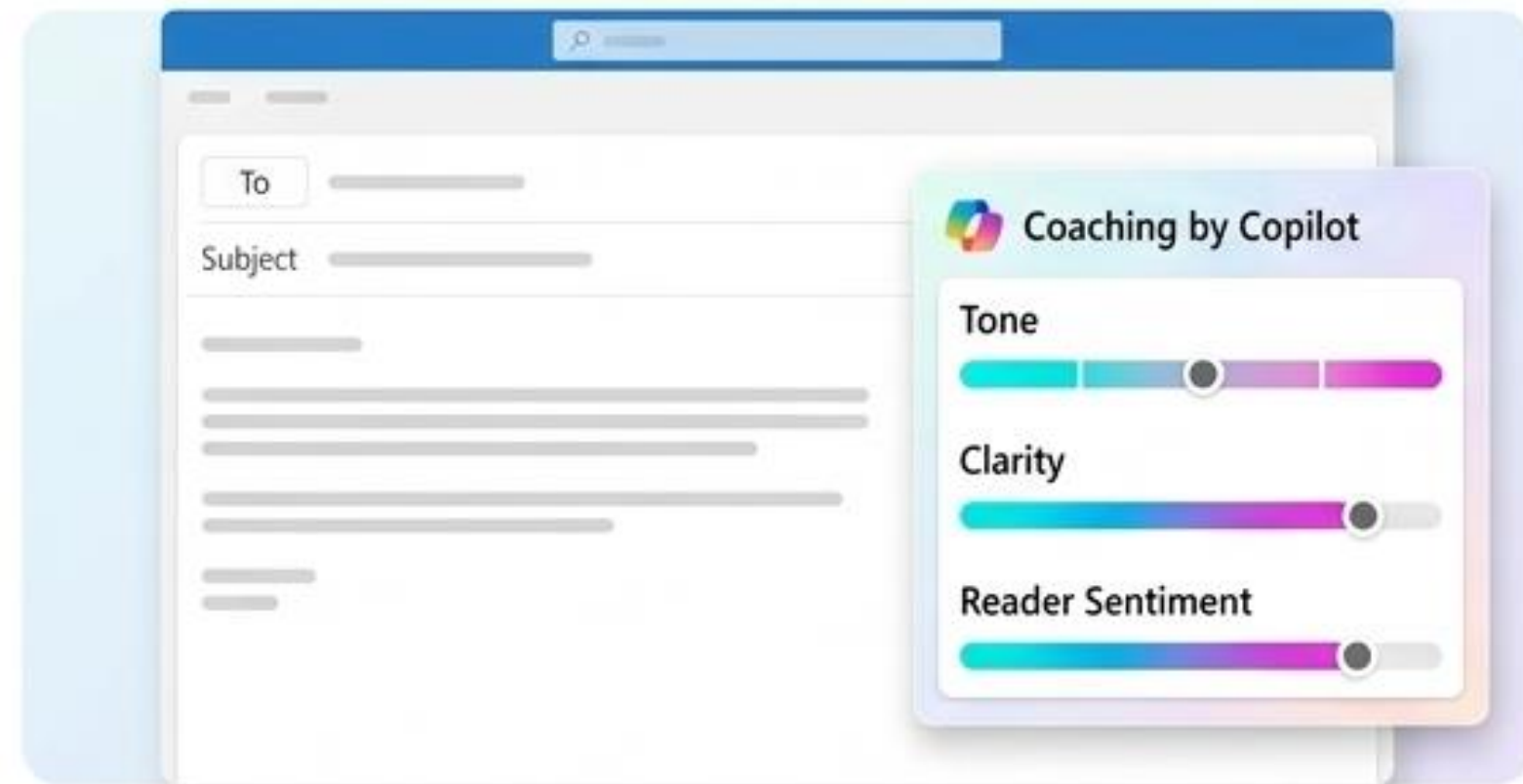
- 🎨 Apply conditional formatting.
- 🎨 Generate and insert new formula columns.
- 🎨 Build complete visual dashboards and charts from raw data in seconds.



Taming Communications in Outlook & Teams



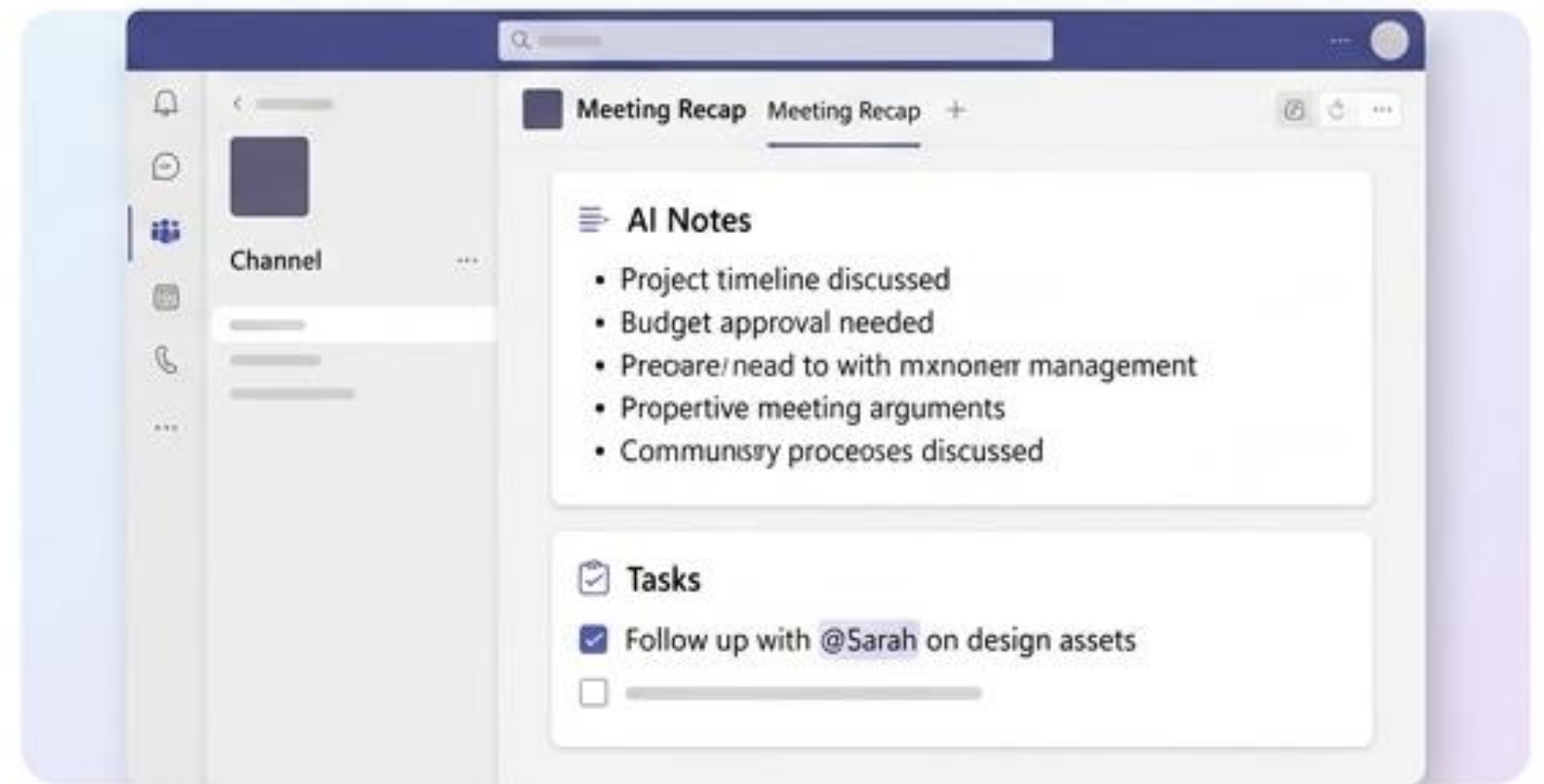
Microsoft Outlook



- ✦ **Draft Replies:** "Draft a polite decline to this meeting, but offer next Tuesday."
- ✦ **Coaching:** Copilot analyzes your drafted email for tone, clarity, and reader sentiment before you hit send.



Microsoft Teams



- ✦ **In-Meeting:** Ask "What did I miss?" if you join late (requires active transcription).
- ✦ **Post-Meeting AI Recap:** Automatically generates meeting notes, lists action items, and tags who committed to specific tasks.

Live Demonstrations: Copilot in Action

1



Demo 1: Chat ContextIQ

Cross-referencing an ongoing email thread with an internal PDF manual.

2



Demo 2: Word & Rewrite

Drafting a department policy and using the Rewrite tool for a formal tone.

3



Demo 3: Excel Agent Mode

Formatting raw financial data and building an instant chart.

4



Demo 4: Teams AI Recap

Pulling actionable tasks from a recorded staff meeting.

How Can YOU Use Copilot Today?



The Lecturer

"Create a 14-week course outline based on these 3 textbook PDFs, including weekly assessment ideas."



The Researcher

"Summarize the methodology of this 40-page paper and compare it to our lab's SOP document."



The Admin Staff

"Draft a professional email to vendors regarding late payments, referencing the attached invoice."



The Manager

"Summarize the Teams chat 'Q3 Budget Planning' and list all outstanding decisions."



Best Practices, Security, and Governance

DO:

- ✦ • Fact-check everything (Always click the citation links).
- ✦ • Use Copilot as a first draft or a brainstorming partner, not the final product.
- ✦ • Verify your Work/Web toggle before handling sensitive institutional data.

DON'T:

- ✦ • Blindly forward AI-generated emails without reviewing them.
- ✦ • Assume 'Temporary Chats' mean zero IT visibility (Retention policies still apply).



The Golden Rule: Copilot only sees what you have permission to see. It cannot bypass IIUM permissions to access restricted files.

Your AI Action Plan



Start Small: Use Web chat for daily questions to build your AI "muscle memory" before tackling complex workflows.



Use the / Command: Ground your requests in real organizational data (Emails, Teams, SharePoint).



Master the Prompt: Remember the RTCF Framework (Role, Task, Context, Format) for high-quality outputs.

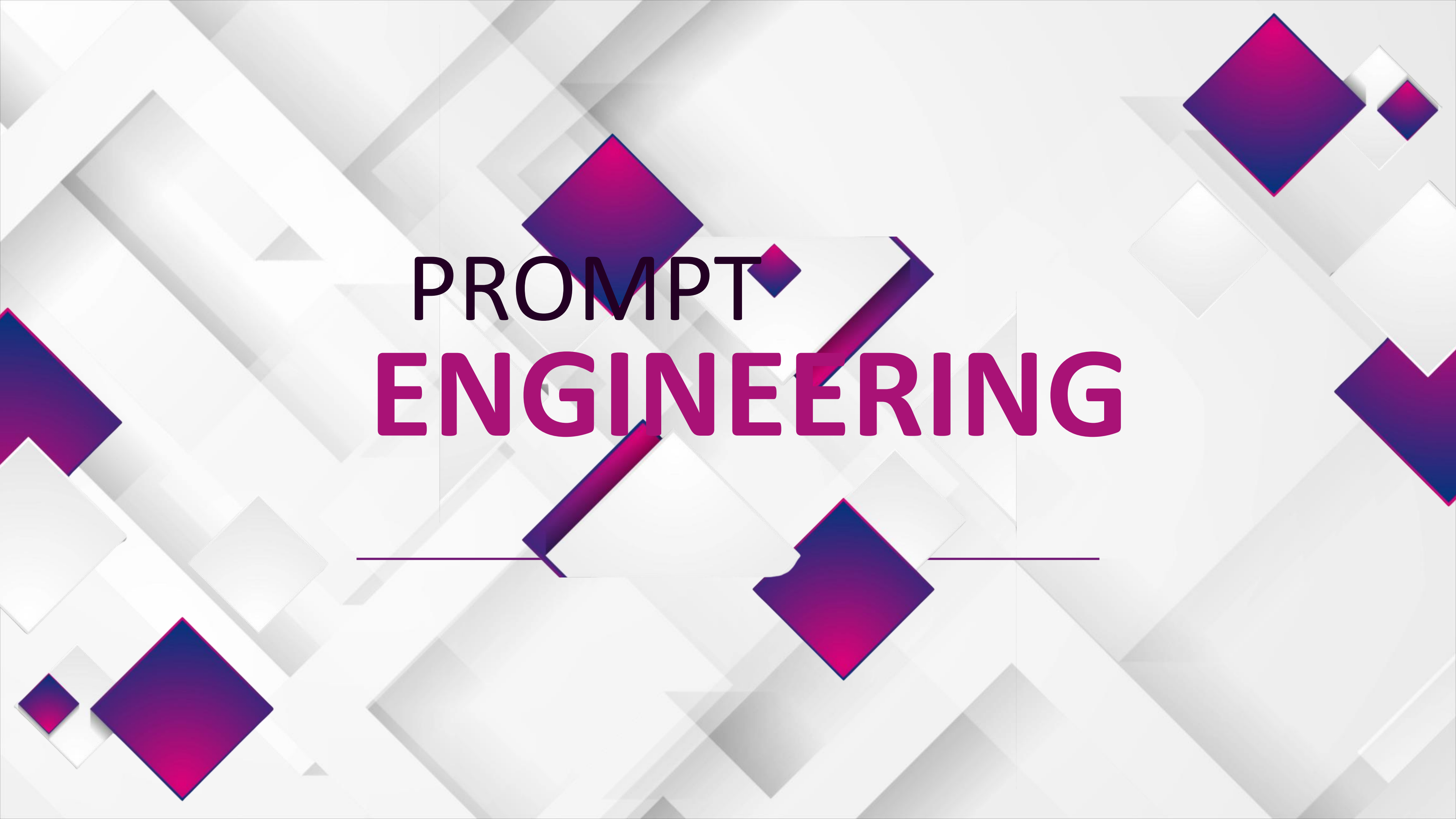


Iterate Relentlessly: Treat AI like an intern. If the first answer is poor, give it direct feedback and ask it to refine the draft.



Scan to Connect & Evaluate



The background features a light gray grid of squares, some of which are slightly offset to create a 3D effect. Overlaid on this are several diamond shapes with a blue-to-purple gradient. A horizontal line is positioned below the word 'ENGINEERING'.

PROMPT ENGINEERING

What We Will Cover



**Understand
what makes
a prompt
effective**



**Apply the
P.A.C.T
framework**



**Improve
prompts
iteratively**



**Use
prompting
for everyday
productivity**

**“English is the
hottest new
programming
language.”
-Andrej Karpathy**



EDUCATION NEXT

Changing (Again)

Example: Sentiment Classification

Software 1.0

```
python
def simple_sentiment(review: str) -> str:
    """Return 'positive' or 'negative' based on a tiny keyword lexicon."""
    positive = {
        "good", "great", "excellent", "amazing", "wonderful", "fantastic",
        "awesome", "loved", "love", "like", "enjoyed", "superb", "delightful"
    }
    negative = {
        "bad", "terrible", "awful", "poor", "boring", "hate", "hated",
        "dislike", "worst", "dull", "disappointing", "mediocre"
    }

    score = 0
    for word in review.lower().split():
        w = word.strip(".,!?:")
        if w in positive:
            score += 1
        elif w in negative:
            score -= 1

    return "positive" if score >= 0 else "negative"
```

Software 2.0

10,000 positive examples
10,000 negative examples
encoding (e.g. bag of words)

train binary classifier

parameters

Software 3.0

You are a sentiment classifier. For every review that appears between the tags

<REVIEW> ... </REVIEW>, respond with exactly one word, either POSITIVE or NEGATIVE (all-caps, no punctuation, no extra text).

Example 1

<REVIEW>I absolutely loved this film—the characters were engaging and the ending was perfect.</REVIEW>

POSITIVE

Example 2

<REVIEW>The plot was incoherent and the acting felt forced; I regret watching it.</REVIEW>

NEGATIVE

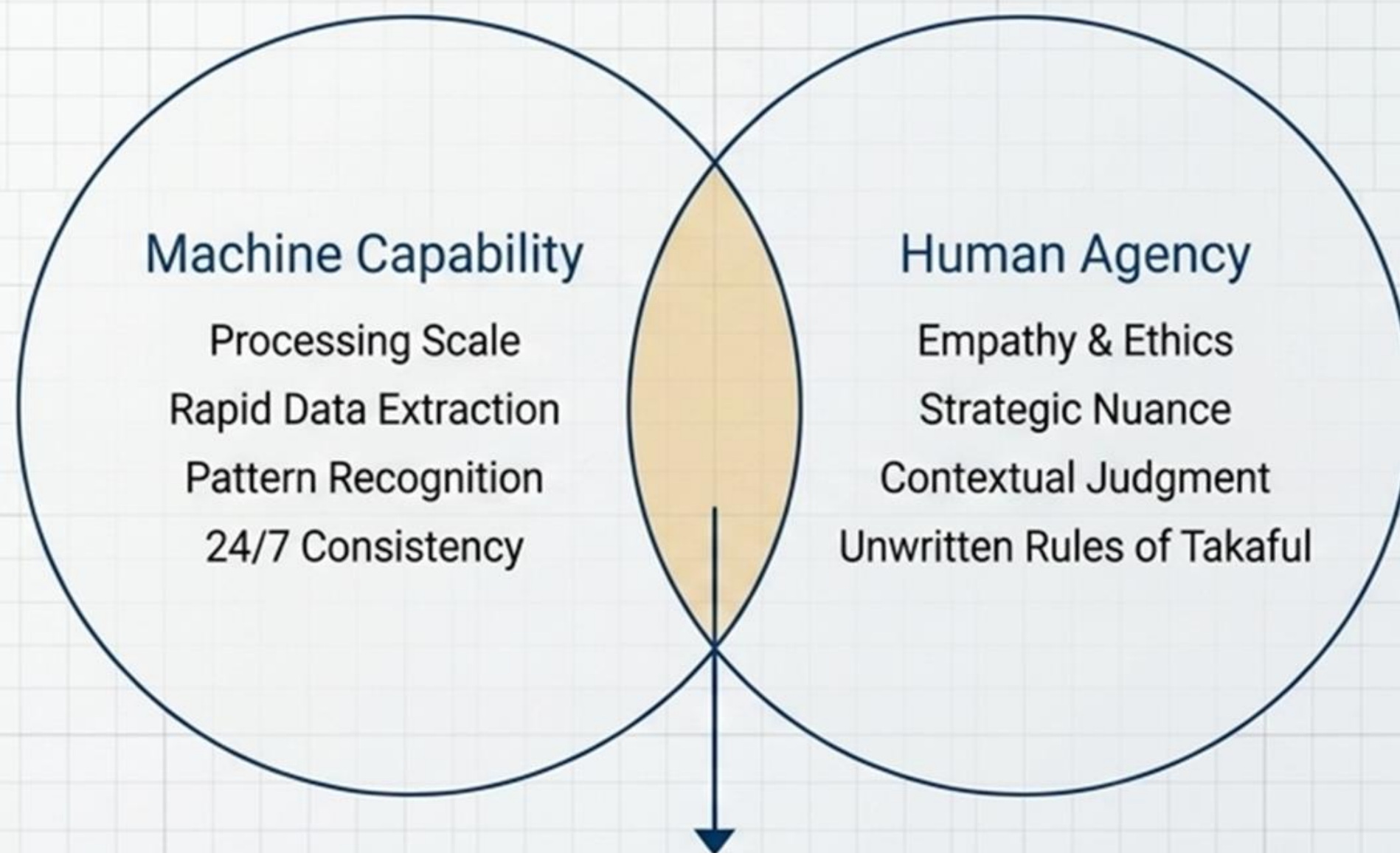
Example 3

<REVIEW>An energetic soundtrack and solid visuals almost save it, but the story drags and the jokes fall flat.</REVIEW>

NEGATIVE

Now classify the next review.

The Future is Human + AI

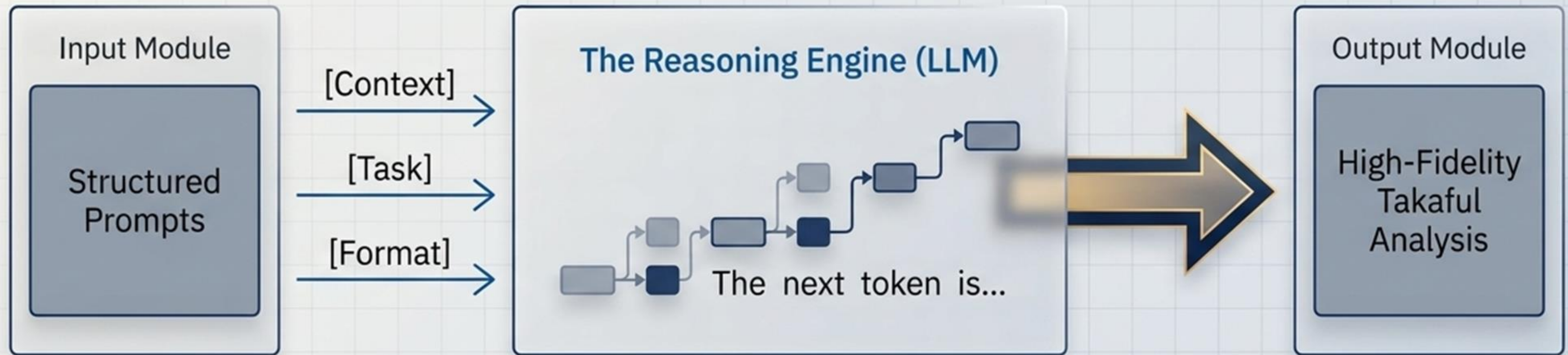


The Synthesis: By ceding basic data processing and summarization to Copilot, Prudential professionals are freed to concentrate entirely on higher-order problem solving, complex adjudications, and genuine customer connection.

The Architecture of Prompt Mastery

Moving beyond basic queries.

Structuring your prompt scientifically reduces cognitive load on the model, eliminates ordering effects, and drastically increases output accuracy.



You are not Googling a question; you are programming a reasoning engine using natural language.

Why Prompting Matters

Same AI Tool



Drastically
Different Results



**Vague Prompt =
Generic Output**



**Good Prompt =
Better Productivity**

AI is like a Brilliant Intern. Highly capable. Has read everything. Knows nothing about your specific business, tone, or needs until you tell it.

The Difference is in the Details



BAD PROMPT

Write an email about the new product.

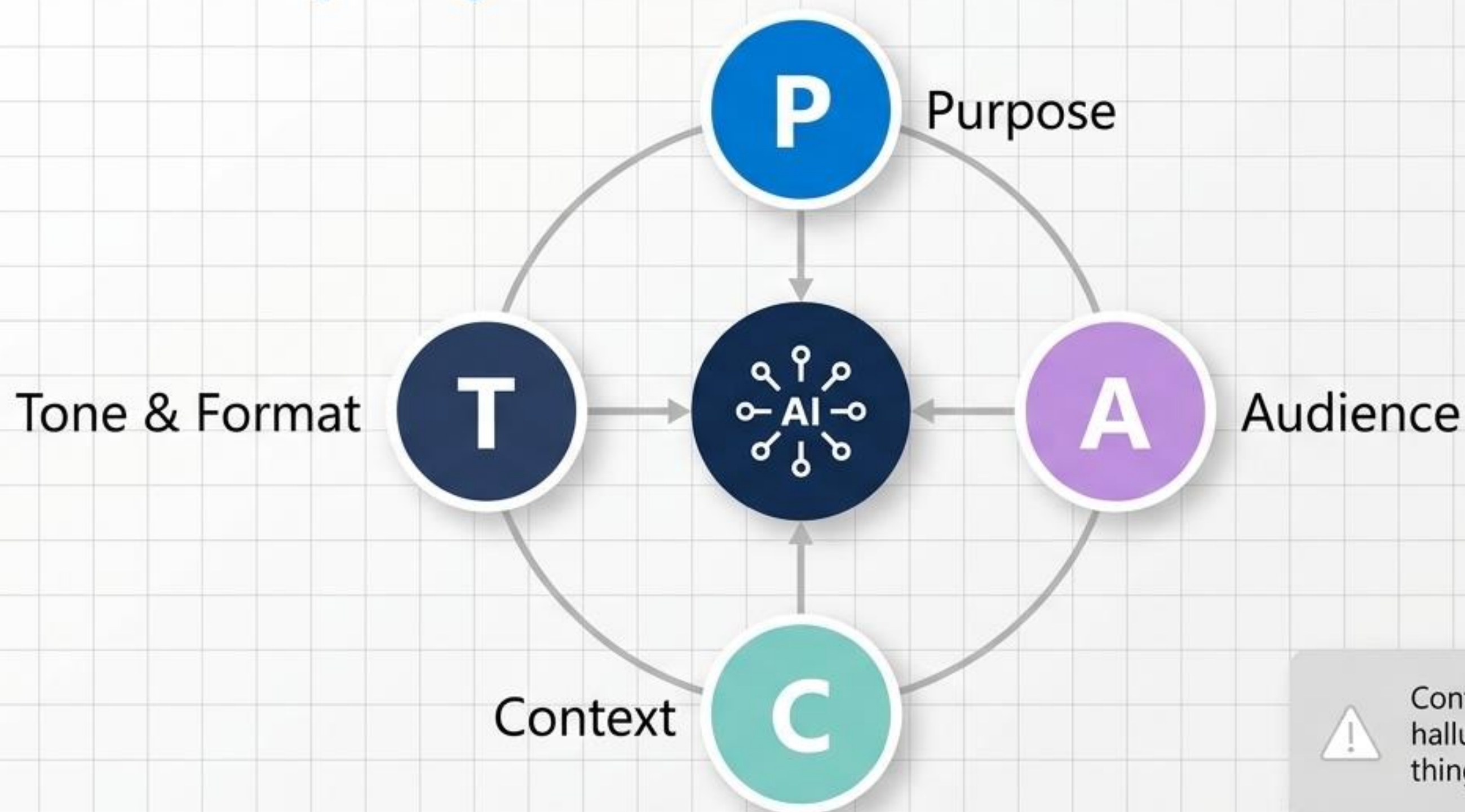


GOOD PROMPT

I am a **SALES MANAGER AT PRUDENTIAL TAKAFUL**. Draft a professional email to our partners announcing our **NEW FAMILY TAKAFUL PLAN LAUNCH IN MAY MAY**. Mention 3 key benefits: **AFFORDABILITY, COMPLIANCE, AND EASY ONLINE ENROLLMENT**. Keep it **UNDER 200 WORDS**. Professional but **WARM TONE**.

The Anatomy of a Good Prompt

Clear instructions replace guesswork.



Context prevents hallucinations (AI making things up).

The P.A.C.T. Framework



P — Purpose
(What do you want?)



A — Audience
(Who are you? Who is reading?)



C — Context
(What does it need to know?)



T — Tone & Format
(How should it look?)

Prompt Battle

Each table improves the same weak prompt, then tests and compares the results.

Weak prompt: “Write an email to a customer about insurance.”

IMPROVE

Add role, context, task, constraints and output format

COMPARE

Which draft would you send? What would a human edit first?

YOUR OUTPUT

One tested prompt and one reason your output is safer

One Image, Three Prompting Decisions

Observe how instructions about subject, composition and style change the result.



1 ORIGINAL



2 ADD A SUBJECT



3 REFINE THE STYLE

From a Weak Prompt to a Creative Brief

Each table rewrites the weak prompt below. The final prompt must cover all six elements.

Weak prompt: “Add a child and make it retro.”

SUBJECT

Who appears, including the child’s age

ACTION

What the model should add or change

COMPOSITION

Placement, pose and framing

STYLE

The intended pop yeh-yeh visual treatment

PRESERVE

Features of the adult and source image that must remain

AVOID

Errors such as extra people, text or distorted anatomy

YOUR OUTPUT

One complete image-editing prompt that another table can follow without explanation

DEBRIEF

What the Prompt Controlled

A strong image prompt works like a clear creative brief.

CONTENT

Who and what appears in the image

COMPOSITION

Where subjects stand, how they pose and what the camera includes

STYLE

Colours, era, photographic treatment and visual references

INVARIANTS

Details that must remain unchanged during editing

NEGATIVE INSTRUCTIONS

Specific errors or unwanted additions to avoid

ITERATION

Evaluate the first result, then request one focused improvement

Subject + Action + Setting + Style + Composition + Constraints

Zero-Shot, One-Shot and Few-Shot

Examples help the model infer the categories, format and decision boundaries you expect.

ZERO-SHOT

Instructions only

Best for clear, familiar tasks

EXAMPLE

Classify this message as Claims, Payment, Coverage or Other.

ONE-SHOT

One worked example

Best when tone or format matters

EXAMPLE

Show one correctly labelled customer message before the new message.

FEW-SHOT

Several worked examples

Best for specialised categories and consistent repeated work

EXAMPLE

Show examples representing the main categories before asking for a label.

More examples can improve consistency, but poor or unrepresentative examples can also steer the model in the wrong direction.

Zero-Shot Prompt

The instruction defines the task and output, but provides no worked example.

Classify the customer message into one category:

Claims enquiry
Payment issue
Coverage question
Other

Customer message:
“My payment was deducted twice this month. Please resolve this urgently.”

Return:
Category:
Reason:

WHAT GUIDES THE MODEL

The category names

The customer’s wording

The requested output fields

Use when the task and categories are already clear.

One-Shot Prompt

One worked example demonstrates both the category and the answer format.

WORKED EXAMPLE

Customer message:
“Has my hospital claim been processed?”

Category: Claims enquiry
Reason: The customer is requesting an update about a submitted claim.

NEW MESSAGE

Now classify:

“My payment was deducted twice this month. Please resolve this urgently.”

Return:
Category:
Reason:

The example demonstrates terminology, response length and the expected level of explanation.

Few-Shot Prompt

Several examples demonstrate category boundaries before the model classifies a new message.

CLAIMS ENQUIRY

“Has my hospital claim been processed?”

PAYMENT ISSUE

“Why was my monthly premium charged twice?”

COVERAGE QUESTION

“Does my policy cover treatment while I am overseas?”

NOW CLASSIFY

“My payment was deducted twice this month. Please resolve this urgently.”

Return the category and a one-sentence reason.

How Examples Influence the AI Response

The model reads the complete prompt as context and predicts a suitable continuation one token at a time.

1

PROMPT BECOMES TOKENS

Words and punctuation are broken into smaller units the model can process.

2

CONTEXT IS INTERPRETED

Instructions and examples indicate the task, labels, tone and output pattern.

3

LIKELY CONTINUATIONS ARE SCORED

The model estimates which next token best fits the patterns in the context.

4

THE RESPONSE IS GENERATED

Tokens are produced sequentially until the answer is complete.

Examples do not retrain the model. They temporarily guide its behaviour within the current conversation.

The model predicts patterns rather than verifying truth, so humans must still check the result.

Customer Message Classification Challenge

Run the same task three times. Compare consistency, format and reasoning.

MESSAGE TO CLASSIFY	“My payment was deducted twice this month. Please resolve this urgently.”
ROUND 1 ZERO-SHOT	Give categories and instructions only.
ROUND 2 ONE-SHOT	Add one correctly classified example.
ROUND 3 FEW-SHOT	Add examples for Claims, Payment and Coverage.
Categories: Claims enquiry, Payment issue, Coverage question or Other	
YOUR OUTPUT	Record the category, format and reason produced in each round

DEBRIEF

When Do Examples Improve the Result?

Compare the three outputs before deciding that more examples are better.

CONSISTENCY

Did the category and response format become more stable?

CLARITY

Did the examples make category boundaries easier to infer?

TRANSFER

Did the model copy irrelevant wording from an example?

BIAS

Were the examples broad enough to represent realistic messages?

EFFICIENCY

Was the extra prompt length justified by a better result?

Use zero-shot for straightforward tasks. Add examples when the organisation's categories, tone or output format need to be demonstrated.

Beyond Zero-Shot and Few-Shot

Choose a strategy according to what the task needs, rather than making every prompt longer.

ROLE

Set the perspective

You are a customer-service quality reviewer.

CONTEXT

Supply relevant background

The customer has contacted the company twice.

CONSTRAINTS

Set boundaries

Do not make assumptions about the policy.

FORMAT

Define a usable output

Return Category, Urgency and Next Action.

AUDIENCE

Adjust language

Explain this to a customer with no insurance background.

CLARIFY FIRST

Handle missing information

Ask up to three questions if essential details are missing.

A short, precise prompt often performs better than a long prompt containing irrelevant instructions.

Grounded Prompts and Structured Outputs

These two strategies reduce unsupported answers and make results easier to review.

GROUNDING PROMPT

Using only the supplied policy document, answer the customer’s question.

Cite the relevant section.

If the document does not contain the answer, state:
“Policy-specific confirmation is required.”

Do not rely on general insurance knowledge.

STRUCTURED OUTPUT

Review the customer message and return:

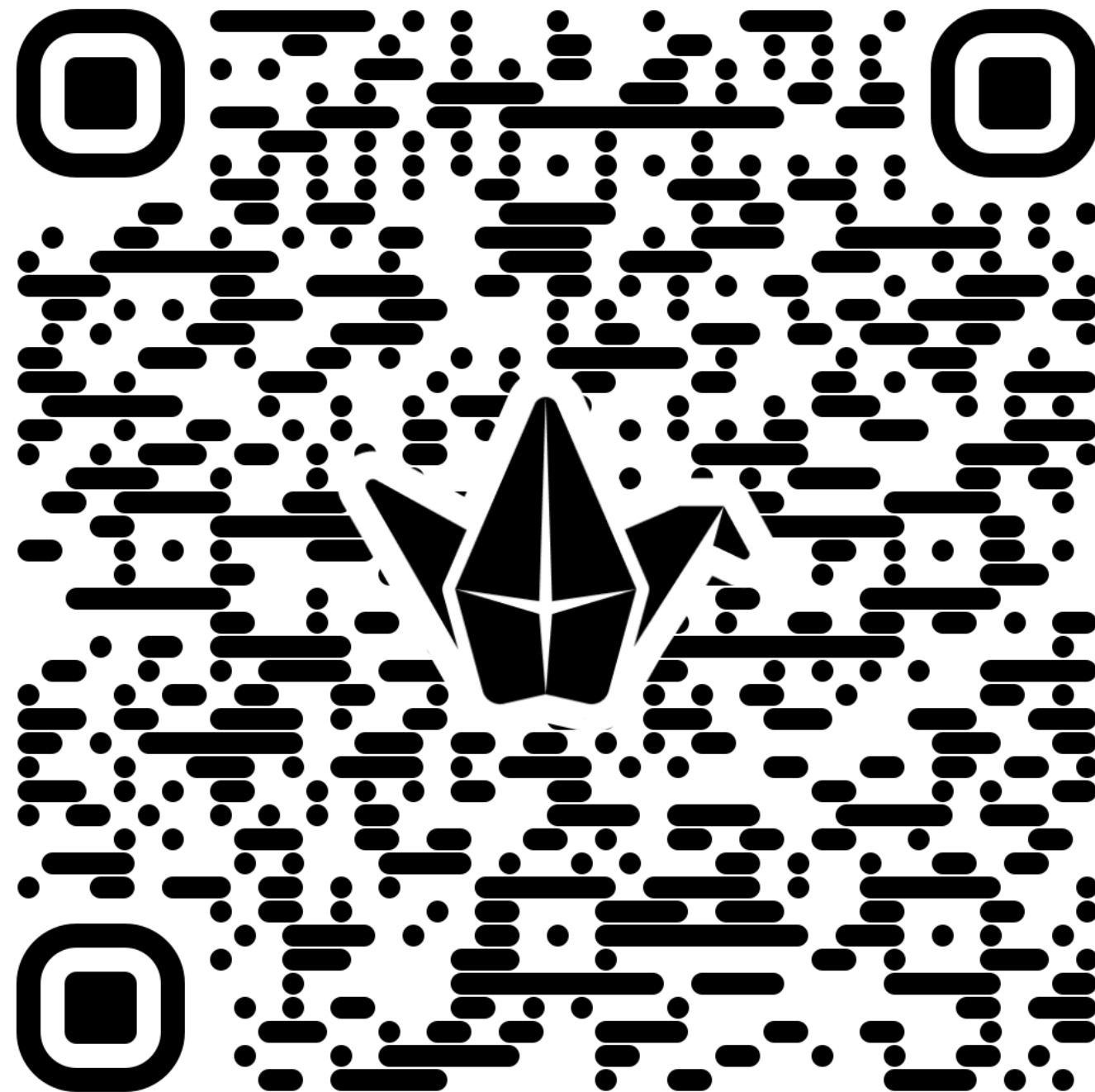
Category:
Urgency: Low, Medium or High
Customer sentiment:
Recommended next action:
Human verification required:

Grounding limits the source. Structure makes the answer consistent and easier to check or transfer into a workflow.

Scenario: You are looking for the best credit card

Padlet

<https://padlet.com/ahmadjazlan/add-your-brainstorming-topic-here-s023rf362tn2nss3c3me>



Scenario:

You are looking for the best credit card.

- **annual salary range**
- **islamic**
- **benefit (cashback vs points)**
- **annual fee waiver**
- **low interest rate**
- **comparison between top 3 banks**
- **you are a financial consultant**
- **cash withdrawal fee**

P is for Purpose

Start with a strong action verb.



Weak: Tell me about... /
Help me with...



Strong: Example: Compare
Product X and Y in a table,
focusing on cost and
coverage.

Draft

Summarize

Analyze

Compare

Translate

Rewrite

Simplify

Generate

A is for Audience & Role

Who are you? Who is reading?

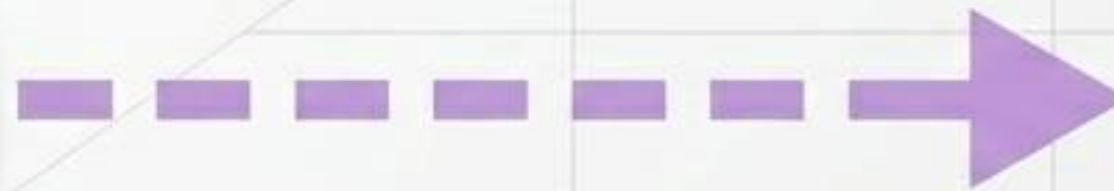
AI adapts its output based on the persona you assign it.



SENDER (Your Role)

"I am a sales agent speaking to a first-time buyer..."

Result: Simple, jargon-free



RECEIVER (The Audience)

"I am an IT manager presenting to the CFO..."

Result: Data-focused, ROI-driven

C is for Context

Give AI the raw materials.

- Provide data, reference an email, or paste a document.
- **Example:** Based on the Q1 data: 1,240 new policies, 18% above target...



Warning: Hallucinations occur when Context is missing!

No context = AI predicts and guesses. Anchor AI to actual facts and data to guarantee accuracy.

T is for Tone & Format

Define exactly how it should look.



tone

Professional

Warm & Empathetic

Urgent

Casual



FORMAT & LENGTH

Bullet points / Numbered list

3-slide presentation outline

Comparison table

Under 100 words

One page maximum

From Weak to Strong (P.A.C.T)

 **BEFORE** (0/4 PACT elements)

Help me with a presentation.

 **AFTER** (4/4 PACT elements)

[P] Create an outline for a 10-slide presentation.

[A] I am the Head of Operations presenting to senior management.

[C] Base it on the attached Q1 performance data. Include challenges and recommendations.

[T] Use a professional, data-focused tone.

P.A.C.T. in Real Work



Email Drafting

"Reply to this angry customer. I am a service rep. Empathize and offer a 3-day resolution."



Meeting Prep

"I am a sales consultant. Generate 5 discovery questions for a cost-sensitive HR director."



Summary

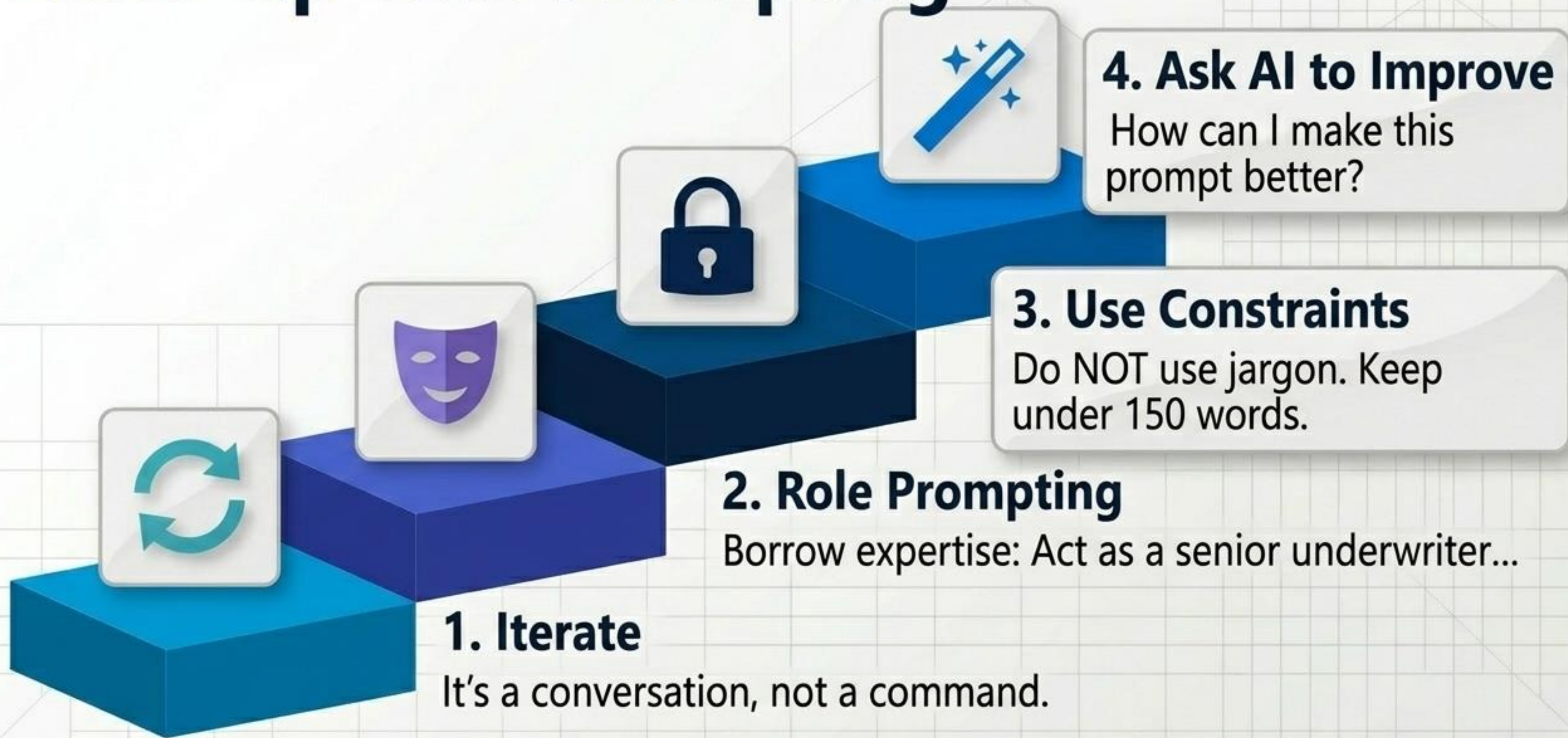
"Summarize this 10-page policy document into a 3-point FAQ for a beginner."



Analysis

"Act as a CFO. Review this Q1 Q1 data and suggest 2 areas to cut costs."

Level-Up Your Prompting



The Copilot Tactic Decision Matrix

Technique	Best For (Takaful Context)	Token Cost / Effort
Chain-of-Thought (CoT) Ask the model to reason step-by-step before answering.	Complex claim adjudications, navigating multi-tiered policy logic, and math.	Token Cost/ cost High output tokens. Takes longer to generate.
Few-Shot Prompting Provide 2-5 worked examples inside the prompt.	Standardizing policy summaries, ensuring consistent tone in customer emails.	Token Cost cost Larger input prompt required.
Self-Consistency Run the identical prompt multiple times, take the majority answer.	High-stakes risk assessments or ambiguous medical reports.	Token Cost cost Multiple Copilot runs needed.
Self-Criticism Ask the model: "What is missing or wrong with this summary?"	Quality-sensitive client communications and final reviews.	Token Cost cost Requires an extra conversational round-trip.

Aisha’s Customer Scenario

Use the fictional profile to prepare for a customer conversation without making product claims.

AISHA, 38

Married, two children, has an existing medical policy and recently asked about additional protection.

TASK

Draft five discovery questions and a short follow-up email

BOUNDARY

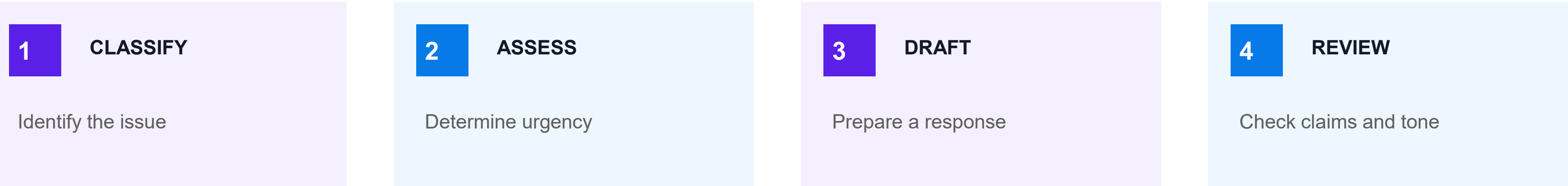
Do not recommend a product or infer sensitive personal information

YOUR OUTPUT

Five questions, one email and two items for human verification

Prompt Chaining and Critique

Break a risky customer-response task into reviewable stages.



FINAL REVIEW PROMPT

Review the draft response. Identify:

- 1. Unsupported claims
- 2. Inappropriate certainty
- 3. Missing empathy
- 4. Information requiring verification

Then produce a revised response and a short verification checklist.

TEAM TASK

Improve the review prompt by adding one Prudential-relevant safety requirement.

Modern Edge Tactics: Repetition & RE2

Prompt Repetition

The Science

Recent Google Research demonstrates that literally duplicating your entire prompt—copying and pasting it so it appears twice—reduces ordering effects. The model processes the instruction twice during the input stage, winning 47 out of 70 benchmarks with zero losses.

The Takaful Tactic

[Paste your full instructions for analyzing a lengthy policy document]
[Paste the exact same instructions again immediately below it]

RE2 (Re-reading)

The Science

A lighter, highly efficient variant of repetition. Research shows that forcing the model to re-focus on the core objective right before generating an answer drastically improves reasoning and reduces hallucination.

The Takaful Tactic

[Your prompt details...]
Read the question again: [Insert your core objective or question here]

Prompt Repetition Improves Non-Reasoning LLMs

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Matan Kalman*
 Google Research
 matank@google.com

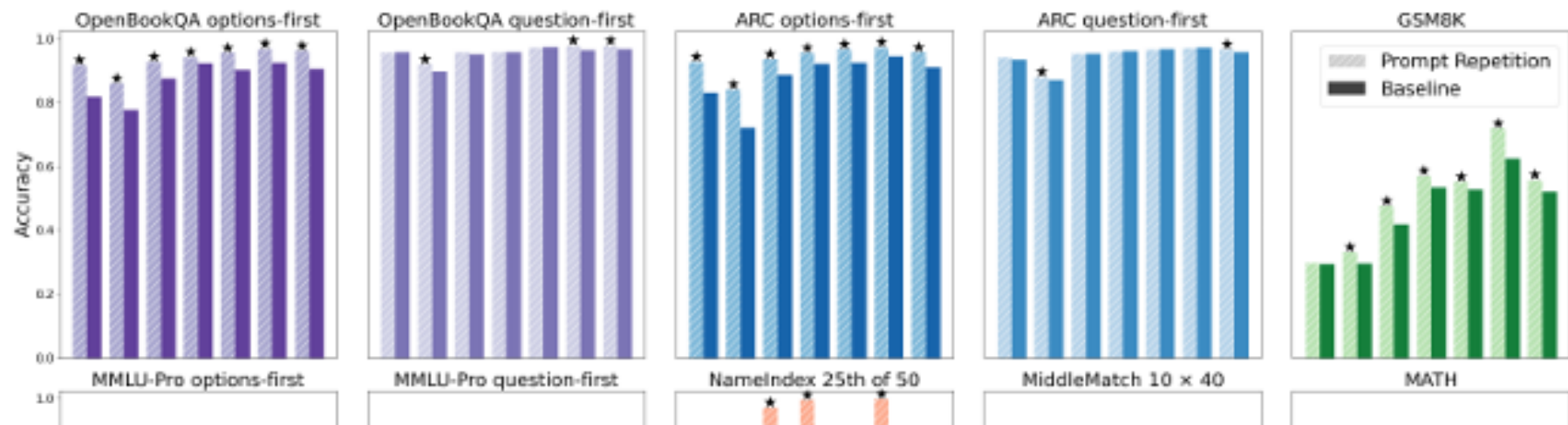
Yossi Matias
 Google Research
 yossi@google.com

Abstract

When not using reasoning, repeating the input prompt improves performance for popular models (Gemini, GPT, Claude, and Deepseek) without increasing the number of generated tokens or latency.

1 Prompt Repetition

LLMs are often trained as *causal* language models, i.e. past tokens cannot attend to future tokens. Therefore, the order of the tokens in a user’s query can affect prediction performance. For example, a query of the form “<CONTEXT> <QUESTION>” often performs differently from a query of the form “<QUESTION> <CONTEXT>” (see *options-first* vs. *question-first* in Figure 1). We propose to *repeat the prompt*, i.e. transform the input from “<QUERY>” to “<QUERY><QUERY>”. This enables each prompt token to attend to every other prompt token, addressing the above. When not using reasoning, **prompt repetition improves the performance of LLMs** (Figure 1) **without increasing the lengths of the generated outputs or latency** (Figures 2 and 3).



Navigating Implications & Risk



Data & IP Bias

AI models inherently carry the biases of their training data. Risk models and customer assessments generated by Copilot must be rigorously audited to ensure fair, unbiased treatment that aligns with Prudential standards.



Homogenized Responses

Avoid the "AI Voice". Relying entirely on Copilot without human editing leads to generic, robotic communications that lack empathy. Always modify the output to maintain the authentic Prudential tone.



The Hallucination Risk

LLMs prioritize sounding plausible over being factual. They will confidently invent numbers if confused. Never cede final judgment on a claim value, medical fact, or policy limit to the model without human verification.

Course-Correct, Don't Restart

Draft a project update.

💬 **Attempt 1**
(Too generic)

💬 **AI Output**
(Generic)

Here is an update on the current project status...

Make it shorter. Put the key risks into a bulleted list. Change the tone to be more urgent.

➡ **The Iteration**

★ **Final Output:**
Perfect,
tailored update.

URGENT: Executive Update

Key Risks:

- Supply chain delay
- Budget variance...

One Skill, Many Tools



Copilot Chat
General tasks,
research



Word
Rewrite,
summarize



Outlook
Email drafts,
threads



Teams
Meeting
recaps, actions



Excel
Data analysis,
formulas



PowerPoint
Deck creation,
redesigns

Rules of the Road: Best Practices



1. Be Specific

Clarity over brevity. Tell it exactly what action to perform.



2. Give Context

Anchor AI to facts and data. Never let it guess the details.



3. Define Format

Don't leave the layout to chance. Specify tables, bullets, or paragraphs.



4. Iterate

Treat it as a conversation. The 2nd and 3rd attempts are always better.

Why Prompts Fail: Common Mistakes



Too vague or brief

Forces the AI to blindly guess your intent.



No context provided

Directly leads to **Hallucinations** (AI inventing facts).



No format specified

Results in massive, unreadable walls of text.



No audience defined

Results in a robotic, generic tone that misses the mark.

Your Daily Template

 Screenshot this!

Act as [role]

Help me [task / action verb]

Context: [paste details / reference doc]

Format: [bullet points / table / email]

Tone: [professional / casual / urgent]

60-Second AI Use-Case Pitch

Choose one business problem and prepare a concise table presentation.

1 DATA

What information do you already have?

2 AI TASK

What should AI classify, summarise, generate or detect?

3 TOOL

What approved tool or prompt would you use?

4 HUMAN

What must a person verify before action?

YOUR OUTPUT

A 60-second pitch with one measurable benefit

SCENARIO

Find **ANY** work-related document(s) in your Onedrive, Sharepoints, Email, etc

Make a nice **infographic** based on the document(s)

Top 3 Takeaways

1

AI is only as good as your brief.

(Use P.A.C.T. every single time)

2

It's a conversation, not a command.

(Iterate and course-correct instead of starting over)

3

Start small.

(Pick one task in one tool, like Outlook, and practice daily)

The Mindset Shift

**“Your job is not to use AI.
Your job is to brief AI well.”**

Q&A

AI in Data Analysis: Empowering PruBSN with Copilot in Excel

From Raw Data to Insights in Minutes



Dr. Dwi Pebrianti
Dr. Aimi Syazwani Ghazali

Module 3

Copilot is your analytical assistant, not a replacement for your expertise

KEEP THE MOMENTUM

Participants Preparation

01

LAPTOP

Microsoft 365 Copilot licence, Excel and Teams

02

RAW DATA

PRUBSN_Tier 1_Business Data.xlsx
<https://github.com/uychan/PruBSN>

03

SANDBOX
CHANNEL

Safe workflow testing



**Microsoft
Copilot**

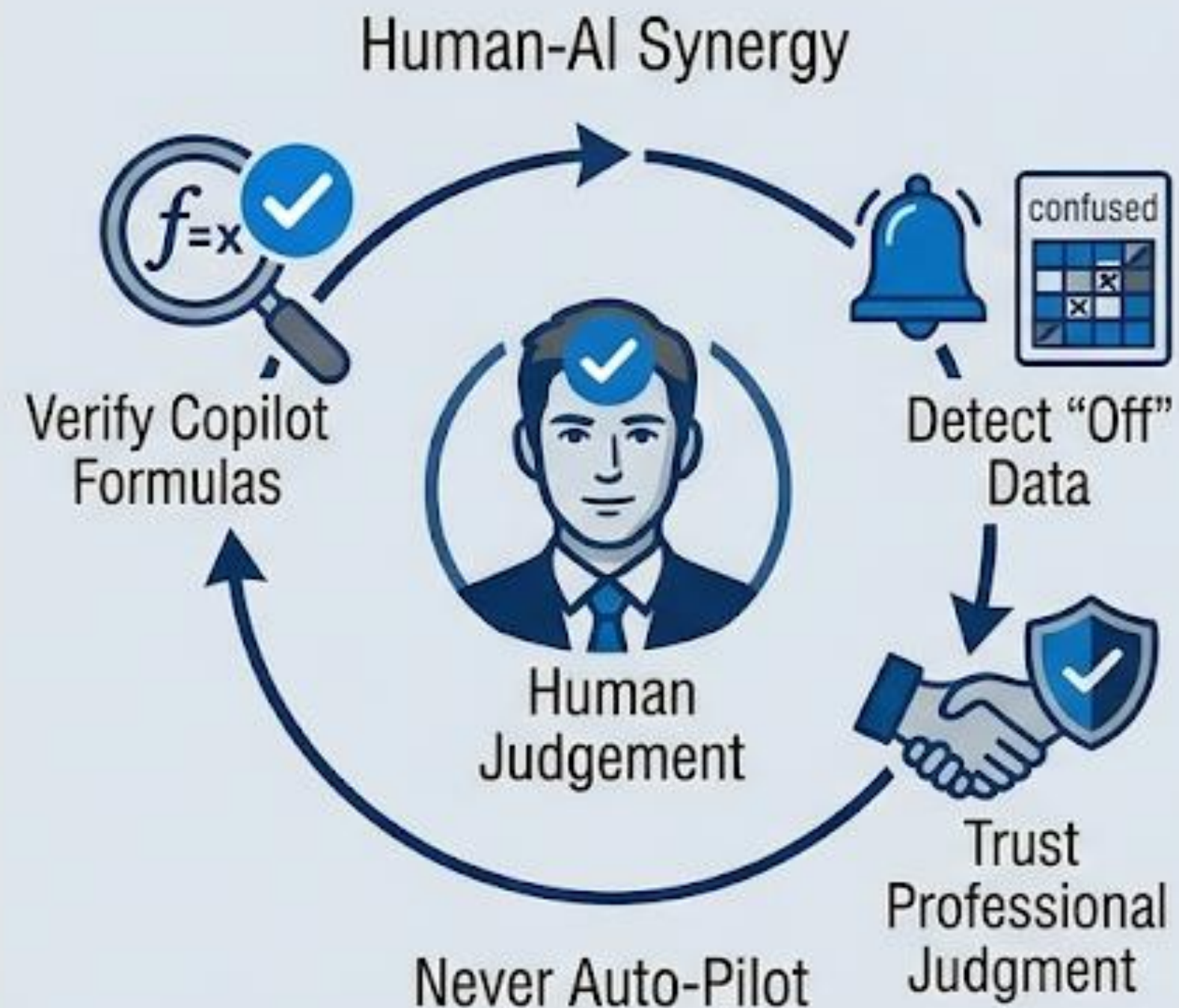


Data Analysis Philosophy: The Copilot Paradigm Shift

The “80/20” Rule: The Historical Shift



Human-in-the-Loop: Trust and Validate



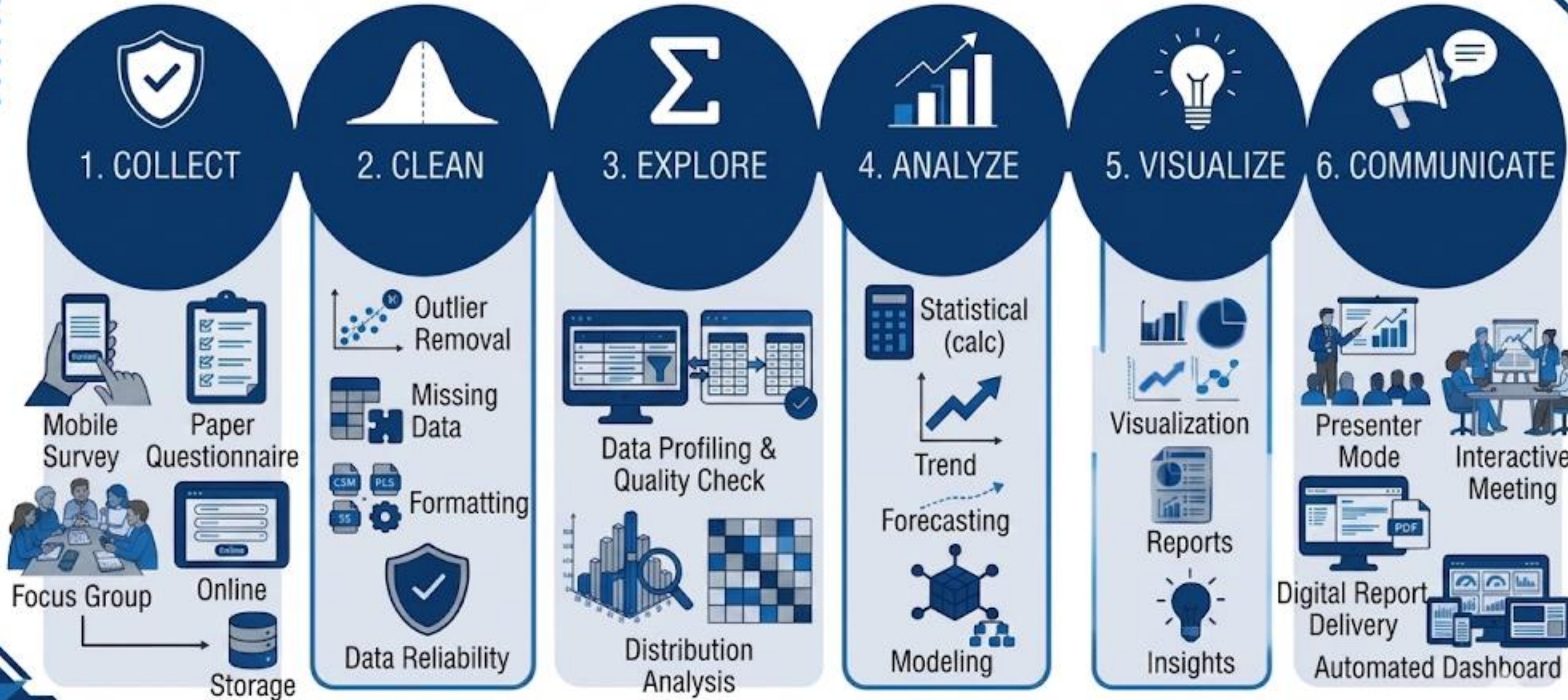
Copilot empowers your expertise. You are the critical, human element for strategic insights.

DATA CYCLE: AN INTEGRATED GUIDE

Foundation

Transformation

Impact





HANDS ON

Step 1: Data Collection



Link to sample file

<https://github.com/uychan/PruBSN>

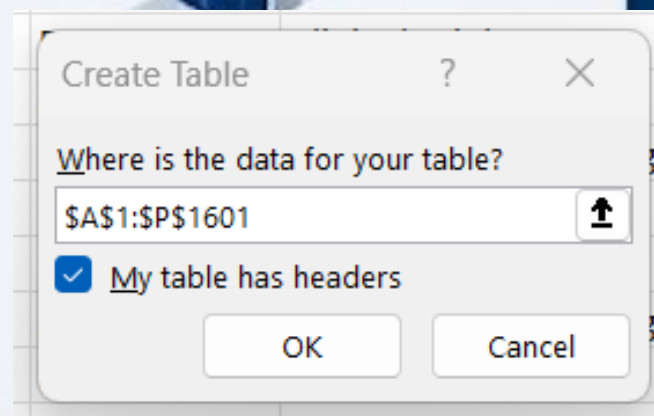
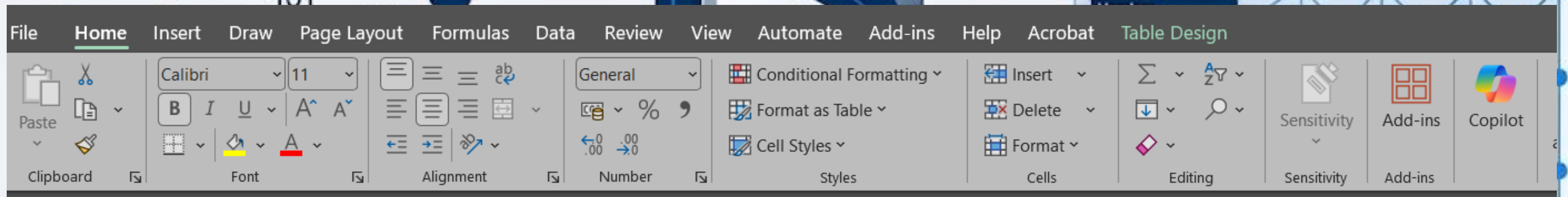
 Facttable.xlsx	Add files via upload	last month
 OrderCalendar.txt	Add files via upload	last month
 PRUBSN_11thMay_BusinessData.xlsx	Add files via upload	2 months ago
 PRUBSN_Tier 1_Business Data.xlsx	Add files via upload	1 minute ago
README		



Ctrl + T is the essential ignition key for your AI assistant.

Unreadable by AI

Headers, data types, and relationships unlocked.



Why? Copilot needs a structured Table to understand the context of your data. It cannot 'read' a loose range of cells.

Relationships	
Cat_ID	Data
21	Namon 1
55	Namon 2
30	Nemon 2

FileHomeInsertDrawPage LayoutFormulasDataReviewViewAutomateAdd-insHelpAcrobat

Clipboard

Font

Alignment

Number

Styles

Cells

Editing

Sensitivity

Add-ins

Adobe Acrobat

officeatwork

Adobe

Claude

A1 : X ✓ fx SR19

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	P	Q	R
1	Customer ID	Column1	Customer Name	Region	Product	Quantity	Unit Price	Discount	Tax Rate	Cost	tal	Reven	Net Income	Payment Status	Payment Method	Delivery Days	
2	C0000	INV-00000	Ali Ahmad	Kuala Lumpur	Data Management	7	684	0.03	0.06	2872.8	4644.36		1771.56	Paid	Credit Card		
3	C0001	INV-00001	Siti Aminah	Penang	Regulatory Consulting	4	1980	0.12	0.06	4752	6969.6		2217.6	Pending	Bank Transfer		
4	C0002	INV-00002	Farah Zulkifli	Johor	Clinical Trial Support	8	1282	0.14	0.06	6153.6	8820.16		2666.56	Overdue	Cash		
5	C0003	INV-00003	Hafiz Rahman	Selangor	Data Management	5	1036	0.11	0.06	3108	4610.2		1502.2	Paid	Credit Card		
6	C0004	INV-00004	Nur Aisyah	Kuala Lumpur	Regulatory Consulting	7	1384	0.04	0.06	5812.8	9300.48		3487.68	Pending	Bank Transfer		
7	C0005	INV-00005	Daniel Ismail	Penang	Clinical Trial Support	3	1208	0.13	0.06	2174.4	3152.88		978.48	Overdue	Cash		
8	C0006	INV-00006	Aina Farhana	Johor	Data Management	7	890	0.23	0.06	3738	4797.1		1059.1	Paid	Credit Card	SR06	
9	C0007	INV-00007	Irfan Hakimi	Selangor	Regulatory Consulting	8	1429	0.02	0.06	6859.2	11203.36		4344.16	Pending	Bank Transfer	SR07	
10	C0008	INV-00008	Nabila Sofea	Kuala Lumpur	Clinical Trial Support	5	1235	0.02	0.06	3705	6051.5		2346.5	Overdue	Cash	SR08	
11	C0009	INV-00009	Amir Haziq	Penang	Data Management	4	689	0.11	0.06	1653.6	2452.84		799.24	Paid	Credit Card	SR09	
12	C0010	INV-00010	Ali Ahmad	Johor	Regulatory Consulting	8	475	0.28	0.06	2280	2736		456	Pending	Bank Transfer	SR10	
13	C0011	INV-00011	Siti Aminah	Selangor	Clinical Trial Support	8	892	0.2	0.06	4281.6	5708.8		1427.2	Overdue	Cash	SR11	
14	C0012	INV-00012	Farah Zulkifli	Kuala Lumpur	Data Management	3	441	0.2	0.06	793.8	1058.4		264.6	Paid	Credit Card	SR12	
15	C0013	INV-00013	Hafiz Rahman	Penang	Regulatory Consulting	6	1439	0.11	0.06	5180.4	7684.26		2503.86	Pending	Bank Transfer	SR13	
16	C0014	INV-00014	Nur Aisyah	Johor	Clinical Trial Support	5	1474	0.18	0.06	4422	6043.4		1621.4	Overdue	Cash	SR14	
17	C0015	INV-00015	Daniel Ismail	Selangor	Data Management	2	325	0	0.06	390	650		260	Paid	Credit Card	SR15	
18	C0016	INV-00016	Aina Farhana	Kuala Lumpur	Regulatory Consulting	8	416	0.19	0.06	1996.8	2695.68		698.88	Pending	Bank Transfer	SR16	
19	C0017	INV-00017	Irfan Hakimi	Penang	Clinical Trial Support	6	1529	0.27	0.06	5504.4	6697.02		1192.62	Overdue	Cash	SR17	
20	C0018	INV-00018	Nabila Sofea	Johor	Data Management	2	1408	0.18	0.06	1689.6	2309.12		619.52	Paid	Credit Card	SR18	
21	C0019	INV-00019	Amir Haziq	Selangor	Regulatory Consulting	5	1262	0.26	0.06	3786	4669.4		883.4	Pending	Bank Transfer	SR19	
22	C0020	INV-00020	Ali Ahmad	Kuala Lumpur	Clinical Trial Support	1	1446	0.22	0.06	867.6	1127.88		260.28	Overdue	Cash	SR20	
23	C0021	INV-00021	Siti Aminah	Penang	Data Management	6	1407	0.04	0.06	5065.2	8104.32		3039.12	Paid	Credit Card	SR21	
24	C0022	INV-00022	Farah Zulkifli	Johor	Regulatory Consulting	9	694	0.27	0.06	3747.6	4559.58		811.98	Pending	Bank Transfer	SR22	
25	C0023	INV-00023	Hafiz Rahman	Selangor	Clinical Trial Support	1	626	0.02	0.06	375.6	613.48		237.88	Overdue	Cash	SR23	
26	C0024	INV-00024	Nur Aisyah	Kuala Lumpur	Data Management	3	1401	0.16	0.06	2521.8	3530.52		1008.72	Paid	Credit Card	SR24	
27	C0025	INV-00025	Daniel Ismail	Penang	Regulatory Consulting	7	1118	0.04	0.06	4695.6	7512.96		2817.36	Pending	Bank Transfer	SR25	
28	C0026	INV-00026	Aina Farhana	Johor	Clinical Trial Support	4	1651	0	0.06	3962.4	6604		2641.6	Overdue	Cash	SR26	
29	C0027	INV-00027	Irfan Hakimi	Selangor	Data Management	9	1517	0.13	0.06	8191.8	11878.11		3686.31	Paid	Credit Card	SR27	
30	C0028	INV-00028	Nabila Sofea	Kuala Lumpur	Regulatory Consulting	3	350	0.09	0.06	630	955.5		325.5	Pending	Bank Transfer	SR28	
31	C0029	INV-00029	Amir Haziq	Penang	Clinical Trial Support	5	583	0.15	0.06	1749	2477.75		728.75	Overdue	Cash	SR29	
32	C0030	INV-00030	Ali Ahmad	Johor	Data Management	3	1861	0.17	0.06	3349.8	4633.89		1284.09	Paid	Credit Card	SR30	
33	C0031	INV-00031	Siti Aminah	Selangor	Regulatory Consulting	7	507	0.01	0.06	2129.4	3513.51		1384.11	Pending	Bank Transfer	SR31	
34	C0032	INV-00032	Farah Zulkifli	Kuala Lumpur	Clinical Trial Support	5	1559	0.04	0.06	4677	7483.2		2806.2	Overdue	Cash	SR32	
35	C0033	INV-00033	Hafiz Rahman	Penang	Data Management	9	945	0.17	0.06	5103	7059.15		1956.15	Paid	Credit Card	SR33	
36	C0034	INV-00034	Nur Aisyah	Johor	Regulatory Consulting	7	1091	0.1	0.06	4582.2	6873.3		2291.1	Pending	Bank Transfer	SR34	
37	C0035	INV-00035	Daniel Ismail	Selangor	Clinical Trial Support	2	574	0.22	0.06	688.8	895.44		206.64	Overdue	Cash	SR35	
38	C0036	INV-00036	Aina Farhana	Kuala Lumpur	Data Management	4	1231	0.2	0.06	2954.4	3939.2		984.8	Paid	Credit Card	SR36	
39	C0037	INV-00037	Irfan Hakimi	Penang	Regulatory Consulting	9	1564	0.25	0.06	8445.6	10557		2111.4	Pending	Bank Transfer	SR37	

Create Table

Where is the data for your table?

SA\$1:SP\$1601

☒ My table has headers

OK Cancel

AutoSaveOff

PRUBSN_Tier 1_Business Data

Search

FileHomeInsertDrawPage LayoutFormulasDataReviewViewAutomateAdd-insHelpAcrobatTable Design

Table Name: Table1

Summarize with PivotTableRemove DuplicatesConvert to Range

Insert SlicerExportRefreshUnlink

Header Row

Total Row

Banded Rows

First Column

Last Column

Banded Columns

Filter Button

Table Style Options

Table Styles

M40

Overdue

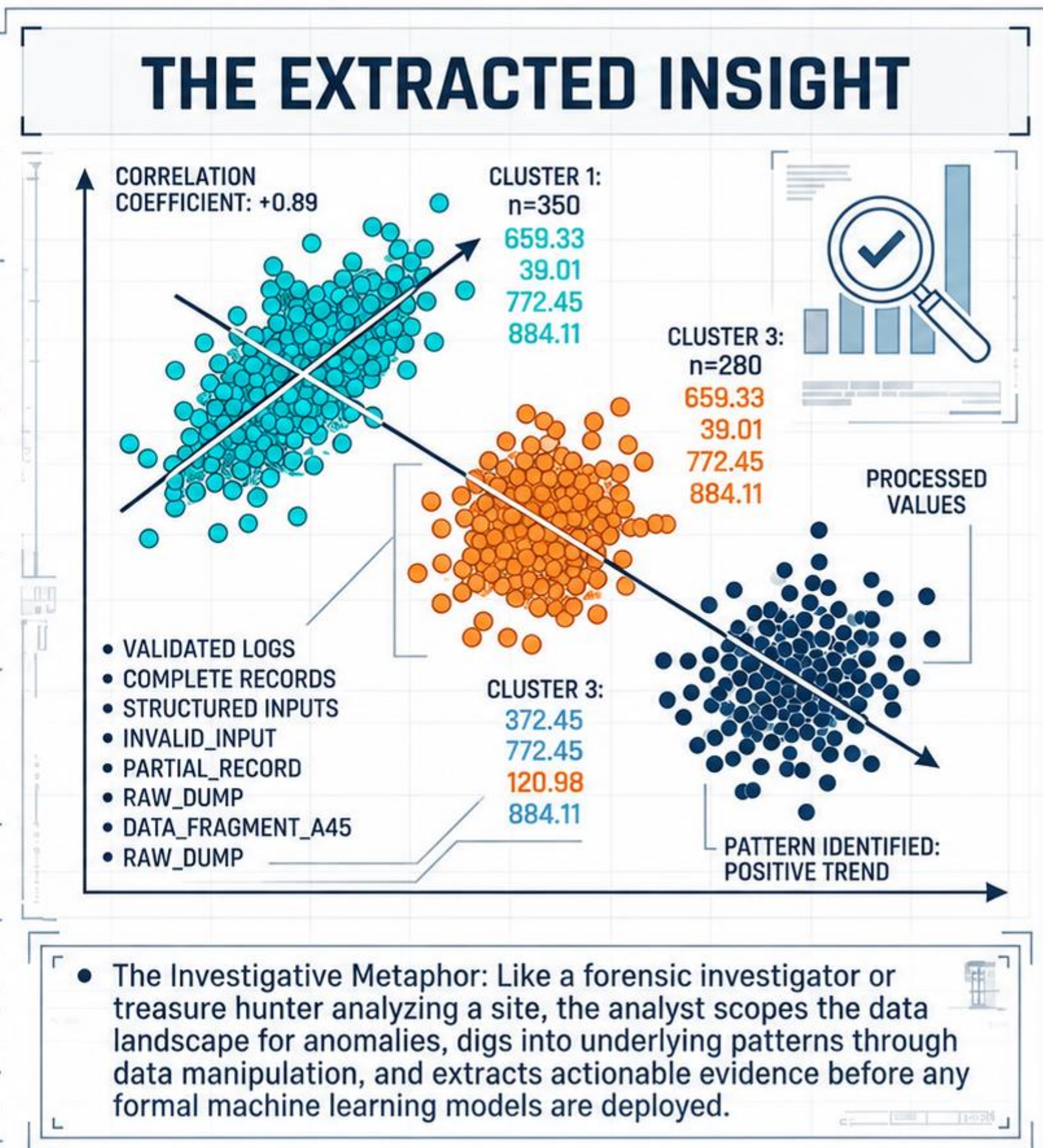
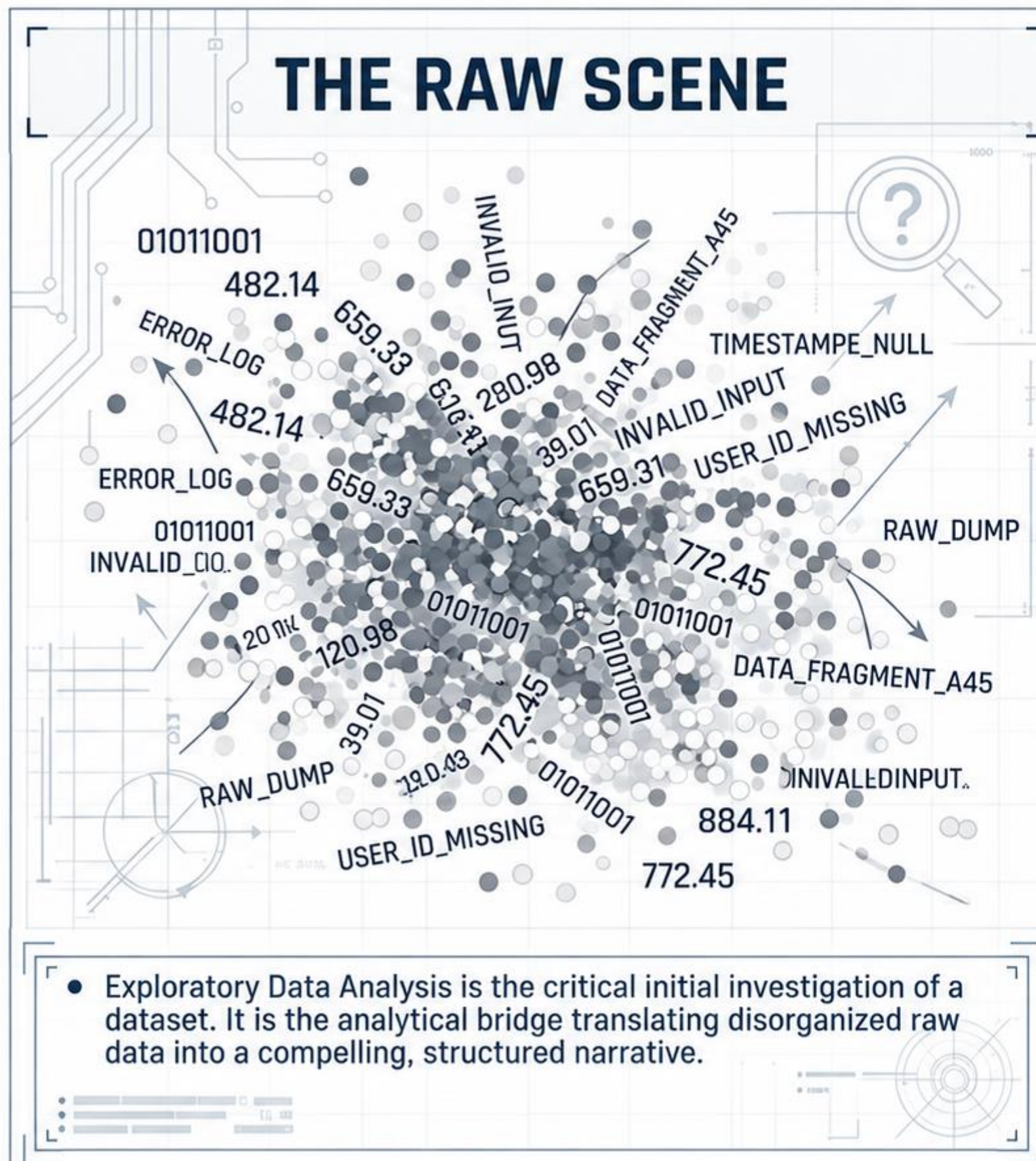
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	Customer_ID	Column1	Customer_Nam	Region	Product	Quantit	Unit_Pric	Discoun	Tax_Rat	Cost	Total_Revenu	Net_Income	Payment_Status	Payment_Method	Sales_Rep_ID	Delivery_Days
2	C0000	INV-00000	Ali Ahmad	Kuala Lumpur	Data Management	7	684	0.03	0.06	2872.8	4644.36	1771.56	Paid	Credit Card	SR00	4
3	C0001	INV-00001	Siti Aminah	Penang	Regulatory Consulting	4	1980	0.12	0.06	4752	6969.6	2217.6	Pending	Bank Transfer	SR01	7
4	C0002	INV-00002	Farah Zulkifli	Johor	Clinical Trial Support	8	1282	0.14	0.06	6153.6	8820.16	2666.56	Overdue	Cash	SR02	9
5	C0003	INV-00003	Hafiz Rahman	Selangor	Data Management	5	1036	0.11	0.06	3108	4610.2	1502.2	Paid	Credit Card	SR03	12
6	C0004	INV-00004	Nur Aisyah	Kuala Lumpur	Regulatory Consulting	7	1384	0.04	0.06	5812.8	9300.48	3487.68	Pending	Bank Transfer	SR04	2
7	C0005	INV-00005	Daniel Ismail	Penang	Clinical Trial Support	3	1208	0.13	0.06	2174.4	3152.88	978.48	Overdue	Cash	SR05	13
8	C0006	INV-00006	Aina Farhana	Johor	Data Management	7	890	0.23	0.06	3738	4797.1	1059.1	Paid	Credit Card	SR06	8
9	C0007	INV-00007	Irfan Hakimi	Selangor	Regulatory Consulting	8	1429	0.02	0.06	6859.2	11203.36	4344.16	Pending	Bank Transfer	SR07	9
10	C0008	INV-00008	Nabila Sofea	Kuala Lumpur	Clinical Trial Support	5	1235	0.02	0.06	3705	6051.5	2346.5	Overdue	Cash	SR08	10
11	C0009	INV-00009	Amir Haziq	Penang	Data Management	4	689	0.11	0.06	1653.6	2452.84	799.24	Paid	Credit Card	SR09	10
12	C0010	INV-00010	Ali Ahmad	Johor	Regulatory Consulting	8	475	0.28	0.06	2280	2736	456	Pending	Bank Transfer	SR10	9
13	C0011	INV-00011	Siti Aminah	Selangor	Clinical Trial Support	8	892	0.2	0.06	4281.6	5708.8	1427.2	Overdue	Cash	SR11	10
14	C0012	INV-00012	Farah Zulkifli	Kuala Lumpur	Data Management	3	441	0.2	0.06	793.8	1058.4	264.6	Paid	Credit Card	SR12	7
15	C0013	INV-00013	Hafiz Rahman	Penang	Regulatory Consulting	6	1439	0.11	0.06	5180.4	7684.26	2503.86	Pending	Bank Transfer	SR13	9
16	C0014	INV-00014	Nur Aisyah	Johor	Clinical Trial Support	5	1474	0.18	0.06	4422	6043.4	1621.4	Overdue	Cash	SR14	9
17	C0015	INV-00015	Daniel Ismail	Selangor	Data Management	2	325	0	0.06	390	650	260	Paid	Credit Card	SR15	3
18	C0016	INV-00016	Aina Farhana	Kuala Lumpur	Regulatory Consulting	8	416	0.19	0.06	1996.8	2695.68	698.88	Pending	Bank Transfer	SR16	4
19	C0017	INV-00017	Irfan Hakimi	Penang	Clinical Trial Support	6	1529	0.27	0.06	5504.4	6697.02	1192.62	Overdue	Cash	SR17	2
20	C0018	INV-00018	Nabila Sofea	Johor	Data Management	2	1408	0.18	0.06	1689.6	2309.12	619.52	Paid	Credit Card	SR18	10
21	C0019	INV-00019	Amir Haziq	Selangor	Regulatory Consulting	5	1262	0.26	0.06	3786	4669.4	883.4	Pending	Bank Transfer	SR19	3
22	C0020	INV-00020	Ali Ahmad	Kuala Lumpur	Clinical Trial Support	1	1446	0.22	0.06	867.6	1127.88	260.28	Overdue	Cash	SR20	11
23	C0021	INV-00021	Siti Aminah	Penang	Data Management	6	1407	0.04	0.06	5065.2	8104.32	3039.12	Paid	Credit Card	SR21	11
24	C0022	INV-00022	Farah Zulkifli	Johor	Regulatory Consulting	9	694	0.27	0.06	3747.6	4559.58	811.98	Pending	Bank Transfer	SR22	5
25	C0023	INV-00023	Hafiz Rahman	Selangor	Clinical Trial Support	1	626	0.02	0.06	375.6	613.48	237.88	Overdue	Cash	SR23	5
26	C0024	INV-00024	Nur Aisyah	Kuala Lumpur	Data Management	3	1401	0.16	0.06	2521.8	3530.52	1008.72	Paid	Credit Card	SR24	6
27	C0025	INV-00025	Daniel Ismail	Penang	Regulatory Consulting	7	1118	0.04	0.06	4695.6	7512.96	2817.36	Pending	Bank Transfer	SR25	13
28	C0026	INV-00026	Aina Farhana	Johor	Clinical Trial Support	4	1651	0	0.06	3962.4	6604	2641.6	Overdue	Cash	SR26	7
29	C0027	INV-00027	Irfan Hakimi	Selangor	Data Management	9	1517	0.13	0.06	8191.8	11878.11	3686.31	Paid	Credit Card	SR27	7
30	C0028	INV-00028	Nabila Sofea	Kuala Lumpur	Regulatory Consulting	3	350	0.09	0.06	630	955.5	325.5	Pending	Bank Transfer	SR28	1
31	C0029	INV-00029	Amir Haziq	Penang	Clinical Trial Support	5	583	0.15	0.06	1749	2477.75	728.75	Overdue	Cash	SR29	2
32	C0030	INV-00030	Ali Ahmad	Johor	Data Management	3	1861	0.17	0.06	3349.8	4633.89	1284.09	Paid	Credit Card	SR30	12
33	C0031	INV-00031	Siti Aminah	Selangor	Regulatory Consulting	7	507	0.01	0.06	2129.4	3513.51	1384.11	Pending	Bank Transfer	SR31	9
34	C0032	INV-00032	Farah Zulkifli	Kuala Lumpur	Clinical Trial Support	5	1559	0.04	0.06	4677	7483.2	2806.2	Overdue	Cash	SR32	10
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36	C0034	INV-00034	Nur Aisyah	Johor	Regulatory Consulting	7	1091	0.1	0.06	4582.2	6873.3	2291.1	Pending	Bank Transfer	SR34	6
37	C0035	INV-00035	Daniel Ismail	Selangor	Clinical Trial Support	2	574	0.22	0.06	688.8	895.44	206.64	Overdue	Cash	SR35	11
38	C0036	INV-00036	Aina Farhana	Kuala Lumpur	Data Management	4	1231	0.2	0.06	2954.4	3939.2	984.8	Paid	Credit Card	SR36	11
39	C0037	INV-00037	Irfan Hakimi	Penang	Regulatory Consulting	9	1564	0.25	0.06	8445.6	10557	2111.4	Pending	Bank Transfer	SR37	1

Sheet1

ReadyAccessibility: Investigate

Display Settings

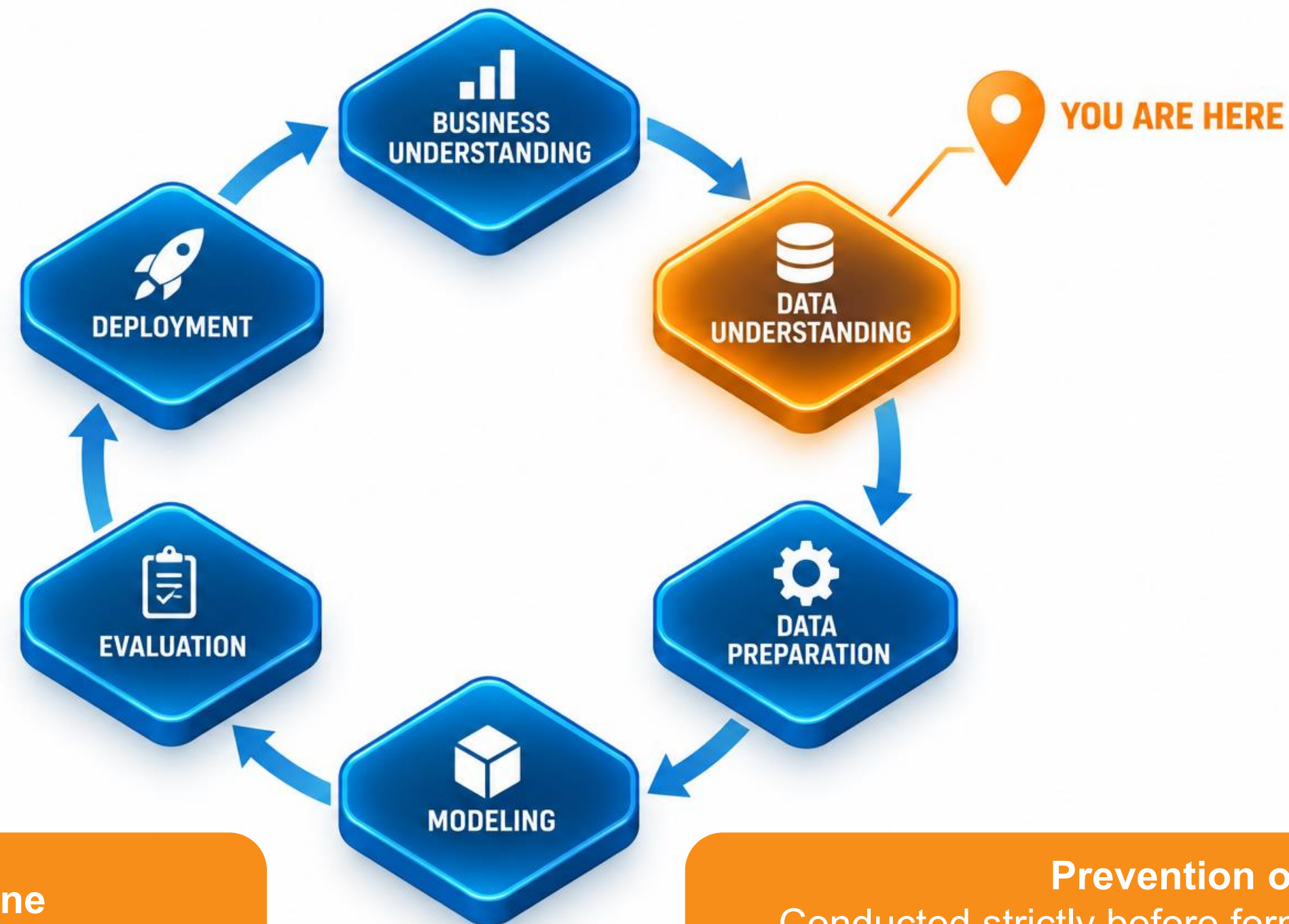
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Case study in Insurance : claim approval, fraud detection, customer segmentation, personalized insurance

CRISP – DM : The Industry Standard Pipeline

Cross-Industry Standard Process for Data Mining



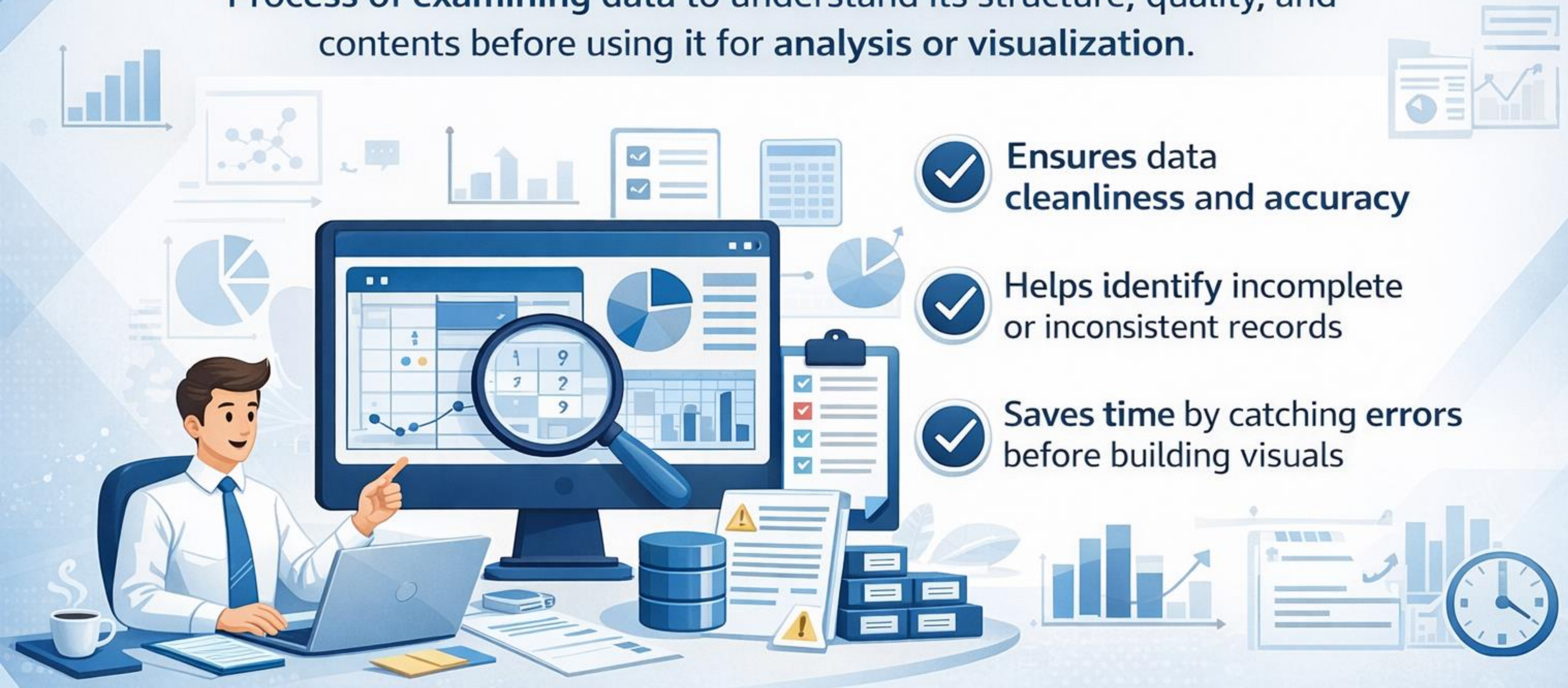
The industry standard pipeline
EDA is not an optional phase. It is the core of the data understanding phase in the CRISP-DM pipeline

Prevention over cure
Conducted strictly before formal hypothesis testing or predictive modeling. Bypassing EDA risks building advanced algorithms on flawed assumption. Garbage in, garbage out

Why Data Exploration Matters

Process of examining data to understand its structure, quality, and contents before using it for analysis or visualization.

- ✓ Ensures data cleanliness and accuracy
- ✓ Helps identify incomplete or inconsistent records
- ✓ Saves time by catching errors before building visuals





HANDS ON

Step 2: Data Cleaning



Data Cleaning Essentials

What is Data Cleaning?

Detecting and correcting inaccurate, incomplete, or inconsistent data in a dataset

Why Clean Data?

To improve data quality — accurate, reliable, and ready for analysis

Key Data Cleaning Steps

- Remove errors, blanks, or duplicates
- Replace incorrect values
- Change or fix data types
- Standardize text formats and column names
- Handle missing or null data



THE DATA ANATOMY MATRIX

NUMERICAL (Continuous & Discrete)	CATEGORICAL (Nominal)	ORDINAL	TIME SERIES
Definition: Measured or counted values. Example: Premium amount, Claim amount, Customer age, Number of claims. Tools: Mean, Median, Standard Deviation.	Definition: Unordered labels or groups. Example: Policy type, Claim type, Payment method, Branch region. Tools: Frequency counts, Proportions, Mode.	Definition: Categorical data with a defined sequence. Example: Risk level (Low-Medium-High), Claim priority, Customer satisfaction rating. Tools: Median, Percentiles, Rank correlation.	Definition: Data collected sequentially over time. Example: Daily claim counts, Monthly premium revenue, Claim trends by month. Tools: Trend analysis, Seasonality checks.

HANDS ON

Common errors



movie_title	num_critic_for_reviews	duration	DIRECTOR_facebook_likes	actor_3_facebook_likes	ACTOR_1_facebook_likes	gross
Avatar?y	723	178	10	855	1000	760505847
Pirates of the Caribbean: At World's End?y	302		563	1000	40000	309404152
Spectre?y	602	148	20	161	11000	200074175
The Dark Knight Rises?y	813		22000	23000	27000	448130642
John Carter?y	462	132	"475"	530	640	73058679
Spider-Man 3?y	392	156	23	4000	24000	336530303
Tangled?y	324		15	284	799	200807262
Avengers: Age of Ultron?y	635	141	10	19000	26000	458991599
Avengers: Age of Ultron?y	635	141	10	19000	26000	458991599
Harry Potter and the Half-Blood Prince?y	375	153	282	10000	25000	301956980
Batman v Superman: Dawn of Justice?y	673	183		2000	15000	330249062
Superman Returns?y	434	169		903	18000	200069408
Quantum of Solace?y	403	106	395	393	451	168368427

Inconsistent headers

Noisy data

Incorrect formatting

Duplication

Missing data





Ask co-pilot:

For data cleaning, analyze the data:
What is wrong with it?
Are there any errors?

Into Copilot
Give the excel file
analyze this data the summarize
What is wrong with it? Are there
any errors?

Highlight :
C0100
C0200
C0300
C0400



LET'S FIX THE ERROR:

Challenge 1 - Column Renaming.

Why is this Important? (Benefits)

- **IMPROVES CLARITY AND UNDERSTANDING:** Use meaningful names like CustomerName or SalesAmount instead of vague ones like Col1 or Value.
- **ENSURE CONSISTENCY:** Standardize data formats (e.g., STD_ID) across all tables and databases.

REFLECTS TRUE MEANING: Change generic terms to specific ones, like Vague name... to Total_Sales.



	A	B	C	D	E	F
1	Customer_ID ▾	Column1 ▾	Customer_Name ▾	Region ▾	Product ▾	Quantity ▾
2	C0000	INV-00000	Ali Ahmad	Kuala Lumpur	Data Management	7

LET'S FIX THE ERROR:

Challenge 1 - Column Renaming.

Co-pilot: Find column headers with generic labels and suggest appropriate labels for them.

Inappropriate Column Name	Suggested Appropriate Name
Col1	Customer_ID
A	Order_Date
Data_2023	Sales_Amount

12 cols
to review

LET'S FIX THE ERROR:

Challenge 2 - Duplicate Data.

Identical or repeated records Identical or repeated records appearing more than once in the dataset.

Causes of Duplicate Data:

Main Causes	Implications (Summary)
(i) Repeated data imports or merges (ii) Manual entry errors (iii) Multiple sources collecting the same information (iv) System glitches or unsynchronized databases	• Identical records appear more than once. • Leading to distorted results in analysis and reporting. • Causes discredit and poor decision making. • Increases storage costs.



Prompt: Find duplicate data

✓ How to handle this in Excel (quick steps)

If you want to remove duplicates directly in Excel:

1. Select the entire table
2. Go to Data → Remove Duplicates
3. Select all columns
4. Click OK

Excel will remove exactly one duplicate row
(the C0100 / INV-00100 entry).

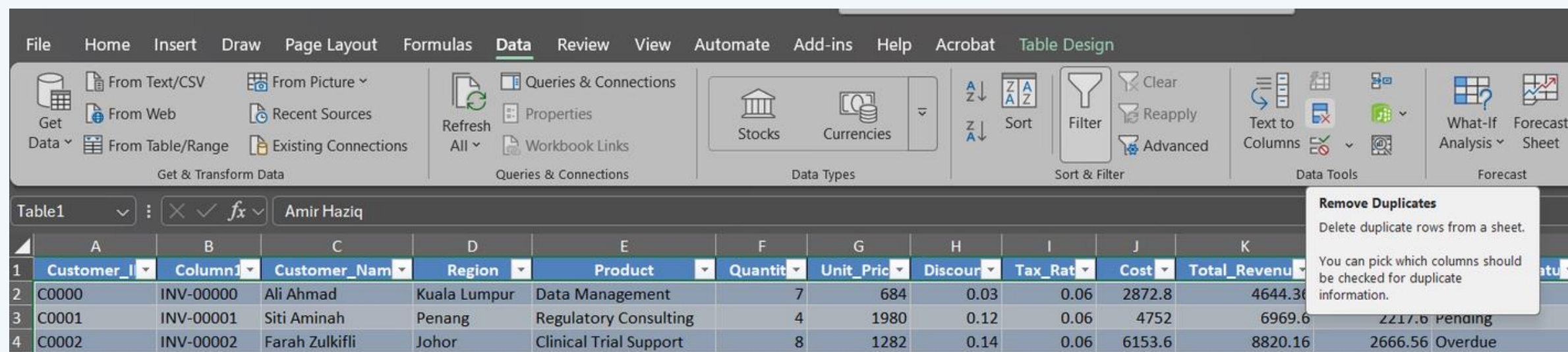


Table1											
	A	B	C	D	E	F	G	H	I	J	K
1	Customer_ID	Column1	Customer_Name	Region	Product	Quantity	Unit_Price	Discount	Tax_Rate	Cost	Total_Revenue
2	C0000	INV-00000	Ali Ahmad	Kuala Lumpur	Data Management	7	684	0.03	0.06	2872.8	4644.36
3	C0001	INV-00001	Siti Aminah	Penang	Regulatory Consulting	4	1980	0.12	0.06	4752	6969.6
4	C0002	INV-00002	Farah Zulkifli	Johor	Clinical Trial Support	8	1282	0.14	0.06	6153.6	8820.16



Data Cycle Guide: Addressing Missing Values (Clean)

Challenge 3: Missing Values

Data gaps where information is blank, null, or unreadable
(e.g., N/A, empty cells)

2. CLEAN

Address structure
& errors

REASON ANALYSIS

(Causes of Data Gaps)

- **Human error**
Incorrect data entry, typos, misreading info
- **System or sensor failure**
Sensor malfunctions, power loss, database connectivity issues
- **Conditional or intentional skipping**
Skip logic in surveys, deliberate non-response
- **Privacy or anonymity constraints**
Data masking, removal for compliance & privacy

REMEDIATION STRATEGIES

(Techniques for Imputation & Handling)

- **Remove rows** – if few or irrelevant
- **Replace values** – with mean, median, or "Unknown"
- **Fill up / down** – for sequential data
- **Flag missing** – keep track for later analysis

Why? Copilot needs structure. Imputation or flagging provides **necessary structure.**

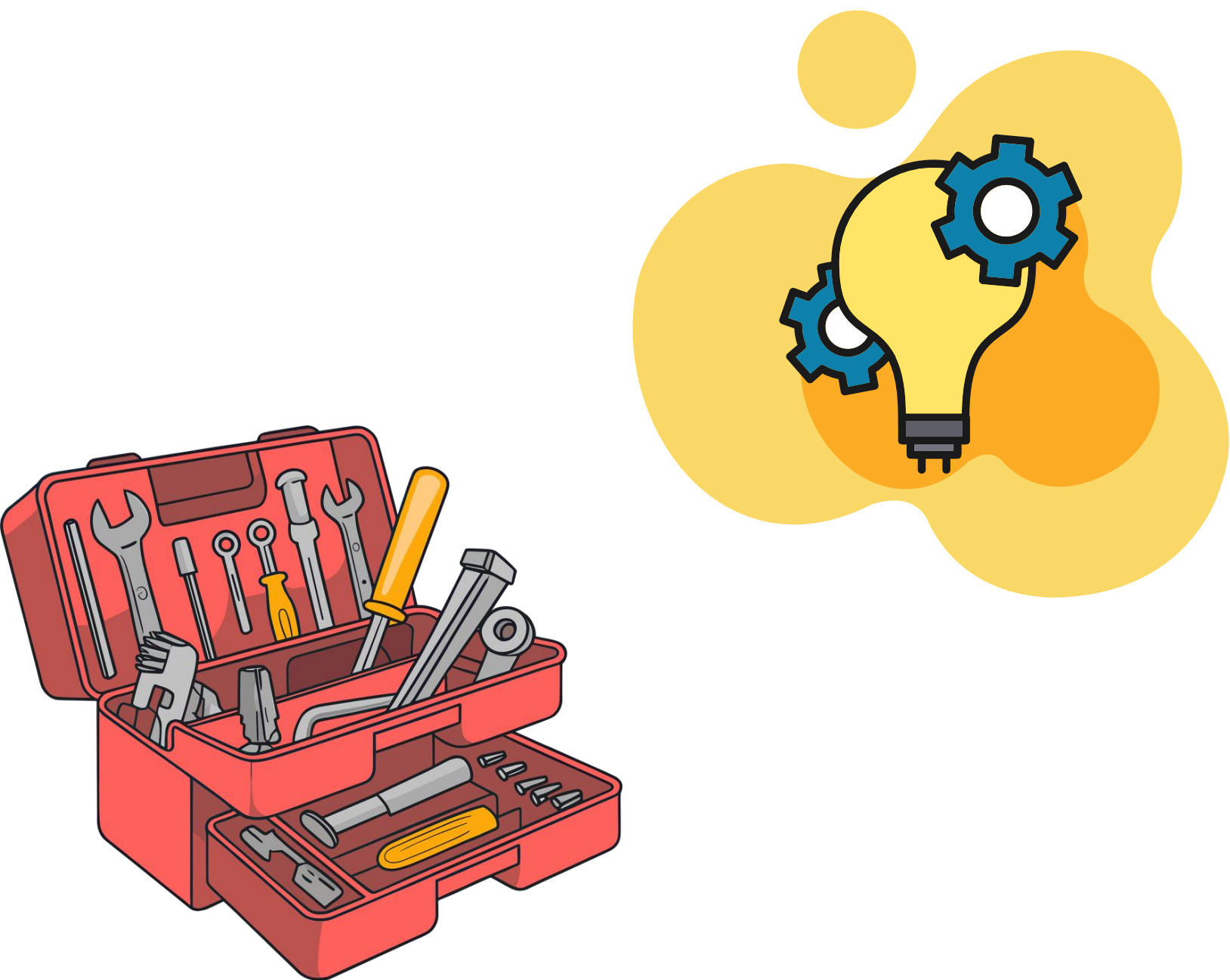


This is *your* expertise!

I have analyzed your data file and can confirm that **Row 402** (which corresponds to Customer ID **C0400**, Invoice **INV-00400**) is indeed missing a value in the **Discount** column.

Verification details for Row 402:

- **Product:** Regulatory Consulting
- **Quantity:** 1
- **Unit Price:** 983
- **Total Revenue:** 707.76
- **Discount:** *Missing (NaN)*



Suggested Best Way to Fill the Value

In your dataset, the values follow a specific mathematical relationship. Instead of using a simple average (mean) or median, the most accurate way is to **reverse-calculate** the discount using the existing financial data in that row.

Based on the other rows in your file, the formula used is:

$$\text{Total Revenue} = (\text{Quantity} \times \text{Unit Price}) \times (1 - \text{Discount})$$

The Calculation:

1. **Gross Amount:** $1 \times 983 = 983$
2. **Revenue Ratio:** $707.76 / 983 = 0.72$
3. **Discount Value:** $1 - 0.72 = 0.28$

Recommended Value: You should enter **0.28** (or 28%) into the Discount cell for Row 402. This perfectly aligns with the Total Revenue and Net Income already recorded for that entry.

How to apply this in Excel/CSV:

1. Locate cell **H402** (assuming Column H is Discount).
2. Type `0.28`.
3. If you want to automate this for any future missing values, you can use this Excel formula in the Discount column:

`=1 - (Total_Revenue / (Quantity * Unit_Price))`

`=1-([@[Total_Revenue]]/([@Quantity]*[@Unit_Price]))`



HANDS ON

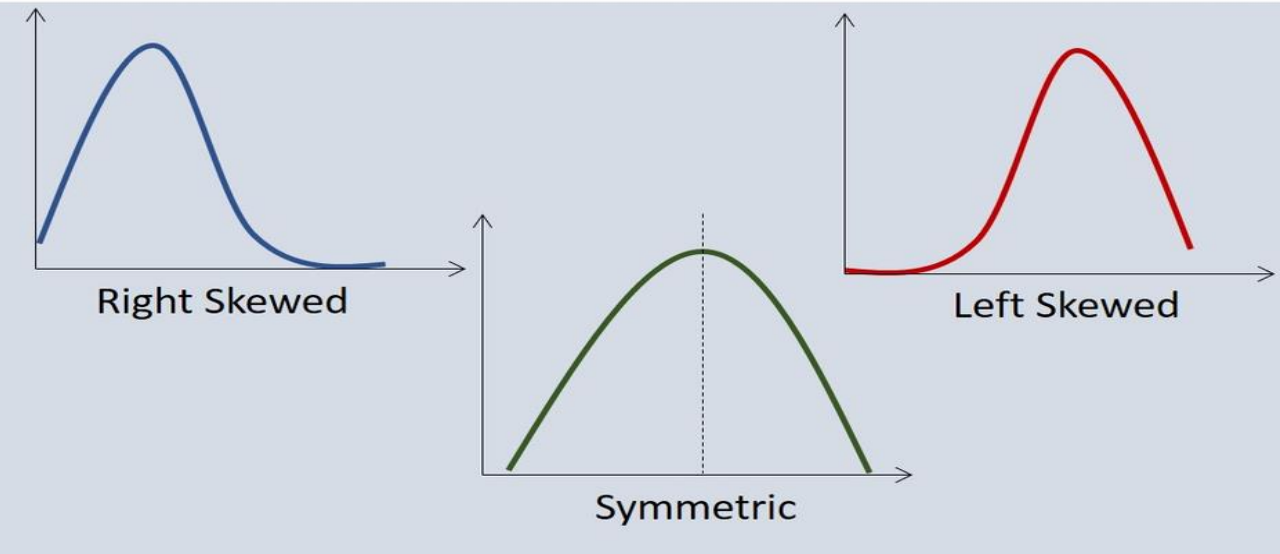
Step 3: Data Exploration

Please perform a distribution analysis.

Share the outputs.

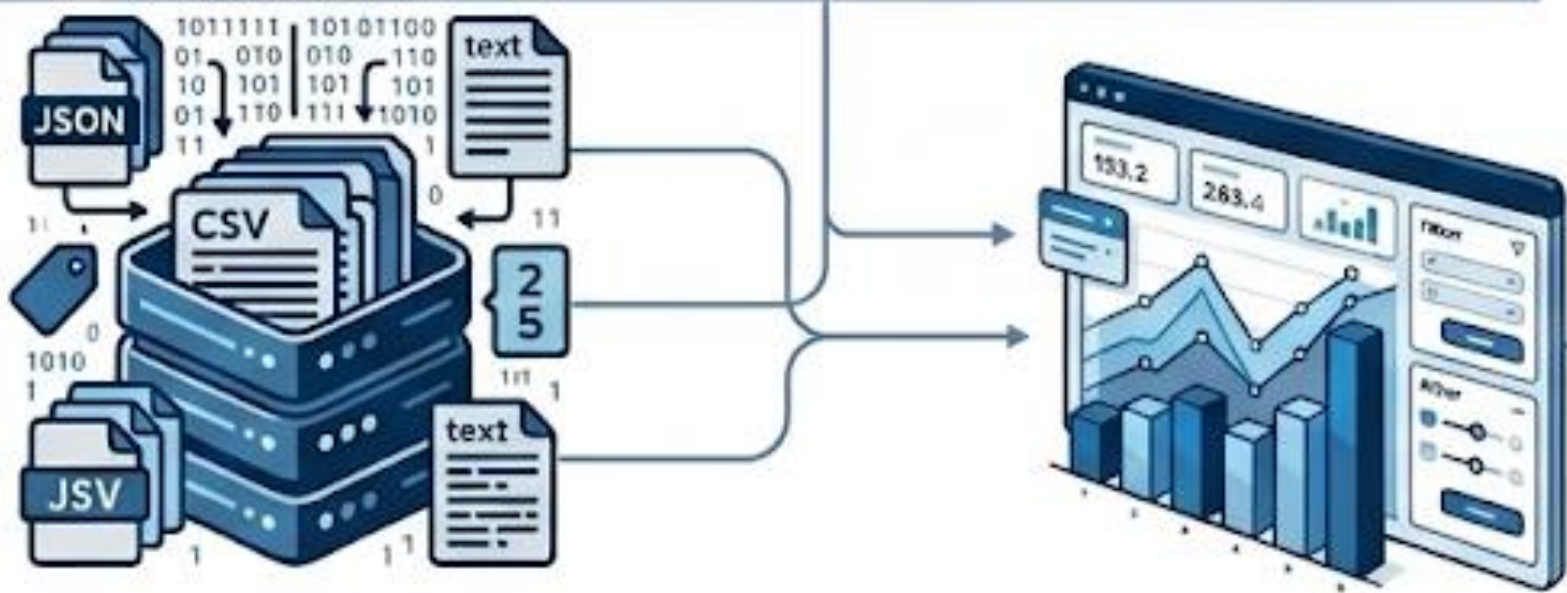


Outcomes:



e.g., Distribution analysis

Field Type	Distribution Shape
Regions	Moderately imbalanced
Products	Balanced (3 categories)
Payment Status	Balanced (rule-based)
Quantity	Right-skewed
Prices & Revenue	Heavily right-skewed
Tax Rate	Narrow, stable
Delivery Days	Compact, uniform



Overall Data Quality Assessment

✓ Strengths

- Rich transactional detail
- High completeness
- Logical relationships preserved
- Suitable for customer, revenue, and operational analysis

✗ Weaknesses

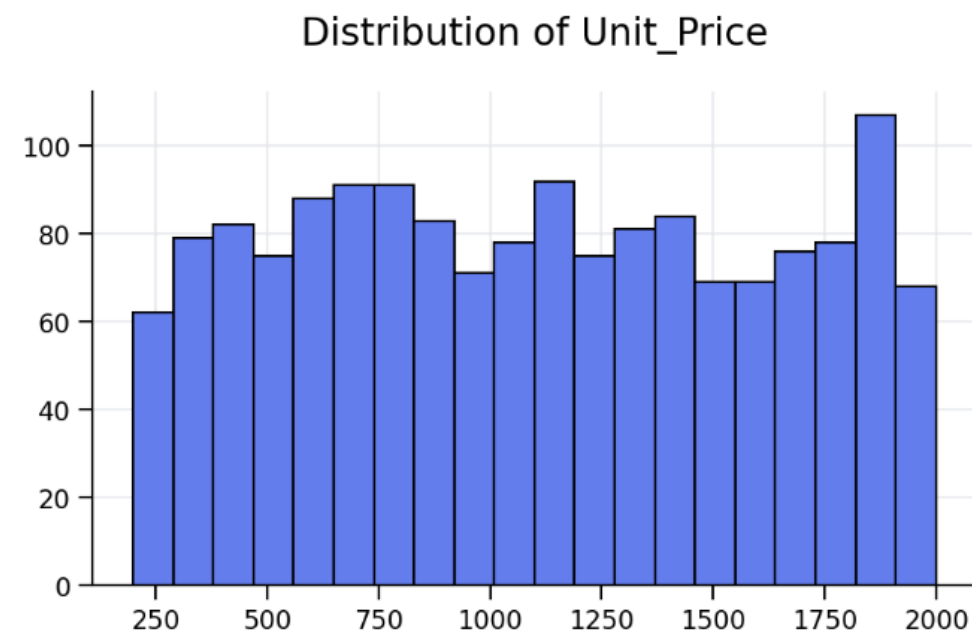
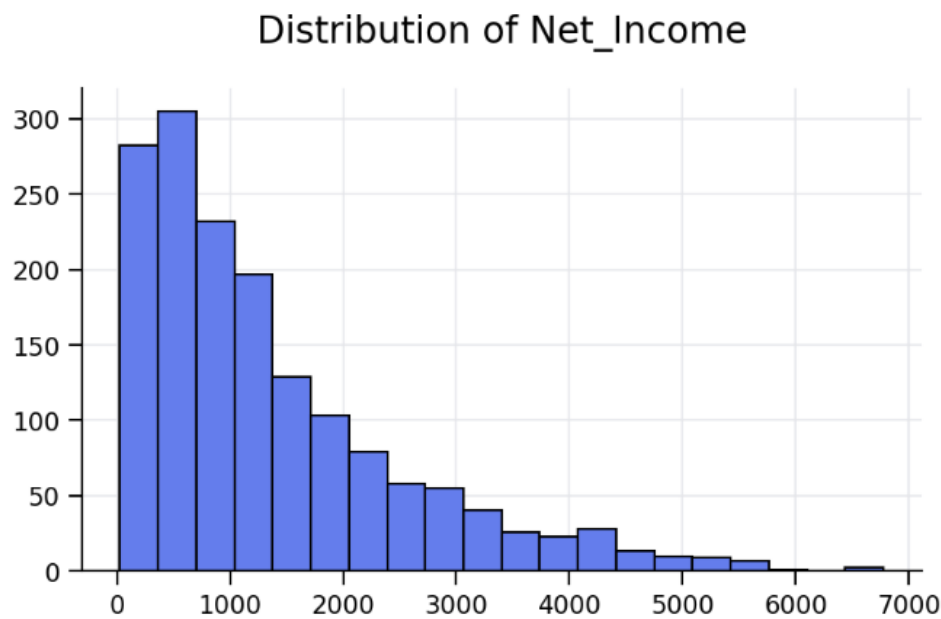
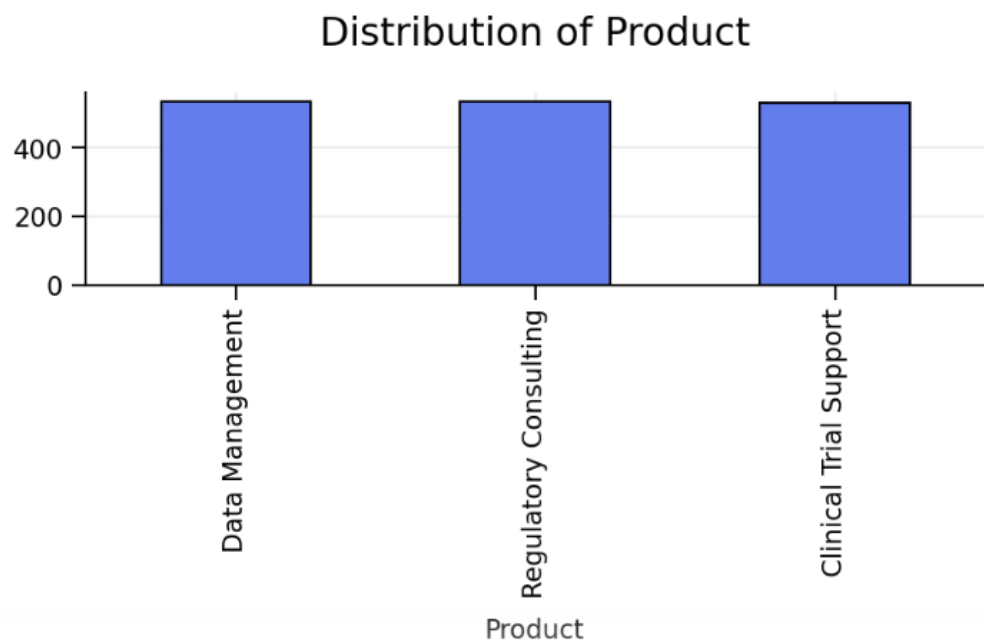
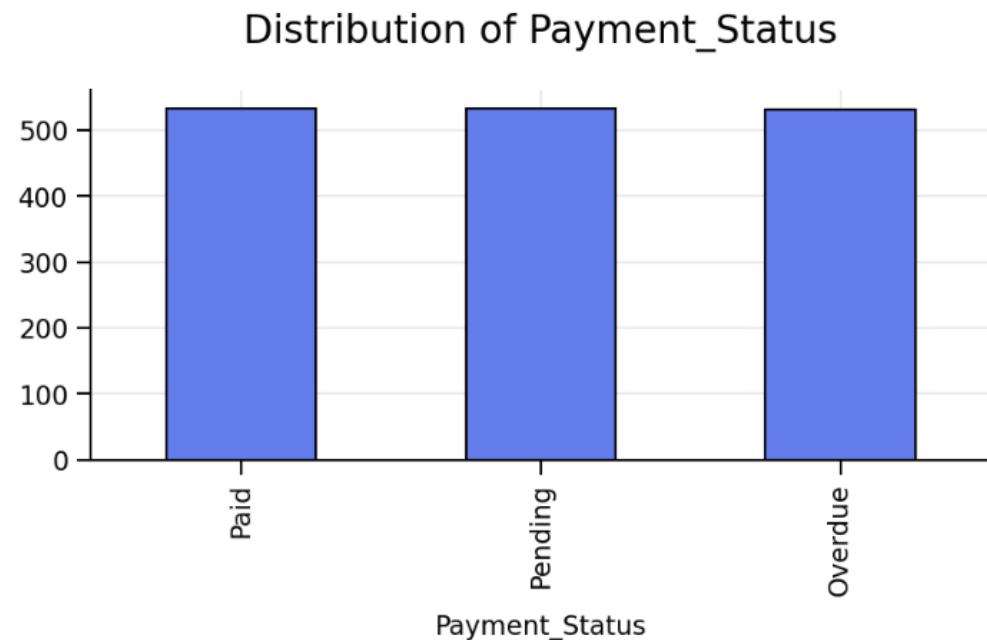
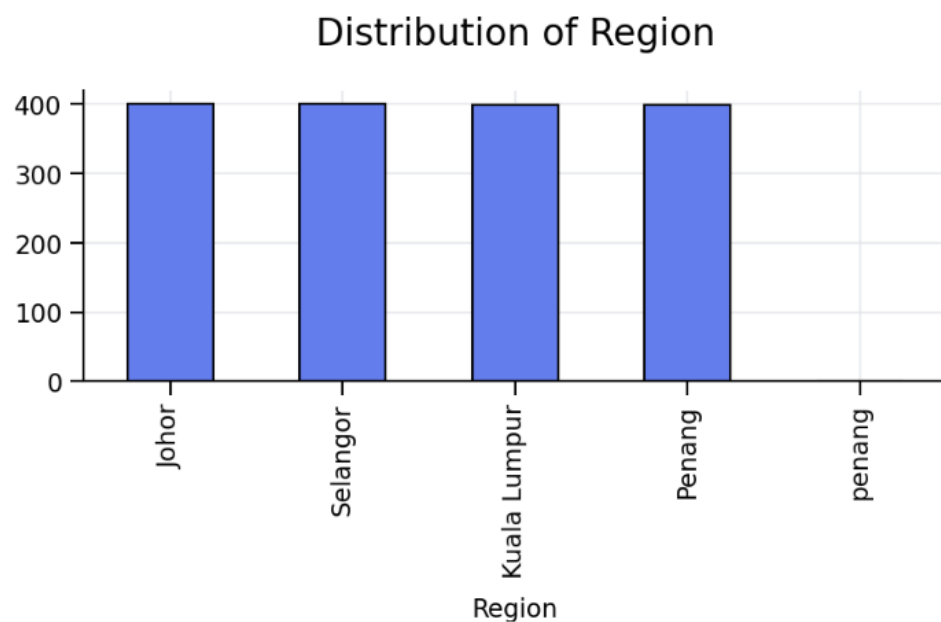
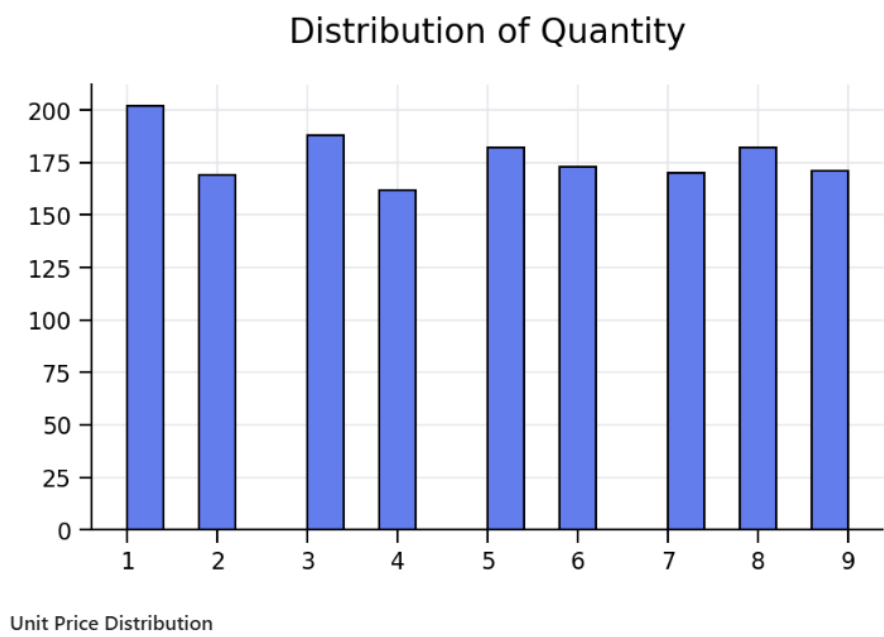
- Numeric formatting issues
- Scientific notation artifacts
- Outliers without flags
- Minor text inconsistencies

Recommended Next Steps

- **Normalize numeric formats:** Convert scientific notation to standard decimals, round monetary fields consistently.
- **Validate outliers:** Apply IQR/Z-score checks on Quantity and Revenue, flag extreme invoices for review.
- **Clean categorical text:** Standardize capitalization, trim and correct corrupted strings.
- **Recompute financials:** Recalculate Cost, Tax, Revenue, Net_Income from base fields, compare against stored values.
- **Prepare for analytics:** Create derived metrics (ARPU, overdue rate, avg delivery time), segment customers by region/service.



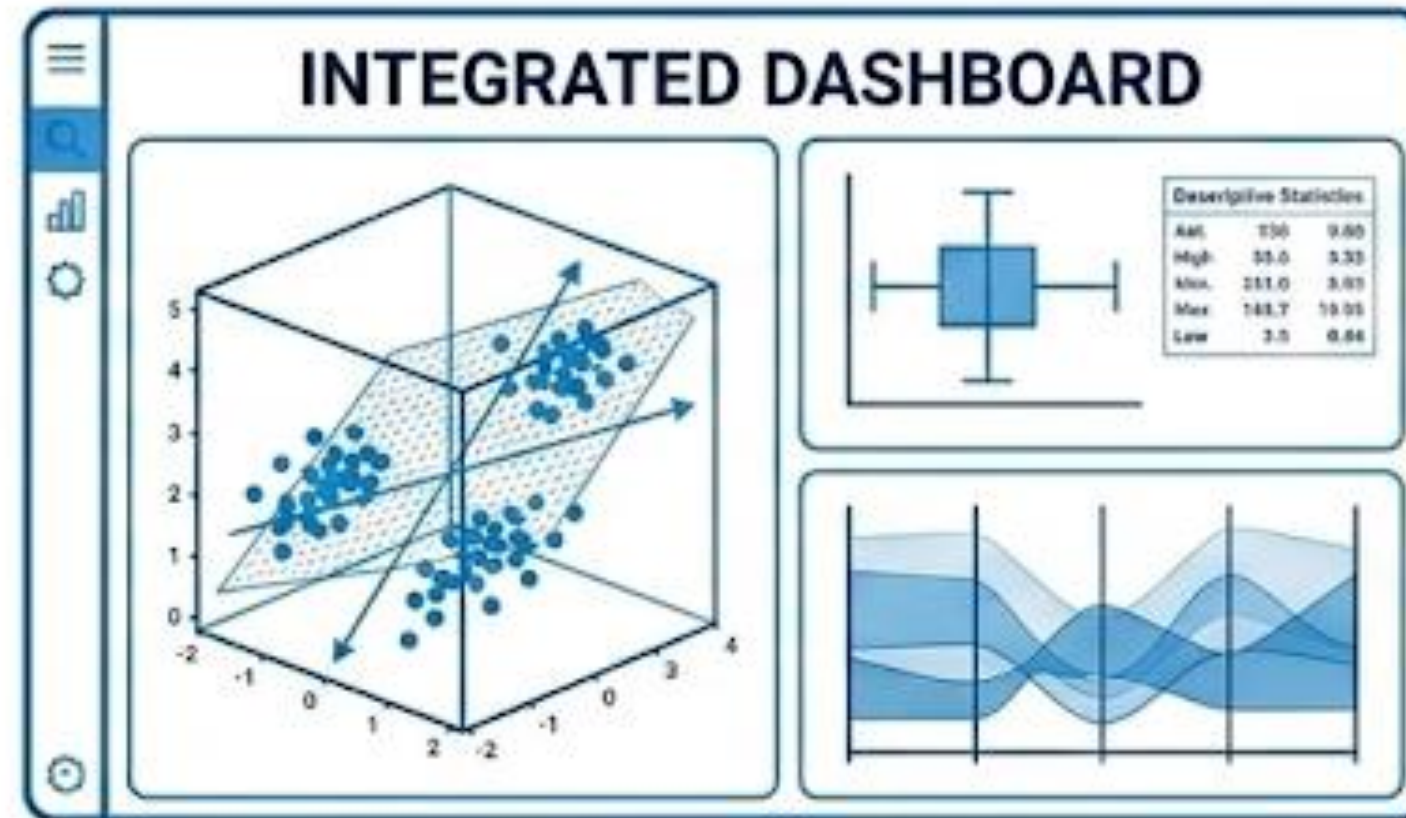
for each of the feature, generate the graph of the distribution





HANDS ON

Step 4 & 5: Data Analysis & Visualization (Focus: Applied Statistics & Advanced Visuals)





HANDS ON



Statistical Model & Forecasting

[Task 1] Revenue Distribution Analysis by Product Type

Analyze how Total_Revenue differs across:

- ProductData Management
- Regulatory Consulting
- Clinical Trial Support

Methods

Descriptive statistics:

- Mean, median, standard deviation of Total_Revenue

Visualization:

- Boxplots or histograms

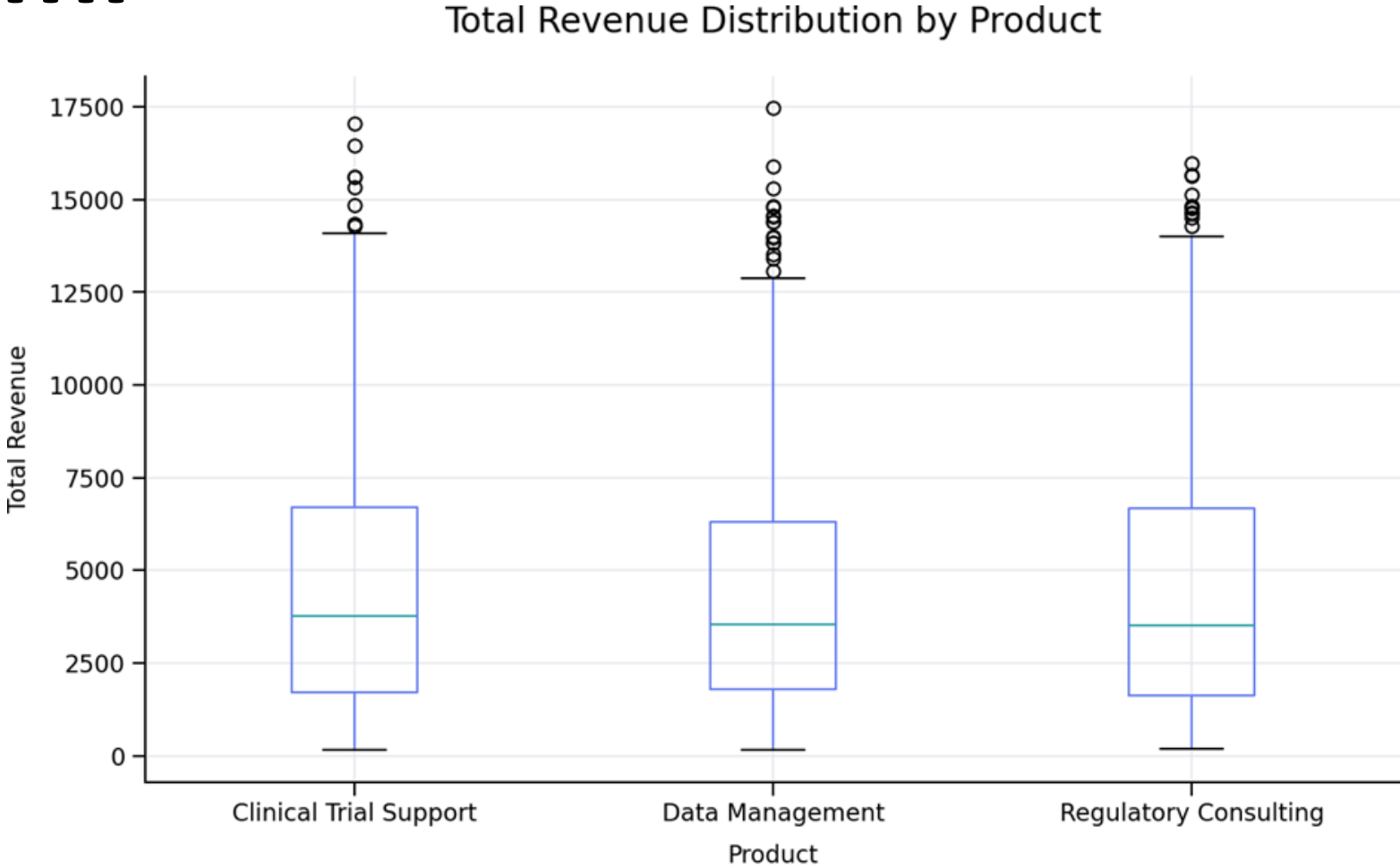


Task 1 – Result Generated by Dr. Aimi

Product	Mean	Median	Standard Deviation
Clinical Trial Support	4,653.96	3,774.19	3,504.52
Data Management	4,575.54	3,544.72	3,505.44
Regulatory Consulting	4,560.62	3,522.26	3,521.73

🔍 Interpretation

- Clinical Trial Support has the highest mean and median revenue, indicating slightly stronger revenue performance.
- Median values are consistently lower than means, suggesting right-skewed distributions (presence of high-value outliers).
- Standard deviation is high and similar across all products, indicating:
 - Large variation in invoice sizes
 - Comparable revenue volatility across services



🔍 Boxplot Insights

- All three products show:
 - Wide interquartile ranges
 - Multiple high-value outliers
- Clinical Trial Support exhibits:
 - Slightly higher upper quartile
 - More extreme upper outliers
- Revenue distributions across products are similar in shape, indicating comparable pricing structures with different deal sizes.

Task 1 – Result Generated by Dr. Dwi

Product Category	Records	Total Revenue	Average Revenue	Median Revenue	Std. Dev.
Clinical Trial Support	532	2,475,907.57	4,653.96	3,774.19	3,504.52
Data Management	534	2,443,338.05	4,575.54	3,544.72	3,505.44
Regulatory Consulting	533	2,431,503.48	4,561.92	3,513.51	3,524.91

Key Findings

1. Clinical Trial Support

- Generates the **highest total revenue: 2.476 million**.
- Has the **highest average revenue per transaction: 4,653.96**.
- Also has the **highest median revenue: 3,774.19**.
- This suggests Clinical Trial Support tends to produce slightly larger transactions than the other services.

2. Data Management

- Total revenue: **2.443 million**.
- Average revenue: **4,575.54**.
- Performs slightly below Clinical Trial Support but above Regulatory Consulting.
- Revenue distribution is very similar to the other categories.

3. Regulatory Consulting

- Total revenue: **2.432 million**.
- Average revenue: **4,561.92**.
- Has the lowest average and total revenue among the three categories, though the difference is relatively small.

Revenue Ranking

Based on **total revenue**:

1. **Clinical Trial Support** → 2,475,907.57
2. **Data Management** → 2,443,338.05
3. **Regulatory Consulting** → 2,431,503.48

Based on **average revenue per transaction**:

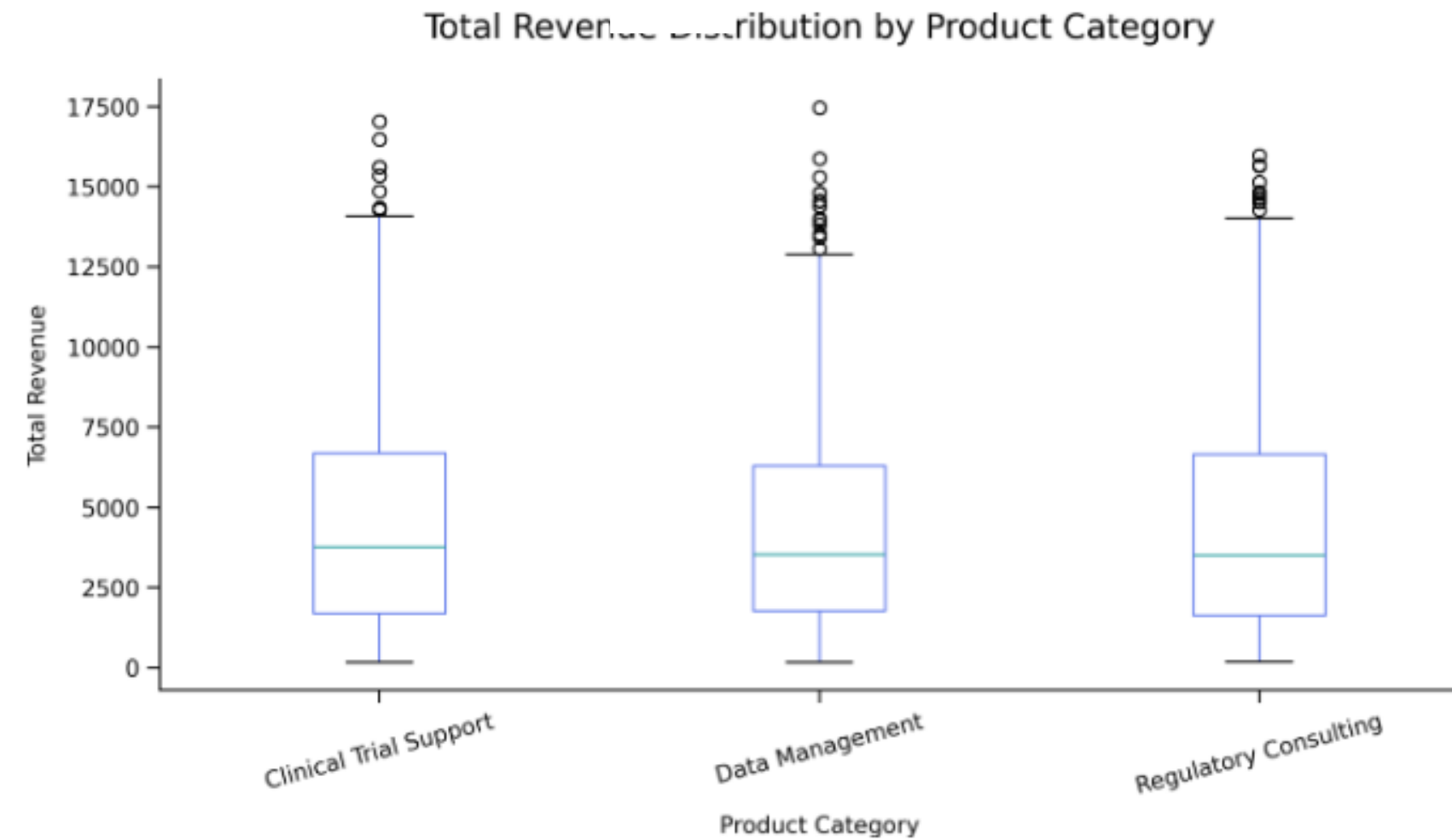
1. **Clinical Trial Support** → 4,653.96
2. **Data Management** → 4,575.54
3. **Regulatory Consulting** → 4,561.92

Conclusion

The three product categories perform very similarly, but **Clinical Trial Support** is the strongest revenue contributor, generating about:

- **1.33% more total revenue than Data Management**
- **1.83% more total revenue than Regulatory Consulting**

The standard deviations are nearly identical (~3,500), indicating that revenue variability is similar across all three services. The main difference is that Clinical Trial Support consistently achieves slightly higher transaction values.



Interpretation

- **Clinical Trial Support** has the highest median revenue (center line of the box), consistent with the earlier statistical summary.
- The **interquartile ranges (IQRs)** are very similar across all three categories, indicating comparable revenue dispersion.
- All three categories exhibit numerous **high-value outliers**, suggesting occasional exceptionally large transactions.
- The overall spread and variability are nearly identical, which aligns with the similar standard deviations (~3,500) observed in the descriptive statistics.

Overall, the boxplot confirms that while **Clinical Trial Support** tends to generate slightly higher revenues, the three product categories have very similar revenue distributions.

AutoSave OffPRUBSN_Tier 1_Business Data_modified

FileHomeInsertDrawPage LayoutFormulasDataReviewViewAutomateHelpAcrobat

CutCopyFormat PainterClipboard

Calibri11A⁺⁺
B I U
Font

Wrap TextMerge & Center
Alignment

General
\$ %
Number

Conditional FormattingFormat as TableCell Styles
Styles

InsertDeleteFormat
Cells

AutoSumFillClear
Editing

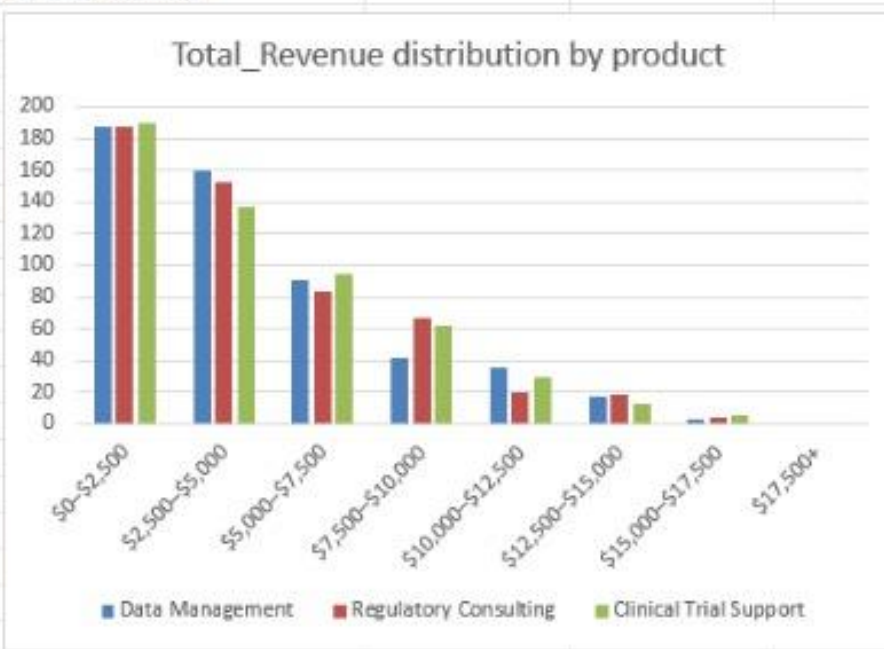
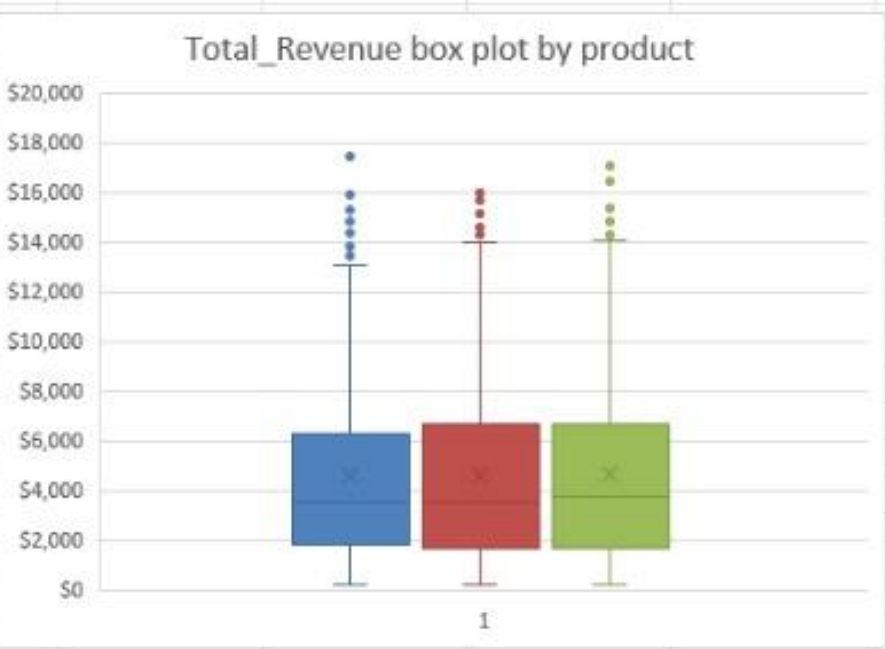
Sort & FilterFind & Select
Sensitivity

Add-ins
Add-ins

Create PDF and Share linkCreate PDF and Share via Outlook
Adobe Acrobat

Claude
Claude

H15

	A	B	C	D	E	F	G	H	I	J
12	2. Histogram of Total_Revenue by product (bin = lower ≤ x < upper)							3. Extracted Total_Revenue values (feeds the chart)		
13	Bin (from)	to (<)	Bin label	Data Management	Regulatory Consulting	Clinical Trial Support	All		Data Management	Regulatory Consulting
14	\$0	\$2,500	\$0-\$2,500	187	187	190	564		\$4,644	\$6,970
15	\$2,500	\$5,000	\$2,500-\$5,000	160	152	137	449		\$4,610	\$9,300
16	\$5,000	\$7,500	\$5,000-\$7,500	91	84	95	270		\$4,797	\$11,203
17	\$7,500	\$10,000	\$7,500-\$10,000	41	67	62	170		\$2,453	\$2,736
18	\$10,000	\$12,500	\$10,000-\$12,500	35	20	30	85		\$1,058	\$7,684
19	\$12,500	\$15,000	\$12,500-\$15,000	17	19	13	49		\$650	\$2,696
20	\$15,000	\$17,500	\$15,000-\$17,500	3	4	5	12		\$2,309	\$4,669
21	\$17,500	\$20,000	\$17,500+	0	0	0	0		\$8,104	\$4,560
22	Total			534	533	532	1599		\$3,531	\$7,513
23									\$11,878	\$956
24	4. Visualisations								\$4,634	\$3,514
25	Total_Revenue distribution by product  Total_Revenue box plot by product  <td></td> <td>\$7,059</td> <td>\$6,873</td>				\$7,059	\$6,873				
				\$3,939	\$10,557					
				\$9,506	\$7,996					
				\$5,733	\$8,411					
				\$3,917	\$214					
				\$2,067	\$3,550					
				\$2,392	\$3,343					
				\$989	\$6,484					
				\$3,555	\$1,747					
				\$4,022	\$9,607					
				\$5,014	\$1,620					
				\$9,864	\$13,318					
				\$1,161	\$10,994					
				\$4,486	\$13,578					
				\$8,318	\$5,707					
47		\$1,552	\$3,870							
48		\$2,168	\$1,252							

Sheet1DistributionRevenue by Product

ReadyAccessibility: Good to go

Claude

chart.

- Region frequency — four near-equal bars (~400 each)
- Quantity distribution — flat/uniform across 1-9
- Total_Revenue distribution — clear right-skewed decay from the \$0-\$2,500 bin down to \$17,500+

The tables in sections 1-4b were verified earlier by value/text read-back; the chart region I have now confirmed by rendered image.

by generating new sheet, analyze how total_revenue differs across
1. productdata Management
2. Regulatory Consulting
3. clinical Trial support
use methods descriptive statistic
mean, median, standard deviation of the Total_revenue
use visualization boxplot or histogram

Creating a new sheet with per-product descriptive stats, a grouped histogram, and a box plot.

Ran 6 scripts, edited cells >

H15 selected

Reply, or /skillify to save this process

+ Opus 5



HANDS ON



Statistical Model & Forecasting

[Task 2] Forecastable Trend: Future Total Revenue Trend

Trend to forecast:

Total Revenue over invoice sequence or time proxy
(If no explicit date column exists, invoice number order can be used as a proxy for time.)

Forecasting approach

Aggregate:

- Total revenue per invoice batch or period

Apply time-series methods:

- Moving average
- Exponential smoothing
- Linear trend projection

Visualization:

- Actual aggregated revenue
- Moving average
- Exponential smoothing
- Linear trend projection

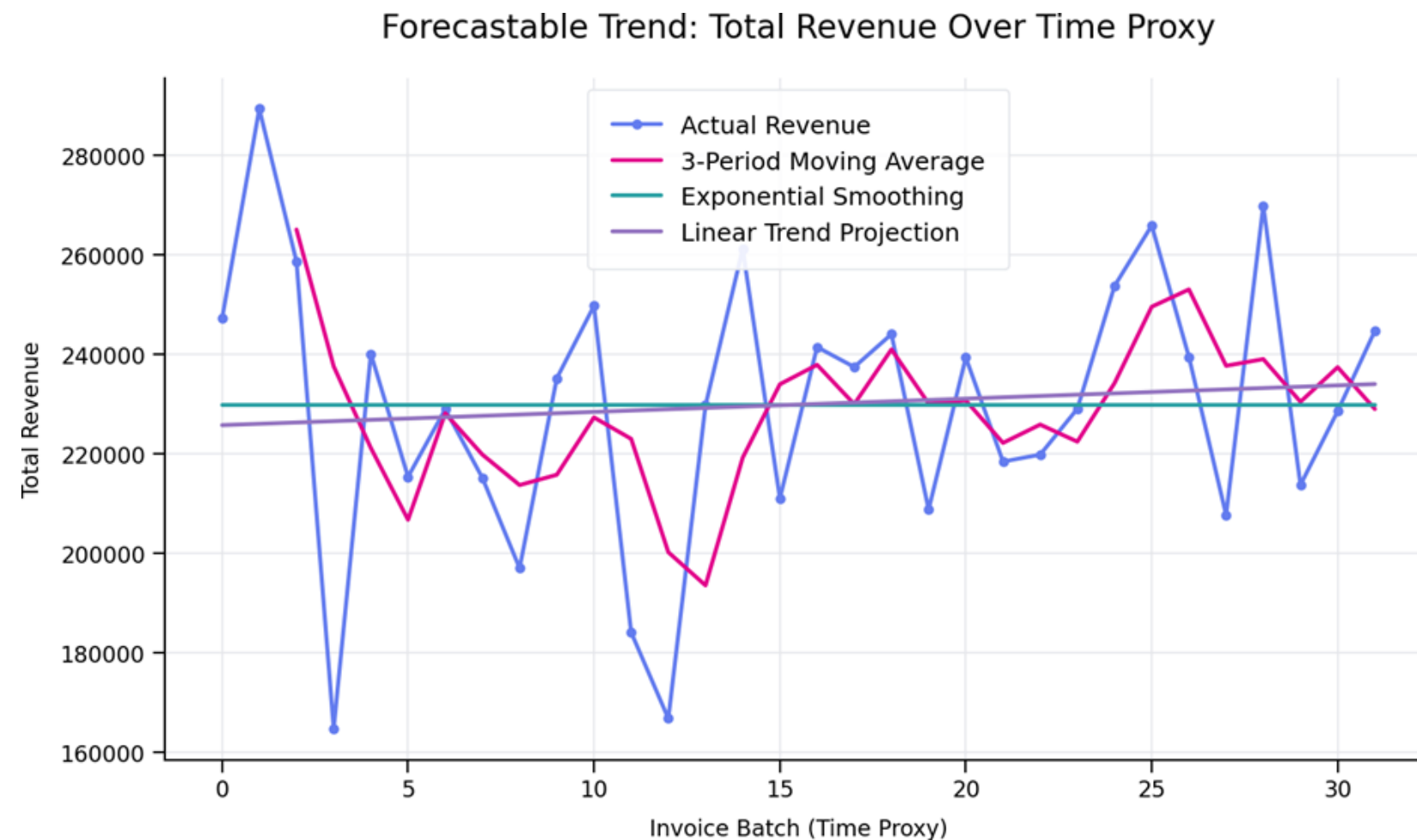
PROMPT

now generate new sheet, forecast trend. trend to forecast: total revenue over invoice sequence or time proxy (if no explicit date column exists, invoice number order can be used as a proxy of time) forecasting

approach aggregate : total revenue per invoice batch or period apply time series method 1) moving average 2) exponential smoothing 3) linear trend projection

visualization actual aggregated revenue moving average exponential smoothing linear trend projection

Task 2 – Result from Dr Aimi



✓ Interpretation of Results

🔍 Key Observations

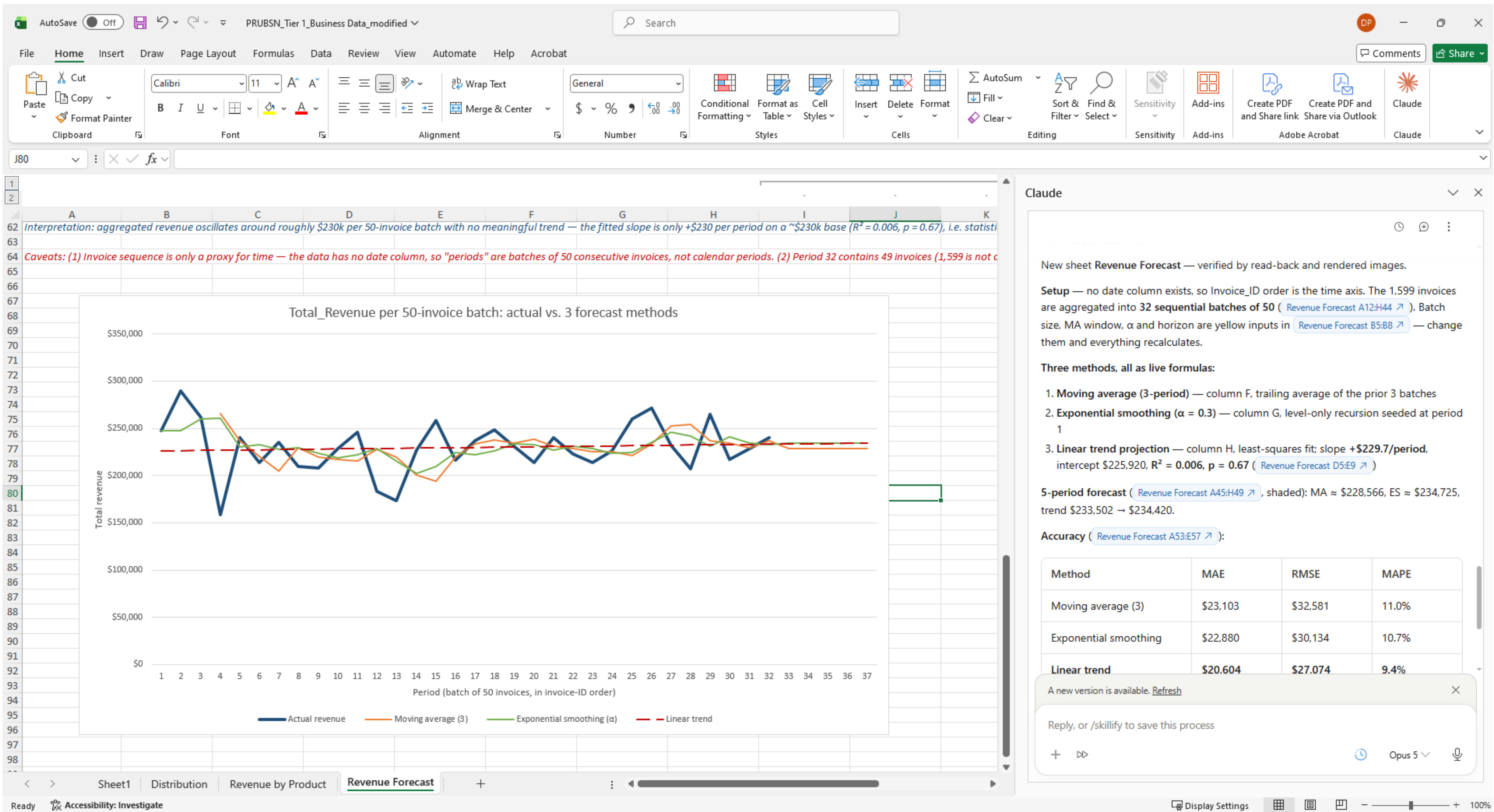
Revenue is generally stable with mild growth
Outliers exist, but aggregation reduces their impact
No evidence of sustained revenue decline
Linear trend suggests incremental long-term growth

🧠 What Can Be Forecasted

Near-term revenue using exponential smoothing
Medium-term revenue trajectory using linear trend
Expected revenue per upcoming invoice batches

✓ Conclusion (Sample)

Using invoice sequence as a time proxy, total revenue exhibits a stable pattern with gradual positive growth. Moving average and exponential smoothing models indicate short-term stability, while linear trend projection suggests modest long-term revenue expansion. These findings support the use of trend-based forecasting for budgeting and sales planning in the absence of explicit temporal data.



Make your data work for you, not the other way around.

~~Manual-labor more.~~
~~Analyze less.~~
Analyze more.
Manual-labor less.

AI for Productivity: Automation Using Microsoft Teams Workflows

Work smarter and eliminate
repetitive manual tasks.



Trainers: Assoc. Prof. Ts. Dr. Hasan Firdaus Bin Mohd Zaki
Dr. Dwi Pebrianti, Eng. Tech., EPE (IPU), ASEAN. Eng.

Center for Unmanned Technology, International Islamic University Malaysia



Core Objectives for Today's Session



Understand Microsoft Teams Workflows

Discover the no-code engine powered by Power Automate.



Automate Repetitive Tasks

Let Microsoft handle manual follow-ups, scheduling, and notifications.



Create Your First Workflow

Build and test a working automation live in this session.



Apply to Real Scenarios

Identify and deploy automations specific to your role at Prudential Takaful.

Why Automation is a Competitive Advantage



The Core Logic: Trigger, Action, Result



Built natively in Teams. Powered by Microsoft Power Automate. 100% No-code / Low-code.

Locating the Workflows App Inside Teams

1

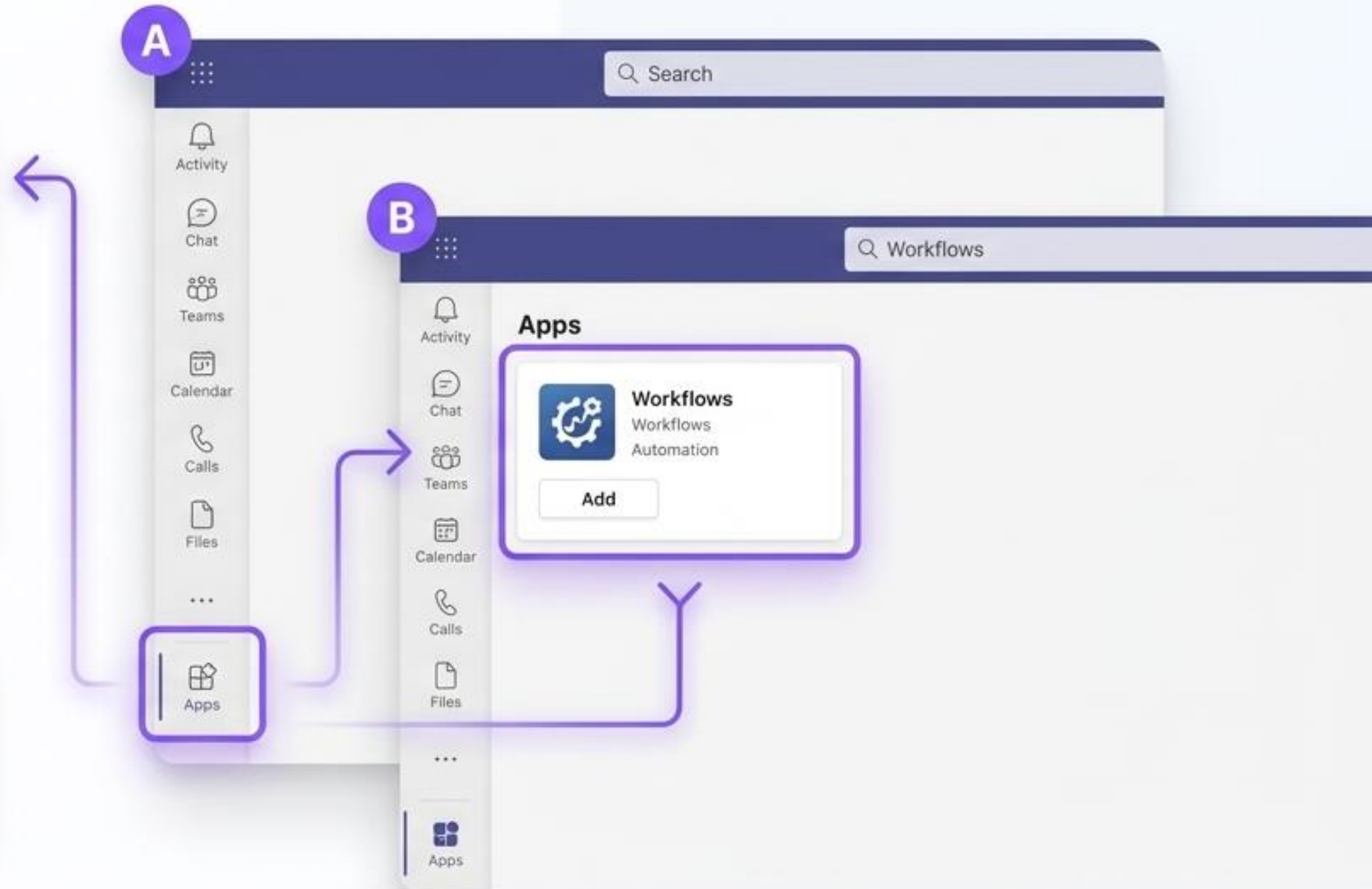
Open Microsoft Teams and click 'Apps' (bottom left corner of the sidebar).

2

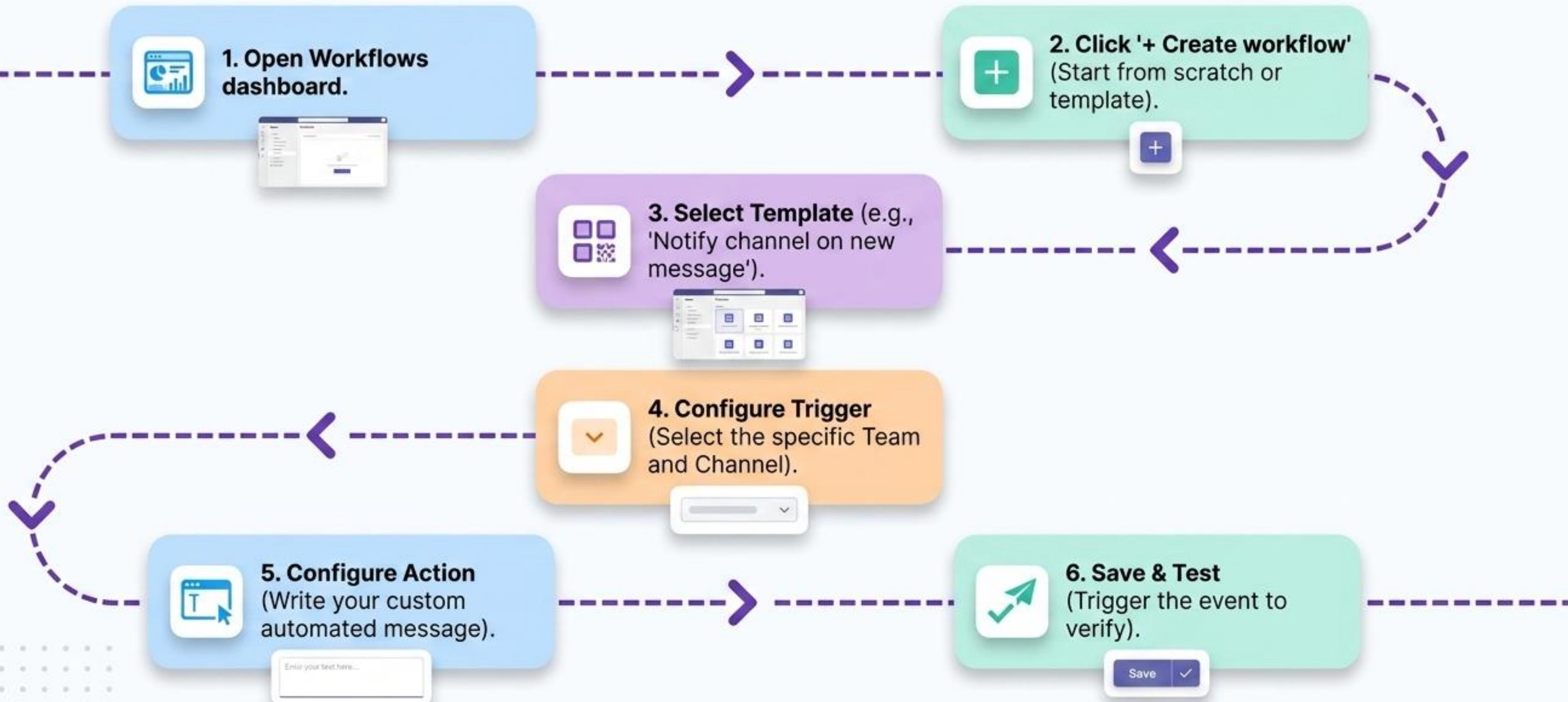
Search for 'Workflows' in the top app search bar.

3

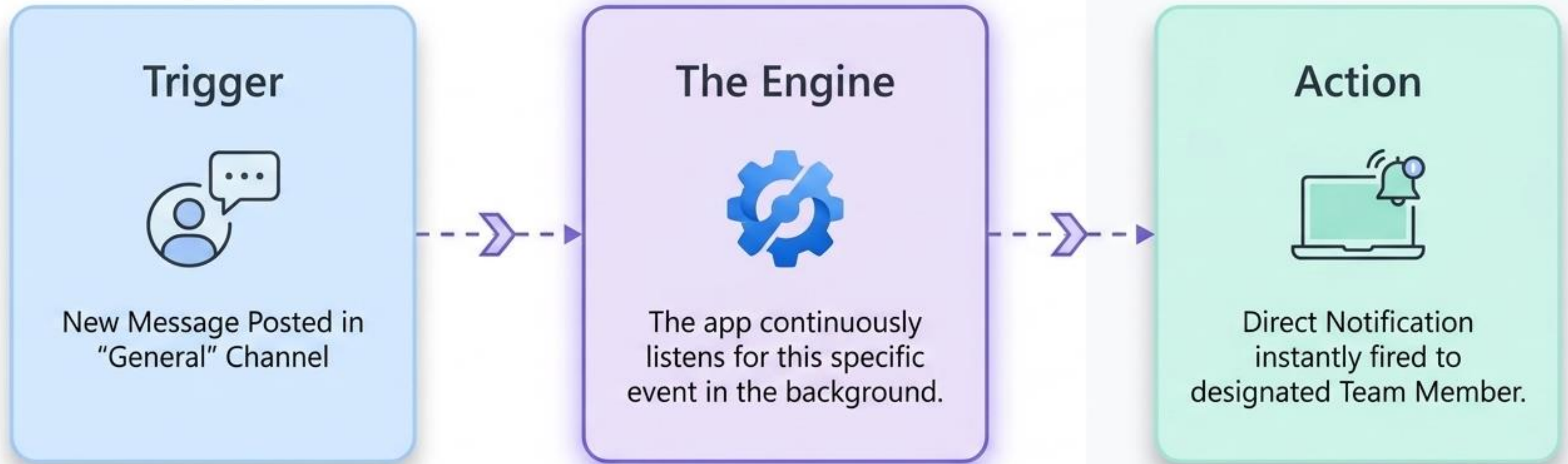
Click the Workflows icon to open the new v2 dashboard (featuring Home, Templates, and Chat tabs).



The Path to Your First Automation



Blueprint: Channel Notification Automation



[General | MECH 2326 AEROSPACE DYNAMICS](#)

Live Demonstration: Watch the Screen



The Demo Scenario

Trainer will build: "Help me prepare for my day" template.

Result: An automated 8:00 AM daily briefing of meetings and tasks sent via Teams chat.

What to Observe

- ✓ Browsing the "Templates" vs "Build from scratch" tabs.
- ✓ Setting the correct Time Zone (crucial for scheduled triggers).
- ✓ Renaming the workflow before saving so it is easily identifiable.
- ✓ Approving app connection permissions ("Sign in").

Hands-On Exercise: Build Your Channel Notification

Your Mission:

When someone posts in a selected channel, send me a direct Teams notification.

1. Open Teams -> Apps -> Workflows.
2. Search the template library for: "Notify me when..."
3. Configure the drop-downs to match your specific testing channel.
4. Save the workflow.
5. Test it: Post a real message in that channel and wait for the ping!



**Estimated time:
5–10 minutes.**

Preparation

Create the team and channel.

The manual uses "PRU BSN AI Training" with a channel called "Sandbox". If you use different names, change them in the document. The channel must be a **standard channel**, because Workflows can't post messages to private channels.

Do one dry run on a PRU BSN account.

Microsoft renames templates and buttons often, so the manual uses flexible wording (for example, "Add workflow / Create / Save"). After your dry run, you can replace these with the exact names and add a screenshot to each step.

Warn participants about the delay.

The notification can take 1–3 minutes to arrive, so tell them not to give up after a few seconds. The manual says this too.



Activity



Chat



Teams



Assignments



Calendar



Calls



OneDrive



Store





<

>

Search (Ctrl+E)

...





< All teams

PB

General

Posts

Shared

Page

Power Automate



Meet now

...

+ New flow

Search

Flows



You don't have any flows

Choose from 100s of templates to start automating your workflows.

Switch to your personal developer environment

Teams templates you might like



Forward new posts in a channel to Outlook



Send webhook alerts to a channel



Forward your emails to a channel

The Result

Activity

Chat

Teams

Assignments

Calendar

Calls

OneDrive

Copilot

Store

All teams

PB

PRU BSN AI Training

Home page

Class Notebook

Classwork

Assignments

Grades

Insights

Reflect

Main Channels

General

General

Posts

DWI PEBRIANTI

9:44 AM

I added a tab at the top of this channel. Check it out!

Power Automate

Reply

DWI PEBRIANTI

9:56 AM

Testing my workflow - Dwi Pebrianti

Just testing

Reply

Post in channel

Mail - DWI PEBRIANTI - Outlook

Ask Gemini

https://outlook.office.com/mail/inbox/id/AAQkADZmYTJiNzM2LTlkODMtNGJhNC1h...

Outlook

Search

File

Home

View

Help

New mail

Translator for Outlook

iPlanner Pro Office 365

Adobe Acrobat for Micros...

Viva

Favorites

Inbox 562

Sent Items

Drafts

Deleted Items 2

dwipebrianti@live.i...

Inbox 562

Drafts

Sent Items

Deleted Items 2

Junk Email [1]

Notes

Archive

Conversation Hi...

RSS Feeds

Search Folders

Go to Groups

Focused

Other

Today

Last Month

July

DWI PEBRIANTI

New Teams message in G...

9:58 AM

New Teams message in General Subject...

MA

Microsoft Power Automate

1 of your flow(s) have failed

Tue 8/25

The flow(s) listed failed in the past wee...

MA

Microsoft Power Automate

5 of your flow(s) have failed

8/2/2026

The flow(s) listed failed in the past wee...

DP

DWI PEBRIANTI ; IMAN ...

Application to Register ...

7/29/2026

Salam Dr Azam and Dr Fazrina, May thi...

UG12-Applicati...

DWI PEBRIANTI

Insurance Claim Summ...

7/27/2026

AI PRU INSURANCEClaim Summary...

DWI PEBRIANTI

Insurance Claim Summ...

7/27/2026

AI PRU INSURANCEClaim Summary...

DWI PEBRIANTI

Status of Your Claim - POL...

7/27/2026

Dear Dear Aimi, Thank you for...

DWI PEBRIANTI

Claim POL98766 Needs...

7/27/2026

Dear Claim for Nasmah Ammount RM...

New Teams message in General

Summarize this email

DWI PEBRIANTI

To: +1 other

Sun 9/20/2026 9:58 AM

New Teams message in General

Subject: Testing my workflow - Dwi Pebrianti

From: DWI PEBRIANTI

Just testing

View message in Teams

Reply

Forward

Live Troubleshooting & Workflow Tips



Common Mistakes

- Forgetting to set the correct Time Zone in scheduled workflows (defaults to UTC).
- Selecting the wrong Team/Channel in the dropdown menus.
- Forgetting to rename the workflow from its default string.



Where to Click

Look for the blue “Add connection” or “Sign in” buttons.

Teams requires you to explicitly grant permission the very first time you use Power Automate.



Pro Tip

Always click the “Run History” tab inside your workflow details.

It will explicitly tell you if a trigger succeeded or failed, and exactly why.

Expanding the Ecosystem: Advanced Integrations



Microsoft Forms

Turn static survey responses into actionable Teams tasks or channel announcements the moment they are submitted.



SharePoint

Track specific document library changes. Never manually check if a file has been updated again.



Approvals

Build multi-step approval chains that allow managers to click 'Approve' or 'Reject' directly inside a Teams chat bubble.



AI & Copilot

For users with Copilot licenses, build complex Workflows simply by typing natural language prompts in the 'Chat' tab.

Essential Rules for Teams Workflows



Workflows Live Inside Teams

There is no need to open web browsers or secondary software. The Home, Templates, and Chat tabs inside the Teams app handle everything.



Templates Are Your Best Friend

Never start from scratch unless necessary. The vast catalog of pre-built triggers covers 90% of business use cases.

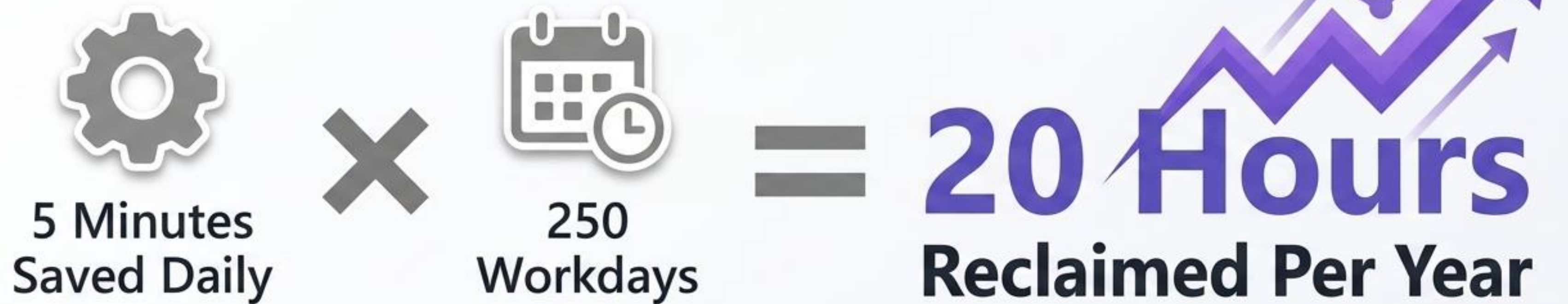


Think "Trigger -> Action"

Train your brain to recognize repetitive friction. If a process starts with "When X happens..." it can be automated.



The Compound Effect of the AI Mindset



The diagram illustrates the compound effect of an AI mindset. It features a gear icon representing a task, a large 'X' for multiplication, a calendar icon representing workdays, an equals sign, and a large upward-pointing arrow representing reclaimed time. The text '5 Minutes Saved Daily' is positioned below the gear, '250 Workdays' below the calendar, and '20 Hours Reclaimed Per Year' to the right of the equals sign.

$$\begin{array}{ccccc} \text{5 Minutes} & \times & \text{250} & = & \text{20 Hours} \\ \text{Saved Daily} & & \text{Workdays} & & \text{Reclaimed Per Year} \end{array}$$

You cannot break Microsoft Teams by experimenting with Workflows.
Look at your calendar tomorrow. Find one recurring task. Automate it.
Welcome to the future of work at Prudential Takaful.

CHALLENGES

Choose Your Level

Start at Level 1. Finished? Move up to the next level.

LEVEL 1



Meeting Reminder



Easy

Scheduled reminder
before stand-up

LEVEL 2



File Upload Alert



Easy

Ping when a new file
lands in SharePoint

LEVEL 3



Form Submission



Medium

Chat summary when a
form is submitted

LEVEL 4



Task Visibility



Medium

Announce completed
Planner tasks

Tier 2 Power Automate

LEVEL 1 OF 5



Automatic Meeting Reminder

★☆☆ Easy

“Please provide an update on your tasks before the 10 AM stand-up.”

HINT 1

WHEN

A schedule, e.g. every weekday at 9:45 AM → trigger Recurrence

DO

Post a message → Post message in a chat or channel

HINT 2



Search keyword: “Schedule” or “Recurrence”

HINT 3



Key setting: Time zone = (UTC+08:00) Kuala Lumpur, Singapore



Test tip: Set the time to 5 minutes from now, wait for it, then change it back to the real time.

Task 1

Activity

Chat

Teams

Assignments

Calendar

Calls

OneDrive

Copilot

Workflows

Store

...

Workflows

Home

Templates

Chat

Present now

Templates > Every Sunday on 10:35 AM, ...

Every Sunday on 10:35 AM, send a prompt to Updates in General

Send recurring prompts to an agent in Teams to keep work moving. Turn conversations into consistent execution with insights, and actions delivered where your team collaborates.

Active Turn off Run Edit ...

Next run: 9/27/2026 10:35:00 AM Last modified: 9/20/2026 10:34:59 AM

Run history

Start time	Status
9/20/2026 10:34:59 AM (1 m ago)	Succeeded

Owners

Adding an owner gives them full control of this flow. They'll be able to add or remove other users as owners, access the run history, and can update, edit or delete this flow. [Learn more](#)

DWI PEBRIANTI

Owner

Add a co-owner

Enter names, email or user groups

Mail - DWI PEBRIANTI - Outlook

Ask Gemini

https://outlook.office.com/mail/inbox/id/AAQkADZmYTJiNzM2LTfkODMtNGJhNC1h...

Outlook

Search

File Home View Help

New mail

Translator for Outlook

iPlanner Pro Office 365

Adobe Acrobat for Micros...

Viva

Favorites

Inbox 561

Sent Items

Drafts

Deleted Items 2

dwipebrianti@live.i...

Inbox 561

Drafts

Sent Items

Deleted Items 2

Junk Email [1]

Notes

Archive

Conversation Hi...

RSS Feeds

Search Folders

Go to Groups

Focused Other

Today

DWI PEBRIANTI

New Teams message in ... 10:36 AM

New Teams message in General From:...

DWI PEBRIANTI

New Teams message in Gen... 9:58 AM

New Teams message in General Subject...

Last Month

MA Microsoft Power Automate

1 of your flow(s) have failed Tue 8/25

The flow(s) listed failed in the past wee...

MA Microsoft Power Automate

5 of your flow(s) have failed 8/2/2026

The flow(s) listed failed in the past wee...

July

DP DWI PEBRIANTI ; IMAN RA...

Application to Register for... 7/29/2026

Salam Dr Azam and Dr Fazrina, May thi...

UG12-Applicati...

DWI PEBRIANTI

Insurance Claim Summ... 7/27/2026

AI PRU INSURANCEClaim Summary...

DWI PEBRIANTI

Insurance Claim Summ... 7/27/2026

AI PRU INSURANCEClaim Summary...

DWI PEBRIANTI

Status of Your Claim - POL... 7/27/2026

Dear Dear Aimi, Thank you for...

New Teams message in General

Summarize this email

DWI PEBRIANTI

To: +1 other Sun 9/20/2026 10:36 AM

New Teams message in General

From: DWI PEBRIANTI

Updates update on project

[View message in Teams](#)

Reply Forward

LEVEL 2 OF 5



File Upload Alert

★☆☆ Easy

Get pinged when a client drops a new policy document into a SharePoint folder.

HINT 1



When a file is created
(properties only)



Post message in a chat or
channel

HINT 2



Search keyword: “SharePoint” or “file”

HINT 3



Shortcut: Your channel's Files tab is a SharePoint folder: pick the team's site and the Sandbox channel folder.



Test tip: Upload any small file to the channel's Files tab.

Task 2

Activity

Chat

Teams

Assignments

Calendar

Calls

OneDrive

Copilot

Store

< All teams

PB

PRU BSN AI Training

Home page

Class Notebook

Classwork

Assignments

Grades

Insights

Reflect

Main Channels

General

PB

General

DWI PEBRIANTI 9:44 AM

I added a tab at the top of this c

Power Automate

Reply

DWI PEBRIANTI 9:56 AM

Testing my workflow - D

Just testing

Reply

DWI PEBRIANTI 10:35 AM

Updates update on project

Reply

DWI PEBRIANTI 11:07 AM

This message has been de

Post in channel

Activity

Chat

Teams

Assignments

Calendar

Calls

OneDrive

Copilot

Store

< All teams

PB

PRU BSN AI Training

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Insights

Reflect

Main Channels

General

PB

General

DWI PEBRIANTI 9:56 AM

Testing my workflow - Dwi Pebrianti

Just testing

Reply

DWI PEBRIANTI 10:35 AM

Updates update on project

Reply

DWI PEBRIANTI 11:07 AM

This message has been deleted. [Undo](#)

Post in channel

Documents > General

A..

Class Materials

CV Dwi Pebrianti 20260809.pdf

certificate-speaker-ttd.pdf

Loop paragraph.loop

Uploaded CV Dwi Pebrianti 20260809.pdf to General

Activity

Chat

Teams

Assignments

Calendar

Calls

OneDrive

Copilot

Store

< All teams

PB

PRU BSN AI Training

Home page

Class Notebook

Classwork

Assignments

Grades

Insights

Reflect

Main Channels

General

PB

General

DWI PEBRIANTI 9:56 AM

Testing my workflow - Dwi Pebrianti

Just testing

Reply

DWI PEBRIANTI 10:35 AM

Updates update on project

Reply

DWI PEBRIANTI 11:07 AM

This message has been deleted. [Undo](#)

Post in channel

Workflows

11:41 AM

New

Hello

There is new file uploaded on our shared folder

Reply

LEVEL 3 OF 5



Form Submission Notification

★★★☆☆ Medium

Receive a direct chat summary when an internal HR or IT support form is submitted.

HINT 1

⚡ WHEN

When a new response is submitted (Forms)

⚙ DO

Get response details, then Post message in a chat or channel

HINT 2



Search keyword: "Forms"

HINT 3



What you need first: A short Microsoft Form with 2 questions



Common mistake: Skipping Get response details: without it, the message can't show what the person answered.



Test tip: Fill in your own form and check your chat.

Task 3

Create Microsoft Form

Question 1:

Employee Name

Question 2:

Request Type

- Policy issue
- Claims issue
- IT support
- Customer enquiry

Question 3:

Describe the issue

[PRU BSN 21 September 2026 – Fill out form](#)



Mail - DWI PEBRIANTI - Outlook x PRU BSN 21 September 2026 x + Ask Gemini

https://forms.cloud.microsoft/Pages/ResponsePage.aspx?id=9nbOGPSZf0WfYqzWXZuXd1txm0RhxdDlX9jC1EhadUMTISMzg1N0Y5VFo5VE41WEIzMEFIMDI1NSQIQCN0PWcu&origin=Invitation&channel=0

PRU BSN 21 September 2026

When you submit this form, it will not automatically collect your details like name and email address unless you provide it yourself.

1. Employee Name

Enter your answer

2. Request Type

☐ Policy Issue

☐ Claims Issue

☐ IT support

☐ Customer enquiry

3. Describe the issue

Enter your answer

Submit

Never give out your password. [Report abuse](#)

Task 3

When a new response is submitted

* Form Id

9nbOGPSZf0WfYqzWXZuXd1txm0RhxcdDlx9jC1EhadUMTISMzg1N0Y5V
Fo5VE41WEIzMEFIMDI1NSQJQCNO0PWcu

Post message in a chat or channel

* Post as

Flow bot

* Post in

Channel

* Team

PRU BSN AI Training

* Channel

General

* Message

Font 12 B I U 
Hello
there is someone who filled in the form

PRU BSN 21 September 2026

1. Employee Name

Tati

2. Request Type

☐ Policy Issue

☒ Claims Issue

☐ IT support

☐ Customer enquiry

Activity

Chat

Teams

Assignments

Calendar

Calls

OneDrive

Copilot

Store

< >

Search (Ctrl+E)

< All teams

PB

PRU BSN AI Training

Home page

Class Notebook

Classwork

Assignments

Grades

Insights

Reflect

Main Channels

General

PB General

Posts

Reply

Last read

Workflows 12:31 PM New

Hello

there is someone who filled in the form

Reply

Workflows 12:34 PM New

Hello

there is someone who filled in the form

Reply

Post in channel

Workflows

Quickly automate tasks using smart templates and AI

App features

Tabs

Add as a tab at the top of a group chat, channel, or meeting

Bots

Complete tasks, find info, and chat using prompts

Messages

Insert content from the app directly into messages

Personal app

Keep track of important content and info

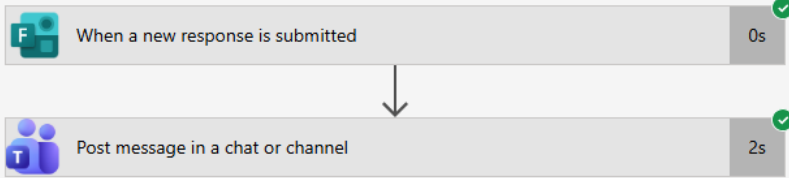
Created by: Microsoft Corporation

Permissions

Receive messages and data that I provide to it.

Send me messages and notifications.

By using Workflows, you agree to the [privacy policy](#), [terms of use](#), and [permissions](#).



LEVEL 4 OF 5



Task Assignment Visibility

★★★ Medium

Notify a channel when a Planner / To-Do task is marked complete.

HINT 1

 **WHEN**

When a task is completed
(Planner)

 **DO**

Post message in a chat or
channel

HINT 2



Search keyword: “Planner”

HINT 3



What you need first: A Planner plan with at least one task (add a Planner tab to your channel)



Test tip: Tick a task as complete and wait a few minutes.

Task 4

Planner Task marked as completed

PRU BSN AI Training

To do

- Prepare training slide
- Set up demo environment
- Invite participants

Completed ✓ 1

- Create registration form

Task marked as complete

Microsoft Teams Automatic notification in channel

PRU BSN AI Training

General

Power Automate 10:24 AM

Task Completed

A Planner task has been marked as complete.

Task: Create registration form
Plan: Workshop Tasks
Completed by: Dwi Pebrianti
Completed at: 19 Sept 2026, 10:24 AM

Great job! 🎉

Reply

Post in channel

Power Automate Flow

2

Microsoft Planner When a task is completed

Group Id: PRU BSN AI Training

Plan Id: Workshop Tasks

Microsoft Teams Post message in a chat or channel

Post as: Flow bot

Post in: Channel

Team: PRU BSN AI Training

Channel: General

Message

Task Completed

Task: {Title}
Plan: {Plan name}
Completed by: {Completed by}
Completed at: {Completed date time}

Great job! 🎉

LEVEL 5 OF 5



Approval Request Automation

★ ★ ★ Harder

Route leave or document approvals automatically to managers via Teams chat.

HINT 1



WHEN

Someone submits a request, e.g. a Microsoft Form → When a new response is submitted



DO

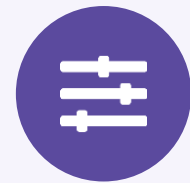
Start and wait for an approval, then post the result

HINT 2



Search keyword: “Approval”

HINT 3



Key setting: For practice, set yourself as the approver



Test tip: You'll get Approve / Reject buttons in Teams. Click one and check that the workflow continues.



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NEXT LEVEL

See you in the next level

See you in the next level of the training
course.

Thank you for your participation!

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